

# A STRIP INEQUALITY FOR PROJECTIONS, TWO-VARIABLE BMV POSITIVITY, AND OPTIMAL CGLMP MEASUREMENTS IN EVERY DIMENSION

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**ABSTRACT.** We prove that for every number of outcomes  $d \geq 2$  the measurements of Durt, Kaszlikowski and Żukowski (DKZ) maximise the CGLMP Bell expression  $I_d$  over all projective measurements on maximally entangled states of any local dimension  $D$ , the maximal value being  $I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d}$ , and that they are the only maximisers: equality forces  $d \mid D$  and, up to a local unitary  $u \otimes \bar{u}$ , the strategy is the DKZ strategy tensored with the identity. For projective measurements, this proves the first clause of Part B of Open Quantum Problem 27 of IQOQI Vienna.

The proof has an analytic part and a computer-assisted part. The analytic part is an inequality for a projection  $B$  and a Hermitian matrix  $g$  of arbitrary size: the sum of an explicit harmonic function  $h_\lambda$  of the strip  $0 < \operatorname{Re} z < 1$  over the eigenvalues of  $B + ig$  dominates  $\operatorname{Tr}[(P(g - \lambda)P)_+]$ ,  $P = \mathbf{1} - B$ , with an exact formula for the defect. It follows from a two-variable positivity theorem of Bessis–Moussa–Villani (Stahl) type for projections with an explicit density  $\frac{1}{2\pi} \sum_i |\operatorname{Im} \xi_i(s, \tau)|$ , where the  $\xi_i$  are the roots of  $\det(g - s - \xi(P - \tau))$ ; the proof uses the Fourier-slice mechanism of Heinävaara’s tracial joint spectral measures. Through a noncommutative continuum version of the Stein–Weiss rearrangement theorem and a family of cell embeddings, the strip inequality reduces the CGLMP problem to a statement about Clausen kernels: an explicit vector lies in a convex cone generated by explicit cell vectors. We verify this cone condition for every  $d$  by interval arithmetic, with linear-programming certificates on rational cells for  $d \leq 200$ , certified Gaussian-modulated Dirichlet constructions for  $201 \leq d \leq 2000$ , and a uniform interval Taylor-model argument for  $d \geq 2001$ .

We also give a second, distribution-free proof of the two-variable positivity theorem, through a Radon identity for matrix pencils whose proof rests on a complex inviscid Burgers equation for the pencil roots. The argument has been formalised in Lean 4 with Mathlib: for  $2 \leq d \leq 20$  the formal proof of optimality and rigidity is complete, including the cone condition; for  $d \geq 21$  it is complete except for the cone condition, which is established by the interval-arithmetic certificates; and the strip inequality and the positivity theorem are formalised in full.

We also give partial answers to the two other clauses of Part B. The noise-resistance clause holds for every  $d$  for the violation of the CGLMP inequality under white noise, with the DKZ measurements as the unique optimum. For the violation of any Bell inequality under uniformly random outcomes, it fails for every  $d \geq 4$  on the maximally entangled state of two  $d$ -level systems once measurements may have outcomes that never occur, by a projective strategy whose critical visibility we compute exactly; for complete von Neumann measurements, for which this noise is white noise on the state, it remains open for  $d \geq 4$ . The Kullback–Leibler clause fails for every  $d \geq 4$  and remains open for  $d = 3$ .

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## 1. INTRODUCTION

**1.1. The problem.** The CGLMP inequalities, introduced by Collins, Gisin, Linden, Massar and Popescu [10], are Bell inequalities for two parties with two measurement settings each and  $d$  outcomes per measurement. Write  $A_1, A_2$  and  $B_1, B_2$  for the outcomes (elements of  $\mathbb{Z}_d$ ) of Alice’s and Bob’s measurements and  $P(A_x = B_y + k)$  for the probability that  $A_x - B_y \equiv k \pmod{d}$  when the settings  $x$  and  $y$  are used. The CGLMP expression is

$$(1.1) \quad I_d = \sum_{k=0}^{\lfloor d/2 \rfloor - 1} \left(1 - \frac{2k}{d-1}\right) \left( [P(A_1 = B_1 + k) + P(B_1 = A_2 + k + 1) + P(A_2 = B_2 + k) + P(B_2 = A_1 + k)] - [P(A_1 = B_1 - k - 1) + P(B_1 = A_2 - k) + P(A_2 = B_2 - k - 1) + P(B_2 = A_1 - k - 1)] \right)$$

(this is the reading of [10] under which the local bound is derived and the optimal measurements of [10] give the value (1.2) below; see [30, Section 2]). Local hidden variable models satisfy  $I_d \leq 2$  [10]. For  $d = 2$ ,  $I_2$  is the CHSH expression; for  $d = 3$  see also the Clauser–Horne form of [23].

A *projective strategy on a maximally entangled state* consists of a local dimension  $D \geq 1$ , the state  $\Phi_D = D^{-1/2} \sum_{i=1}^D |ii\rangle \in \mathbb{C}^D \otimes \mathbb{C}^D$ , and projection-valued measures  $\{P_a^x\}_{a \in \mathbb{Z}_d}$  (Alice,  $x = 1, 2$ ) and  $\{Q_b^y\}_{b \in \mathbb{Z}_d}$  (Bob,  $y = 1, 2$ ) on  $\mathbb{C}^D$ ; it gives  $P(a, b|x, y) = \langle \Phi_D | P_a^x \otimes Q_b^y | \Phi_D \rangle$ . The measurements found numerically by Kaszlikowski et al. [22] and described analytically by Durt, Kaszlikowski and Żukowski [12] (diagonal phase shifts followed by a discrete Fourier transform, then a measurement in the computational basis; we call this the *DKZ strategy*,  $D = d$ ; the explicit bases are recalled in Section 2.2) give

$$(1.2) \quad I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d},$$

for instance  $I_{\text{ME}}(3) = 2.8729\dots$ ,  $I_{\text{ME}}(4) = 2.8962\dots$ ,  $I_{\text{ME}}(20) = 2.9547\dots$ , and  $I_{\text{ME}}(d) \rightarrow 32G/\pi^2 = 2.9698\dots$  as  $d \rightarrow \infty$ , where  $G$  is Catalan’s constant (Corollary 8.1). For  $d \geq 3$  the maximally entangled state is not the optimal state for  $I_d$  [1].

Open Quantum Problem 27 of IQOQI Vienna, “The power of CGLMP inequalities” [36], goes back to questions of Gill [15]. Part A asks whether every facet of the local polytope

of the scenario with two settings and  $d$  outcomes per party, other than the positivity facets, is of CGLMP type; this was answered negatively by Bancal, Gisin and Pironio [4]. Part B records that the observables which numerically maximise the CGLMP violation on a maximally entangled state are measurements in the computational basis transformed by a discrete Fourier transform and diagonal unitaries, and asks to show that this is necessarily the case; it also asks to show that these measurements give the highest noise resistance and the best Kullback–Leibler discrimination from local models [42]. Theorems A and B below prove the first clause of Part B for projective measurements, and Section 9 gives partial answers to the other two.

**1.2. Main results.** Our first two results answer the first clause of Part B for projective measurements.

**Theorem A** (optimality; computer-assisted). *Let  $d \geq 2$  and  $D \geq 1$ . For every projective strategy on the maximally entangled state  $\Phi_D$ ,*

$$I_d \leq I_{\text{ME}}(d),$$

*and the DKZ strategy attains  $I_{\text{ME}}(d)$ .*

**Theorem B** (rigidity; computer-assisted). *Let  $d \geq 2$ ,  $D \geq 1$ , and suppose that a projective strategy on  $\Phi_D$  attains  $I_d = I_{\text{ME}}(d)$ . Then  $d$  divides  $D$ , and there is a unitary  $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$  such that  $uP_a^x u^* = P_a^{x, \text{DKZ}} \otimes \mathbf{1}$  and  $\bar{u}Q_b^y \bar{u}^* = Q_b^{y, \text{DKZ}} \otimes \mathbf{1}$  for all  $x, y, a, b$ ; moreover  $(u \otimes \bar{u})\Phi_D = \Phi_d \otimes \Phi_{D/d}$ , and  $u$  is unique up to  $u \mapsto (\mathbf{1}_d \otimes u')u$  with  $u'$  unitary. Consequently, if  $d \nmid D$  the maximum of  $I_d$  over projective strategies on  $\Phi_D$  is strictly smaller than  $I_{\text{ME}}(d)$ .*

The proofs of Theorems A and B are computer-assisted only through Theorem E below; all other steps are proved by hand. The argument has also been formalised in the Lean 4 theorem prover [34, 35] (Section 11.5): for  $2 \leq d \leq 20$  the formal proof of Theorems A and B is complete, including Theorem E, and for  $d \geq 21$  it is complete except for the input Theorem E. Thus Theorems A and B are proved for every  $d$ , with a computer-assisted step (interval arithmetic) for  $d \geq 21$ , and are fully formalised for  $d \leq 20$ . The analytic heart of the argument is a matrix inequality, which we now state. Let  $S = \{z \in \mathbb{C} : 0 < \text{Re } z < 1\}$  and, for  $\lambda \in \mathbb{R}$  and  $z = x + iy \in S$ ,

$$(1.3) \quad h_\lambda(x + iy) = -\frac{1}{\pi^2} \text{Im Li}_2(e^{\pi(y-\lambda)} e^{-i\pi x}),$$

where  $\text{Li}_2$  is the dilogarithm (principal branch, holomorphic on  $\mathbb{C} \setminus [1, \infty)$ ). The function  $h_\lambda$  is harmonic on  $S$  and extends continuously to the closed strip, with boundary values  $(y - \lambda)_+$  on the line  $\text{Re } z = 0$  and 0 on the line  $\text{Re } z = 1$  (Lemma 3.2). For  $0 < x < 1$  let

$$(1.4) \quad K_x(u) = \frac{\sin \pi x}{2(\cosh \pi u - \cos \pi x)} \quad (u \in \mathbb{R}),$$

the Poisson kernel of the strip  $S$  for its left edge. For an orthogonal projection  $E$  and a Hermitian matrix  $X$  we write  $\text{Tr}_E f(EXE)$  for the trace over  $\text{ran } E$  of  $f$  applied to the compression of  $X$  to  $\text{ran } E$ , and  $X_+$  for the positive part of a Hermitian matrix  $X$ .

**Theorem C** (strip inequality). *Let  $M \geq 1$ , let  $B$  be an orthogonal projection on  $\mathbb{C}^M$ ,  $P = \mathbf{1} - B$ , and let  $g$  be a Hermitian  $M \times M$  matrix. For every  $\lambda \in \mathbb{R}$ ,*

$$(1.5) \quad \sum_{\nu \in \text{spec}(B+ig)} h_\lambda(\nu) - \text{Tr}[(P(g - \lambda)P)_+] = \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s - \lambda) \rho(s, \tau) \, ds \, d\tau \geq 0,$$

*where the eigenvalues are counted with algebraic multiplicity and  $\rho \geq 0$  is the density of Theorem D. Equality holds for one  $\lambda$  if and only if it holds for all  $\lambda$ , if and only if  $Bg = gB$ .*

**Theorem D** (two-variable BMV positivity for projections). *In the situation of Theorem C let  $r = \text{rank } B$ , let  $\lambda_{\min}, \lambda_{\max}$  be the extreme eigenvalues of  $g$ , and for  $s \in \mathbb{R}$  and  $0 < \tau < 1$  let  $\xi_1(s, \tau), \dots, \xi_M(s, \tau)$  be the roots, with multiplicity, of the polynomial  $\xi \mapsto \det(g - s - \xi(P - \tau))$  of degree  $M$ . Put*

$$\rho(s, \tau) = \frac{1}{2\pi} \sum_{i=1}^M |\text{Im } \xi_i(s, \tau)| \quad (0 < \tau < 1), \quad \rho(s, \tau) = 0 \quad (\tau \notin (0, 1)).$$

*Then  $\rho$  is continuous on  $\mathbb{R} \times (0, 1)$ , vanishes unless  $\lambda_{\min} < s < \lambda_{\max}$ , satisfies  $\rho(s, \tau) \leq \frac{M}{2\pi} \|g - s\| (\tau(1 - \tau))^{-1/2}$ , hence is integrable, and for all  $(a, t) \in \mathbb{C}^2$*

$$(1.6) \quad \mathcal{D}(a, t) := \text{Tr } e^{ag - tP} - e^{-t} \text{Tr}_P e^{aPgP} - \text{Tr}_B e^{aBgB} = a^2 \int_0^1 \int_{\mathbb{R}} e^{as - t\tau} \rho(s, \tau) \, ds \, d\tau.$$

*Moreover  $\int_{\mathbb{R}} \rho(s, \tau) \, ds = \|BgP\|_F^2$  for every  $\tau \in (0, 1)$ .*

Taking  $a = 1$  in (1.6) shows that  $t \mapsto \text{Tr } e^{g - tP}$  is the Laplace transform of the positive measure  $\text{Tr}_B(e^{BgB}) \delta_0 + \text{Tr}_P(e^{PgP}) \delta_1 + (\int_{\mathbb{R}} e^s \rho(s, \tau) \, ds) \, d\tau$  on  $[0, 1]$ , which is Stahl's theorem [40] (the former Bessis–Moussa–Villani conjecture [5]) for a projection, with an explicit density. Theorem D is stronger: it asserts positivity of a measure in two variables, after the two pinched terms are subtracted and the result is divided by  $a^2$ . Neither modification can be omitted:  $\mathcal{D}(0, t) = 0$  for all  $t$ , so  $\mathcal{D}$  itself is the Laplace transform of a positive measure only if  $\mathcal{D} \equiv 0$ , and  $a^{-2} \text{Tr } e^{ag - tP}$  is not entire.

The last ingredient is a condition on explicit Clausen kernels. Let  $N = 4d$  and let  $\text{Cl}_2(\theta) = \sum_{n \geq 1} n^{-2} \sin n\theta$  be the Clausen function. Put

$$\Xi(u) = -\frac{N^2}{2\pi^2} \text{Cl}_2\left(\frac{2\pi u}{N}\right), \quad \Xi^\sharp(u) = \Xi(u) + \Xi(2d - u) \quad (0 \leq u \leq 2d),$$

so that  $(\Xi^\sharp)''(u) = 2 \csc \frac{\pi u}{2d}$  on  $(0, 2d)$ . Let  $\Delta_d = \{\ell \in \mathbb{R}^d : \ell_r \geq 0, \sum_r \ell_r = d\}$  and  $\Delta_d^+ = \{\ell \in \Delta_d : \ell_r > 0 \text{ for all } r\}$ . For  $\ell \in \Delta_d$ , extended  $d$ -periodically, let  $S_r(m) = \ell_r + \ell_{r+1} + \dots + \ell_{r+m-1}$  and define the *cell vector*  $u^\ell \in \mathbb{R}^{d-1}$  by

$$(1.7) \quad u_m^\ell = \Delta_m^2 \left[ \frac{1}{d} \sum_{r=0}^{d-1} \Xi^\sharp(S_r(m)) \right] \quad (1 \leq m \leq d-1),$$

where  $\Delta_m^2 f = f(m+1) - 2f(m) + f(m-1)$ . Let  $v \in \mathbb{R}^{d-1}$ ,  $v_m = 2 \csc \frac{\pi m}{2d}$ , and  $t_s = e_s + e_{d-s}$  ( $1 \leq s \leq d-1$ ), where  $(e_m)$  is the standard basis.

**Theorem E** (cone condition; computer-assisted). *For every  $d \geq 2$  there are a finite positive Borel measure  $\mu$  on  $\Delta_d$  with  $\mu(\Delta_d^+) = \mu(\Delta_d) > 0$  and numbers  $w_s \geq 0$  such that*

$$(1.8) \quad v = \int_{\Delta_d} u^\ell \mu(d\ell) + \sum_{s=1}^{\lfloor d/2 \rfloor} w_s t_s.$$

For  $2 \leq d \leq 200$  the measure  $\mu$  is a finite sum of point masses at rational cell vectors and  $w = 0$ ; the certificates are verified in interval arithmetic, and for  $d \leq 20$  also by the Lean kernel in exact rational interval arithmetic. For  $d \geq 201$ ,  $\mu$  is the law of an explicit random cell vector (a Dirichlet vector with Gaussian-modulated parameters), and the verification is by interval arithmetic for each  $201 \leq d \leq 2000$  and by a single interval Taylor-model computation, uniform in  $d$ , for  $d \geq 2001$  (Section 6).

**1.3. Status of Problem 27.** Table 1 lists every clause of Problem 27 as originally posed [36] and its status after this work. Part A was answered negatively by Bancal, Gisin and Pironio [4]. Theorems A and B prove the clause of Part B on optimality and uniqueness, for projective measurements. Section 9 gives partial answers to the two other clauses of Part B. The answer to the noise clause depends on how noise resistance is measured and on the class of measurements allowed. If it refers to the violation of the CGLMP inequality, with uniformly random outcomes (equivalently, for complete von Neumann measurements on  $\Phi_d$ , white noise on the state) as noise, the clause holds for every  $d$ , and the DKZ measurements are the unique optimum (Corollary 9.1). In Gill’s literal reading, in which the noisy correlations must violate some Bell inequality, it fails for every  $d \geq 4$  on  $\Phi_d$  once measurements may have outcomes that never occur (Theorem 9.2); it holds for  $d = 2$ , and for  $d = 3$  it holds on  $\Phi_3$  but fails on  $\Phi_2$  (Proposition 9.11). For complete von Neumann measurements on  $\Phi_d$ , for which Gill’s noise is white noise on the state, it remains open for  $d \geq 4$  (Section 9.4); this is the formulation of Formal Conjectures issue #3444 [33]. The Kullback–Leibler clause fails for every  $d \geq 4$  (Theorem 9.14); for  $d = 3$  the DKZ measurements are a strict local maximum (Proposition 9.16), and global optimality remains open. The results of this paper have not yet been reviewed by outside experts.

**1.4. Strategy of the proof.** The proof proceeds in six steps; Sections 2–7 follow this order.

(1) *The clock model.* By an exact reduction from [30], recalled in Section 2,  $I_d = 4F(V)/(d(d-1))$  for a  $d$ -tuple  $V$  of unitaries of order four on some  $\mathbb{C}^M$ , and  $I_{\text{ME}}(d) = 4F_{\text{DKZ}}/(d(d-1))$ . The spectral projections of the  $V_k$  form a family  $(Q_x)_{x \in \mathbb{Z}_{4d}}$  of projections, each residue class modulo  $d$  being a four-outcome projection-valued measure, and  $F(V) - F_{\text{DKZ}} = \frac{1}{2}\langle v, \delta \rangle$ , where  $\delta \in \mathbb{R}^{d-1}$  collects the deviations of  $d-1$  “net pair counts” from their values at the DKZ configuration (Lemma 2.5).

(2) *Two-variable BMV positivity and the strip inequality* (Section 3): Theorems D and C, proved by hand. Appendix B gives a second proof of both, without distribution theory or the Paley–Wiener–Schwartz theorem, through a Radon identity for matrix pencils.

(3) *A noncommutative continuum theorem* (Section 4): for projection-valued step fields  $B$  on the circle, fields  $0 \leq A \leq \mathbf{1} - B$ , and the conjugate function  $g$  of  $B$ , the pairing  $\int \tau(Ag)$  is at most the value of the classical (commutative) problem. This is a matrix version of the circle Stein–Weiss rearrangement theorem [41, 25]; the strip inequality enters pointwise, through a harmonic function on the disc.

(4) *Cell embeddings* (Section 5): placing the projections  $Q_x$  on cells of lengths  $\ell_{x \bmod d}$  on a circle of length  $4d$  produces a step field to which (3) applies; averaging over rotations gives the linear inequalities  $\langle u^\ell, \delta \rangle \leq 0$ . Consequently Theorem E implies  $\langle v, \delta \rangle \leq 0$ , which is Theorem A.

(5) *The cone condition* (Section 6): Theorem E, computer-assisted.

(6) *Rigidity* (Section 7): in the equality case every inequality of the chain is an equality. Through the equality clause of Theorem C this forces  $[B(\theta), g(\theta)] = 0$  almost everywhere on a cell embedding with all cells of positive length; a residue argument then shows that all  $Q_x$  commute, and the commutative case [30] identifies the strategy.

The continuum problem in (3) is degenerate: its maximum is attained by many configurations, including non-commuting ones (Remark 4.7), and the discrete problem differs from it by terms of order  $d$  located at the junctions between blocks. Step (4) avoids any explicit estimate of these junction terms: the cell inequalities are exact consequences of the continuum theorem, and the discrete kernel is recovered from them by the cone condition. The mechanism behind the cone condition is that the discrete cotangent kernel is the continuum kernel averaged over Poisson-like random cells (Remark 5.6).

**1.5. Relation to previous work.** For  $d = 2$ , Theorem A is Tsirelson’s bound [8] on maximally entangled states and Theorem B is the corresponding rigidity. For  $d = 3$ , Lang, Vértesi and

| Clause                                                                                                                                                                   | Status                                                                                                                               | References                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 27A: every facet of the local polytope that is not contained in a face of the no-signalling polytope is of CGLMP type, possibly lifted                                   | false ( $d = 4$ )                                                                                                                    | Bancal–Gisin–Pironio [4]                                                                              |
| 27B: the DKZ measurements are optimal for $I_d$ on maximally entangled states                                                                                            | proved for every $d$ (projective measurements;<br>computer-assisted for $d \geq 21$ ;<br>formally verified in Lean for $d \leq 20$ ) | Theorem A; earlier: $d = 2$ [8], $d \leq 20$ [31], commuting (equal-link) strategies for all $d$ [30] |
| 27B: they are <i>necessarily</i> the optimal ones                                                                                                                        | proved for every $d$ (projective; computer-assisted for $d \geq 21$ ;<br>formally verified in Lean for $d \leq 20$ )                 | Theorem B; earlier: $d \leq 20$ [31], equal-link strategies [30]                                      |
| 27B: highest noise resistance, (a) violation of the CGLMP inequality, uniform outcome noise                                                                              | proved for every $d$ , DKZ the unique optimum (projective; formally verified, with the scope of Theorems A and B)                    | Corollary 9.1                                                                                         |
| 27B: highest noise resistance, (b) and (c): violation of local realism, complete von Neumann measurements on $\Phi_d$ (uniform outcome noise = white noise on the state) | open for $d \geq 4$ ; true for $d = 2, 3$ ; DKZ optimal in numerical searches for $d \leq 8$                                         | Proposition 9.11, Section 9.4; numerics: [16] ( $d \leq 5$ ), [32]                                    |
| 27B: highest noise resistance, (b) Gill’s reading: violation of local realism, uniform outcome noise, measurements with unused outcomes or POVMs                         | false for every $d \geq 4$ on $\Phi_d$ (computer-assisted for $4 \leq d \leq 9$ ); $d = 3$ : true on $\Phi_3$ , false on $\Phi_2$    | Theorem 9.2, Remark 9.12, Proposition 9.11; mechanism [1]; numerics on $\Phi_4, \Phi_{16}$ [3]        |
| 27B: best Kullback–Leibler discrimination                                                                                                                                | false for every $d \geq 4$ (computer-assisted; complete von Neumann competitors); $d = 3$ : open, DKZ a strict local maximum         | Theorem 9.14, Proposition 9.16; $d = 4$ first by Zhang [44]; $4 \leq d \leq 9$ [31]                   |
| Related: maximal value of $I_d$ over all states                                                                                                                          | exact for $d = 3, \dots, 8$                                                                                                          | Ioannou–Rosset [20] ( $d = 3, 4$ ), [31] ( $d = 5, \dots, 8$ )                                        |

TABLE 1. The clauses of Open Quantum Problem 27 as originally posed [36], and their status. “Projective” means that POVMs are not covered by Theorems A and B; for  $3 \leq d \leq 8$  optimality and uniqueness also hold for arbitrary POVMs (Section 10, item (1)). The counterexample in the sixth row needs measurements with unused outcomes, or POVMs; the counterexamples to the Kullback–Leibler clause are complete von Neumann measurements.

Navascués [26] showed with a semidefinite relaxation that the maximal value on maximally entangled states agrees with  $I_{\text{ME}}(3)$  to within  $10^{-10}$  (this is the partial result listed in [36]). In [31] we gave exact sum-of-squares certificates proving Theorems A and B for  $3 \leq d \leq 20$ , and in [30] we proved both theorems for every  $d$  for strategies whose reduced family commutes (equivalently, whose four “link” operators coincide). Ioannou and Rosset [20] obtained exact sum-of-squares bounds for the maximal value of  $I_3$  and  $I_4$  over all states. The SATWAP expressions of



Salavrakos et al. [38] are maximally violated by maximally entangled states with measurements of CGLMP type and underlie device-independent self-testing results [39]; they are different Bell expressions.

For the noise clause, the critical visibilities  $2/I_{\text{ME}}(d)$  of the DKZ measurements against all local models were computed numerically by Kaszlikowski et al. [22] ( $d \leq 9$ ) and by Durt, Kaszlikowski and Żukowski [12] ( $d \leq 16$ ), and identified with the threshold of the CGLMP inequality in [10, eq. (23)]; Chen et al. [7] treated  $d = 3$  analytically, and Gruca, Laskowski and Żukowski [16] recomputed the thresholds numerically for  $d \leq 5$  and found that states of Schmidt rank two are the most noise-robust. The mechanism behind Theorem 9.2 goes back to Acín, Durt, Gisin and Latorre [1], and Baek, Ryu and Lee [3] observed numerically that coarse-grained measurements beat DKZ on  $\Phi_4$  and  $\Phi_{16}$  under uniform noise (Remark 9.13). The Kullback–Leibler clause refers to the statistical strength of van Dam, Gill and Grünwald [42]; Zhang [44] found the first counterexample ( $d = 4$ ), and in [31] we gave counterexamples for  $4 \leq d \leq 9$ .

The mechanism of our proof of Theorem D—inverting a trace function of a pencil of Hermitian matrices by a two-dimensional Fourier transform computed along lines, and identifying the density as  $\frac{1}{2\pi} \sum |\text{Im}|$  of the roots of an auxiliary pencil by an integral identity for  $\log |\det|$ —is the one introduced by Heinävaara [17] for tracial joint spectral measures, which yields, among other results, a new proof of Stahl’s theorem [40] (see also [13, 28]). Theorem D is a variant adapted to a projection and to an affine chart: the subtraction of the two pinched terms and the division by  $a^2$  produce a measure without singular part, which is what the strip inequality requires. We found Theorem D independently, but we do not claim novelty for the mechanism, and we give complete proofs. We have not found Theorem C, or an equivalent statement, in the literature. Its commutative case is an identity, and its continuum consequence (Theorem 4.2) contains the circle version of the theorem of Stein and Weiss [41] on the distribution of the conjugate function of an indicator function, in the form of Laeng [25] (see also [6, 11]).

Appendix B gives a second proof of Theorem D that uses no distributions, no Paley–Wiener–Schwartz theorem and no Fourier inversion in two variables. Its key step is a Radon identity (Theorem B.1): the integral over  $\tau \in (0, 1)$  of  $\frac{1}{2\pi} \sum_i |\text{Im } y_i(\tau)|$ , where the  $y_i(\tau)$  are the eigenvalues of  $(P - \tau)^{-1}H$ , equals  $\text{Tr } H_+ - \text{Tr}(BHB + PHP)_+$  for every Hermitian  $H$ . We prove it with a complex inviscid Burgers equation satisfied by the pencil roots. The same argument applies to an arbitrary Hermitian matrix in place of the projection, and then gives a proof of Stahl’s theorem without distribution theory or Fourier analysis (Remark B.8); we make no priority claim about this by-product, and refer to [40, 13, 17] for the history of the theorem and of the mechanism. This elementary route is the one followed by the Lean formalisation.

**1.6. Conventions.**  $\text{Tr}$  is the trace on  $M_M(\mathbb{C})$  and  $\tau = \frac{1}{M} \text{Tr}$ . Spectra are multisets, counted with algebraic multiplicity.  $\|X\|$  is the operator norm and  $\|X\|_F$  the Frobenius norm. For  $f \in L^1(\mathbb{R})$ ,  $\hat{f}(\sigma) = \int_{\mathbb{R}} f(w) e^{-i\sigma w} dw$ ; the Hilbert transform is  $\mathcal{H}f(w) = \frac{1}{\pi} \text{p.v.} \int_{\mathbb{R}} \frac{f(u)}{w-u} du$ , which acts on  $L^2(\mathbb{R})$  as the Fourier multiplier  $-i \text{sgn } \sigma$ . On the circle  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$  we use the normalised measure  $dm = d\theta/2\pi$ , and the conjugate function of  $f \in L^2(\mathbb{T})$  is  $\text{p.v.} \int \cot \frac{\theta-\varphi}{2} f(\varphi) dm(\varphi)$ , the multiplier  $-i \text{sgn } n$ . Catalan’s constant is  $G = \text{Cl}_2(\pi/2) = \sum_{n \geq 0} (-1)^n (2n+1)^{-2}$ , and  $\text{Cl}_2(\theta) = -\int_0^\theta \log |2 \sin \frac{t}{2}| dt$ . For  $\alpha, \beta \geq 0$  with  $\alpha + \beta \leq 1$  we put

$$(1.9) \quad \Phi_c(\alpha, \beta) = \frac{1}{2\pi^2} \left[ \text{Cl}_2(2\pi\alpha) + \text{Cl}_2(2\pi\beta) + \text{Cl}_2(2\pi(1 - \alpha - \beta)) \right], \quad \Phi_c\left(\frac{1}{4}, \frac{1}{4}\right) = \frac{G}{\pi^2}.$$

Results labelled *computer-assisted* depend on computations in rigorous interval arithmetic with outward rounding, or in exact rational or algebraic arithmetic (Section 9); everything else is proved by hand. Numerical observations that are not part of any proof are labelled as such.

The Lean formalisation (Section 11.5) is an independent check; no proof in this paper depends on it.

## 2. THE CLOCK MODEL AND THE Q-CONFIGURATION

Throughout,  $d \geq 2$ ,  $N = 4d$ ,  $w = e^{2\pi i/d}$ ,  $z = e^{2\pi i/N}$  (so  $z^4 = w$  and  $z^d = i$ ), and  $\kappa(u) = \cot(\pi u/N)$  for  $u \in \mathbb{Z}_N \setminus \{0\}$ ,  $\kappa(0) = 0$ .

**2.1. Strategies and the clock model.** Encode a projective strategy on  $\Phi_D$  by the unitaries  $U_x = \sum_a w^a P_a^x$  and  $U'_y = \sum_b w^b Q_b^y$  and put

$$(2.1) \quad R_1 = U_2, \quad R_2 = (U'_2)^T, \quad R_3 = U_1, \quad R_4 = (U'_1)^T,$$

the transpose being taken in the Schmidt basis of  $\Phi_D$ ; then  $R_i^d = \mathbf{1}$ , and conversely every quadruple of unitaries on  $\mathbb{C}^D$  with  $R_i^d = \mathbf{1}$  comes from a projective strategy on  $\Phi_D$ , through spectral projections. A simultaneous conjugation  $R_i \mapsto u R_i u^*$  is the same as applying the local unitary  $u \otimes \bar{u}$ , which fixes  $\Phi_D$ . For unitaries  $V_0, \dots, V_{d-1} \in U(M)$  with  $V_k^4 = \mathbf{1}$  let

$$(2.2) \quad F(V) = \sum_{0 \leq j < k \leq d-1} \operatorname{Re} \left[ h_{k-j} \tau(V_j V_k^*) \right], \quad h_m = \sec \frac{\pi m}{2d} - i \csc \frac{\pi m}{2d},$$

and  $F_{\text{DKZ}} = F(\mathbf{1}, \dots, \mathbf{1}) = \sum_{m=1}^{d-1} (d-m) \sec \frac{\pi m}{2d}$ . Let  $X$  be the cyclic shift  $X|k\rangle = |k+1\rangle$  on  $\mathbb{C}^d$  and define the *covariant strategy*  $R^V$  on  $\mathbb{C}^d \otimes \mathbb{C}^M$  by

$$(2.3) \quad W = \sum_{k=0}^{d-1} |k\rangle\langle k| \otimes z^k V_k, \quad R_1^V = X \otimes \mathbf{1}_M, \quad R_{i+1}^V = W R_i^V W^* \quad (i = 1, 2, 3).$$

**Theorem 2.1** (reduction to the clock model [30, Theorem 2.3]). *Let  $d \geq 2$ .*

- (1) *For every covariant strategy,  $I_d(R^V) = 4F(V)/(d(d-1))$ .*
- (2) *For every projective strategy  $R$  on  $\Phi_D$  there are  $M$  (one can take  $M = 4D$ ) and unitaries  $V_0, \dots, V_{d-1} \in U(M)$  with  $V_k^4 = \mathbf{1}$ , unique up to a simultaneous unitary conjugation, such that  $R$  is unitarily equivalent (by a simultaneous conjugation of  $R_1, \dots, R_4$ ) to a direct summand of  $R^V$  and  $I_d(R) = I_d(R^V) = 4F(V)/(d(d-1))$ .*

*In particular  $I_d \leq I_{\text{ME}}(d)$  holds for all projective strategies on maximally entangled states if and only if  $F(V) \leq F_{\text{DKZ}}$  for all  $M$  and all  $V_0, \dots, V_{d-1} \in U(M)$  with  $V_k^4 = \mathbf{1}$ .*

The family  $V$  in (2) is the *reduced family* of  $R$ ; it is obtained from a twirl over a cyclic symmetry of order  $4d$  followed by the Stone–von Neumann theorem for  $\mathbb{Z}_d$ .

**2.2. The DKZ strategy and the commutative sector.** Let  $W_0 = \operatorname{diag}(z^k)_{k=0}^{d-1}$  and  $R^{(0)} = R^V$  for  $M = 1$ ,  $V \equiv \mathbf{1}$ . The DKZ measurements are the measurements in the orthonormal bases  $|k\rangle_{A,x} = d^{-1/2} \sum_j e^{-2\pi i j(k+\alpha_x)/d} |j\rangle$  and  $|l\rangle_{B,y} = d^{-1/2} \sum_j e^{2\pi i j(l-\beta_y)/d} |j\rangle$  with  $\alpha = (0, \frac{1}{2})$  and  $\beta = (\frac{1}{4}, -\frac{1}{4})$  (the measurements of [10, eqs. (12)–(15)]), and their chain (2.1) is  $GR^{(0)}G^*$  with  $G = W_0^{-2}$  [30, Section 2.4]. Hence  $I_d(\text{DKZ}) = 4F_{\text{DKZ}}/(d(d-1)) = I_{\text{ME}}(d)$ .

**Theorem 2.2** (commutative sector [30, Theorem B, Corollary C, Proposition 2.4]). *Let  $d \geq 2$ .*

- (1) *If  $V_0, \dots, V_{d-1} \in U(M)$  commute pairwise and  $V_k^4 = \mathbf{1}$ , then  $F(V) \leq F_{\text{DKZ}}$ , with equality if and only if every joint eigenvector  $v$  of the  $V_k$  satisfies  $V_k v = i^{-a_k} v$  with a one-step configuration  $a_k = c + [k \geq r]$  ( $c \in \mathbb{Z}_4$ ,  $0 \leq r \leq d-1$ ;  $c$  and  $r$  may depend on  $v$ ).*
- (2) *A projective strategy on  $\Phi_D$  has commuting reduced family if and only if its four link operators coincide,  $R_1^n R_2^{-n} = R_2^n R_3^{-n} = R_3^n R_4^{-n} = w^{-n} R_4^n R_1^{-n}$  for all  $n$ .*
- (3) *If a projective strategy on  $\Phi_D$  with commuting reduced family attains  $I_d = I_{\text{ME}}(d)$ , then  $d \mid D$  and there is a unitary  $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$  with  $u R_i u^* = R_i^{\text{DKZ}} \otimes \mathbf{1}$  for  $i = 1, \dots, 4$ .*



### 2.3. Q-configurations.

**Definition 2.3.** A *Q-configuration* on  $\mathbb{C}^M$  is a family  $(Q_x)_{x \in \mathbb{Z}_N}$  of orthogonal projections on  $\mathbb{C}^M$  such that for every  $k \in \{0, \dots, d-1\}$  the four projections  $Q_{k-da}$ ,  $a \in \mathbb{Z}_4$ , are pairwise orthogonal with sum  $\mathbf{1}$ . We put  $A_x = Q_x$  and  $B_x = Q_{x+d}$ , and for  $j \in \mathbb{Z}_N$  we let  $(\text{rot}_j Q)_x = Q_{x-j}$ .

Every element of  $\mathbb{Z}_N$  is uniquely of the form  $k - da$  with  $0 \leq k \leq d-1$  and  $a \in \mathbb{Z}_4$ . Given a family  $V$  as above, let  $Q_{k-da}$  be the spectral projection of  $E_k := V_k^*$  for the eigenvalue  $i^a$ ; this is a Q-configuration,  $E_k = \sum_{a \in \mathbb{Z}_4} i^a Q_{k-da}$ , and conversely every Q-configuration arises from a unique family  $V$  in this way. The following properties are immediate:  $A_x B_x = 0$  (two outcomes of one measurement); for every  $d$ -periodic function  $\ell$  on  $\mathbb{Z}_N$ ,

$$(2.4) \quad \sum_{x \in \mathbb{Z}_N} \ell_x B_x = \sum_{x \in \mathbb{Z}_N} \ell_x A_x = \left( \sum_{r=0}^{d-1} \ell_r \right) \mathbf{1};$$

and every rotation  $\text{rot}_j Q$  is again a Q-configuration. The Q-configuration of  $V \equiv \mathbf{1}$  ( $M = 1$ ) is the *window*  $Q_x = \mathbf{1}_{[0,d)}(x)$ ; its rotations are the windows  $\mathbf{1}_{[c,c+d)}$ .

**Lemma 2.4** (projection form). *For every family  $V$  and its Q-configuration,*

$$F(V) = \frac{1}{2} \langle A, B \rangle_\tau - \frac{d}{2}, \quad \langle A, B \rangle_\tau := \sum_{x, y \in \mathbb{Z}_N} \kappa(x - y) \tau(A_x B_y).$$

*Proof.* This is [30, eq. (13)]; we include the short proof. For  $j < k$ ,  $m = k - j$  and  $a, b \in \mathbb{Z}_4$  put  $x = j - da$ ,  $y = k - db$ . Then  $\tau(V_j V_k^*) = \tau(E_j^* E_k) = \sum_{a, b} i^{b-a} \tau(Q_x Q_y)$  and  $\text{Re}[h_m i^s] = \sec(\frac{\pi m}{2d} - \frac{\pi s}{2})$  for  $s \in \mathbb{Z}_4$  (check  $s = 0, 1, 2, 3$ ), which equals  $\sec \frac{\pi(y-x)}{2d}$  because  $y - x = m - d(b - a)$  and  $u \mapsto \sec \frac{\pi u}{2d}$  is even with period  $4d$ . Since  $\tau(Q_x Q_y) = \tau(Q_y Q_x)$ ,

$$F(V) = \frac{1}{2} \sum_{x \not\equiv y \pmod{d}} \tau(Q_x Q_y) \sec \frac{\pi(x - y)}{2d}.$$

The identity  $\sec t = \frac{1}{2} [\cot(\frac{t}{2} + \frac{\pi}{4}) - \cot(\frac{t}{2} - \frac{\pi}{4})]$  with  $t = 2\pi(x - y)/N$  gives  $\sec \frac{\pi(x - y)}{2d} = \frac{1}{2} [\kappa(x - y + d) - \kappa(x - y - d)]$  for  $x \not\equiv y \pmod{d}$ , and the two halves give equal sums (exchange  $x$  and  $y$  and use that  $\kappa$  is odd). Hence  $F(V) = \frac{1}{2} \sum_{x \not\equiv y \pmod{d}} \tau(Q_x Q_y) \kappa(x - y + d) = \frac{1}{2} \sum_{x \not\equiv y' \pmod{d}} \tau(A_x B_{y'}) \kappa(x - y')$  with  $y' = y - d$ . In the full sum  $\langle A, B \rangle_\tau$  the pairs with  $x \equiv y' \pmod{d}$  contribute only for  $x = y' + d$  (otherwise  $Q_x Q_{y'+d} = 0$ , or  $\kappa(0) = 0$  when  $x = y'$ ), and these contribute  $\kappa(d) \sum_x \tau(Q_x) = d$ . So  $\langle A, B \rangle_\tau = 2F(V) + d$ .  $\square$

**2.4. Net pair counts.** For a Q-configuration put

$$T(m) = \sum_{y \in \mathbb{Z}_N} \tau(A_{y+m} B_y), \quad \mathcal{N}(m) = T(m) - T(-m), \quad \delta_m = \mathcal{N}(m) - m \quad (1 \leq m \leq d-1),$$

and  $\delta = (\delta_m)_{m=1}^{d-1}$ . For sites  $0 \leq j < k \leq d-1$  let  $\pi_{jk}(t) = \sum_{b \in \mathbb{Z}_4} \tau(Q_{j-db} Q_{k-d(b+t)})$ ,  $t \in \mathbb{Z}_4$ ; this is a probability vector on  $\mathbb{Z}_4$  (nonnegative, since  $\tau(QQ') \geq 0$  for projections, with sum  $\sum_b \tau(Q_{j-db}) = 1$ ).

**Lemma 2.5** (linear form). *For every Q-configuration:*

- (1)  $\mathcal{N}(2d - m) = \mathcal{N}(m)$  for  $1 \leq m \leq 2d - 1$ , and  $\mathcal{N}(d) = d$ ;
- (2)  $F(V) = \sum_{m=1}^{d-1} \csc \frac{\pi m}{2d} \mathcal{N}(m)$ , the window has  $\mathcal{N}(m) = m$ , and

$$F(V) - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle, \quad v_m = 2 \csc \frac{\pi m}{2d};$$

- (3) for  $1 \leq m \leq d-1$ ,  $\mathcal{N}(m) = \sum_{k-j=m} [\pi_{jk}(1) - \pi_{jk}(3)] + \sum_{k-j=d-m} [\pi_{jk}(0) - \pi_{jk}(2)]$ ; consequently  $\langle t_s, \delta \rangle = \mathcal{N}(s) + \mathcal{N}(d-s) - d \leq 0$  for  $1 \leq s \leq d-1$ .

*Proof.* (1) Put  $C(n) = \sum_u \tau(Q_u Q_{u+n})$ ; then  $C(-n) = C(n)$  by cyclicity of the trace, and  $T(m) = \sum_y \tau(Q_{y+m} Q_{y+d}) = C(d-m)$ , so  $\mathcal{N}(m) = C(m-d) - C(m+d)$ . Hence  $\mathcal{N}(2d-m) = C(d-m) - C(3d-m) = C(m-d) - C(m+d) = \mathcal{N}(m)$  (indices modulo  $N$ ), and  $\mathcal{N}(d) = C(0) - C(2d) = \sum_u \tau(Q_u) - 0 = d$ , because  $Q_u$  and  $Q_{u+2d}$  are distinct outcomes of one measurement.

(2) Since  $\kappa$  is odd and  $\kappa(2d) = 0$ ,  $\langle A, B \rangle_\tau = \sum_{m \in \mathbb{Z}_N} \kappa(m) T(m) = \sum_{m=1}^{2d-1} \kappa(m) \mathcal{N}(m)$ , and by (1) this is  $\kappa(d)d + \sum_{m=1}^{d-1} (\kappa(m) + \kappa(2d-m)) \mathcal{N}(m) = d + \sum_m v_m \mathcal{N}(m)$ , because  $\cot \frac{\pi m}{4d} + \tan \frac{\pi m}{4d} = 2 \csc \frac{\pi m}{2d}$ . Lemma 2.4 gives the formula for  $F(V)$ . For the window,  $T(m) = \#\{y \in [-d, 0) : y+m \in [0, d)\} = m$  and  $T(-m) = 0$  for  $1 \leq m \leq d-1$ , so  $\mathcal{N}(m) = m$ ; thus  $F_{\text{DKZ}} = \sum_m m \csc \frac{\pi m}{2d}$  (which agrees with  $\sum_m (d-m) \sec \frac{\pi m}{2d}$  after  $m \mapsto d-m$ ), and the formula for  $F(V) - F_{\text{DKZ}}$  follows.

(3) Write  $y+d = j-db$  with  $0 \leq j \leq d-1$ . If  $j+m < d$ , then  $y+m = (j+m) - d(b+1)$  is the outcome  $b+1$  of site  $k = j+m$ ; if  $j+m \geq d$ , then  $y+m = (j+m-d) - db$  is the outcome  $b$  of site  $k' = j+m-d < j$ . Summing over  $b$  gives  $T(m) = \sum_{k-j=m} \pi_{jk}(1) + \sum_{k-j=d-m} \pi_{jk}(0)$ ; in the same way  $T(-m) = \sum_{k-j=m} \pi_{jk}(3) + \sum_{k-j=d-m} \pi_{jk}(2)$ . Hence  $\mathcal{N}(s) + \mathcal{N}(d-s) = \sum [\pi_{jk}(0) + \pi_{jk}(1) - \pi_{jk}(2) - \pi_{jk}(3)]$ , the sum running over the  $d$  pairs at distance  $s$  or  $d-s$  (counted twice if  $s = d/2$ ), and each term is at most 1.  $\square$

By Theorem 2.1 and Lemma 2.5, Theorem A is equivalent to

$$(2.5) \quad \langle v, \delta \rangle \leq 0 \quad \text{for every Q-configuration, on every } \mathbb{C}^M.$$

The inequalities  $\langle t_s, \delta \rangle \leq 0$  of Lemma 2.5(3) hold pairwise and carry no information about  $\langle v, \delta \rangle$  by themselves; they are the “trivial directions” in Theorem E.

### 3. TWO-VARIABLE BMV POSITIVITY AND THE STRIP INEQUALITY

Throughout this section  $M \geq 1$ ,  $B$  is an orthogonal projection on  $\mathbb{C}^M$  of rank  $r$ ,  $P = \mathbf{1} - B$ ,  $g$  is Hermitian,  $g_d = BgB + PgP$  is the pinching of  $g$ ,  $\lambda_{\min} \leq \lambda_{\max}$  are the extreme eigenvalues of  $g$ , and  $K = [\lambda_{\min}, \lambda_{\max}] \times [0, 1] \subseteq \mathbb{R}^2$ .

#### 3.1. The strip, its Poisson kernel and the functions $h_\lambda$ .

**Lemma 3.1** (strip Poisson kernel). *Let  $0 < x < 1$  and let  $K_x$  be as in (1.4).*

- (1)  $K_x > 0$  and  $\int_{\mathbb{R}} K_x = 1 - x$ . For  $|u| \geq 1$ ,  $K_x(u) \leq 2 \sin(\pi x) e^{-\pi|u|}$ ; if  $x \geq \frac{1}{2}$ , then  $K_x(u) \leq \sin(\pi x) e^{-\pi|u|}$  for all  $u$ .
- (2) For  $u \neq 0$ ,  $K_x(u) = \sum_{n \geq 1} \sin(n\pi x) e^{-n\pi|u|}$ .
- (3) For  $a \in \mathbb{C}$  with  $|\operatorname{Re} a| < \pi$ ,  $\int_{\mathbb{R}} e^{-au} K_x(u) du = \frac{\sin(a(1-x))}{\sin a}$  (interpreted as  $1-x$  at  $a=0$ ).

*Proof.* (1)  $\cosh \pi u \geq 1 > \cos \pi x$ . For  $|u| \geq 1$ ,  $\cosh \pi u - \cos \pi x \geq \cosh \pi u - 1 \geq \frac{1}{2} e^{\pi|u|} - 1 \geq \frac{1}{4} e^{\pi|u|}$ ; for  $x \geq \frac{1}{2}$ ,  $\cos \pi x \leq 0$  and  $\cosh \pi u - \cos \pi x \geq \frac{1}{2} e^{\pi|u|}$ . The mass is (3) at  $a=0$ .

(2) With  $q = e^{-\pi|u|} \in (0, 1)$  and  $\theta = \pi x$ ,  $\sum_n q^n \sin n\theta = \operatorname{Im} \frac{qe^{i\theta}}{1-qe^{i\theta}} = \frac{q \sin \theta}{1-2q \cos \theta + q^2} = \frac{\sin \theta}{2 \cosh \pi u - 2 \cos \theta}$ .

(3) First let  $a = i\kappa$  with  $\kappa \in \mathbb{R} \setminus \{0\}$ , and  $f(u) = e^{-i\kappa u} / (\cosh \pi u - \cos \pi x)$ . Integrate  $f$  over the boundary of the rectangle  $[-L, L] \times [0, 2]$ ; the vertical sides tend to 0 as  $L \rightarrow \infty$ , and  $f(u+2i) = e^{2\kappa} f(u)$ , so the contour integral tends to  $(1 - e^{2\kappa}) \int_{\mathbb{R}} f$ . Inside the rectangle

the denominator vanishes at  $u = ix$  and  $u = i(2 - x)$ , simply, with residues of  $f$  equal to  $e^{\kappa x}/(i\pi \sin \pi x)$  and  $-e^{\kappa(2-x)}/(i\pi \sin \pi x)$ . Hence  $\int_{\mathbb{R}} f = \frac{2(e^{\kappa x} - e^{\kappa(2-x)})}{(1 - e^{2\kappa}) \sin \pi x} = \frac{2 \sinh(\kappa(1-x))}{\sinh(\kappa) \sin \pi x}$ , that is,  $\int e^{-i\kappa u} K_x(u) du = \frac{\sinh(\kappa(1-x))}{\sinh \kappa} = \frac{\sin(a(1-x))}{\sin a}$ . Both sides of (3) are holomorphic on  $|\operatorname{Re} a| < \pi$  (the left side by the exponential decay in (1); on the right the only zero of  $\sin a$  in this strip,  $a = 0$ , is removable), and they agree on  $i\mathbb{R} \setminus \{0\}$ , hence everywhere.  $\square$

For  $\lambda \in \mathbb{R}$  let

$$(3.1) \quad \phi_\lambda(z) = \frac{i}{\pi^2} \operatorname{Li}_2(e^{-i\pi z - \pi\lambda}) \quad (z \in S).$$

For  $z = x + iy \in S$  the argument  $e^{-i\pi z - \pi\lambda} = e^{\pi(y-\lambda)} e^{-i\pi x}$  has argument  $-\pi x \in (-\pi, 0)$ , so it avoids the cut  $[1, \infty)$ ; thus  $\phi_\lambda$  is holomorphic on  $S$ , and  $\operatorname{Re} \phi_\lambda = h_\lambda$  by (1.3). Moreover  $h_\lambda(z) = h_0(z - i\lambda)$ .

**Lemma 3.2** (the strip functions). *Let  $\lambda \in \mathbb{R}$ .*

- (1) *For  $z = x + iy \in S$ ,  $h_\lambda(z) = \int_{\mathbb{R}} (w - \lambda)_+ K_x(y - w) dw$ .*
- (2)  *$h_\lambda$  extends continuously to the closed strip  $\bar{S}$ , with  $h_\lambda(iy) = (y - \lambda)_+$  and  $h_\lambda(1 + iy) = 0$ . (For  $y > \lambda$  the point  $iy$  corresponds to an argument of  $\operatorname{Li}_2$  on the cut  $[1, \infty)$ ; the boundary value is the limit from inside  $S$ ,  $\operatorname{Im} \operatorname{Li}_2(\varrho - i0) = -\pi \log \varrho$  for  $\varrho > 1$ .)*
- (3)  *$0 \leq h_\lambda(x + iy) \leq (1 - x)(y - \lambda)_+ + 2$  on  $\bar{S}$ .*
- (4) *For fixed  $x \in (0, 1)$ ,  $\partial_y^2 h_\lambda(x + iy) = K_x(y - \lambda) > 0$ . For fixed  $y$ ,  $x \mapsto h_\lambda(x + iy)$  is concave on  $[0, 1]$ .*
- (5) *For  $\nu = x + iy \in \bar{S}$  let  $\omega_\nu$  be the measure  $K_x(y - w) dw$  if  $0 < x < 1$ ,  $\delta_y$  if  $x = 0$ , and 0 if  $x = 1$ . Then  $h_\lambda(\nu) = \int (w - \lambda)_+ \omega_\nu(dw)$ , and for  $a \in \mathbb{C}$  with  $|\operatorname{Re} a| < \pi$ ,  $\int e^{aw} \omega_\nu(dw) = e^{ay} \sin(a(1 - x))/\sin a$ .*

*Proof.* (1) Let  $\Phi(z)$  denote the right side. Since  $\operatorname{Re} \cot(A + iB) = \sin 2A/(\cosh 2B - \cos 2A)$ ,  $K_x(y - w) = \frac{1}{2} \operatorname{Re} \cot \frac{\pi(z - iw)}{2}$ , which is harmonic in  $z \in S$ ; differentiation under the integral is justified by Lemma 3.1(1), so  $\Phi$  is harmonic on  $S$ . So is  $h_\lambda = \operatorname{Re} \phi_\lambda$ . For  $y < \lambda$ , the integrand vanishes unless  $w > \lambda > y$ , where by Lemma 3.1(2)  $K_x(y - w) = \sum_n \sin(n\pi x) e^{-n\pi(w-y)}$ , uniformly convergent for  $w \geq \lambda$ ; integrating term by term,

$$\Phi(z) = \sum_{n \geq 1} \sin(n\pi x) \frac{e^{n\pi(y-\lambda)}}{(n\pi)^2} = -\frac{1}{\pi^2} \operatorname{Im} \sum_{n \geq 1} \frac{(e^{\pi(y-\lambda)} e^{-i\pi x})^n}{n^2} = h_\lambda(z).$$

Two harmonic functions on the connected set  $S$  that agree on the open set  $\{y < \lambda\}$  agree everywhere.

(2) As  $x \rightarrow 0$ ,  $K_x(y - \cdot)$  is an approximate identity at  $y$ : it is positive, has mass  $1 - x$ , and  $\int |u| K_x(u) du \rightarrow 0$ . Indeed  $\cosh \pi u - \cos \pi x = 2 \sinh^2 \frac{\pi u}{2} + 2 \sin^2 \frac{\pi x}{2} \geq 2(u^2 + x^2)$  (as  $\sinh t \geq t$  and  $\sin \frac{\pi x}{2} \geq x$  on  $[0, 1]$ ), so  $K_x(u) \leq \frac{\pi x}{4(u^2 + x^2)}$  and  $\int_{|u| \leq 1} |u| K_x(u) du \leq \frac{\pi x}{4} \log(1 + x^{-2}) \rightarrow 0$ , while for  $|u| \geq 1$  we use Lemma 3.1(1). Since  $|(w - \lambda)_+ - (y - \lambda)_+| \leq |w - y|$ ,  $h_\lambda(x + iy') \rightarrow (y - \lambda)_+$  as  $(x, y') \rightarrow (0, y)$ . As  $x \rightarrow 1$ , Lemma 3.1(1) gives  $h_\lambda(x + iy) \leq \sin(\pi x) \int (w - \lambda)_+ e^{-\pi|y-w|} dw \rightarrow 0$ , locally uniformly in  $y$ . The statement about  $\operatorname{Li}_2$  on the cut is the classical  $\operatorname{Im} \operatorname{Li}_2(\varrho \pm i0) = \pm \pi \log \varrho$  for  $\varrho > 1$ .

(3)  $(w - \lambda)_+ \leq (y - \lambda)_+ + |w - y|$ ,  $\int K_x = 1 - x$ , and  $\int |u| K_x(u) du \leq \int_{|u| \leq 1} K_x + 4 \int_1^\infty u e^{-\pi u} du \leq 1 + 1$ .

(4) For fixed  $x$ ,  $y \mapsto h_\lambda(x + iy)$  is the convolution of the convex function  $(\cdot - \lambda)_+$  with the integrable, smooth, exponentially decaying kernel  $K_x$ ; its second derivative is the convolution of  $\delta_\lambda$  with  $K_x$ , that is,  $K_x(y - \lambda)$ . As  $h_\lambda$  is harmonic,  $\partial_x^2 h_\lambda = -K_x(y - \lambda) < 0$  on  $(0, 1)$ , and  $x \mapsto h_\lambda(x + iy)$  is continuous on  $[0, 1]$  by (2).

(5) The first formula is (1) and (2); the second follows from Lemma 3.1(3) after the substitution  $u = y - w$ .  $\square$

**3.2. The pencil.** For  $s \in \mathbb{R}$  and  $0 < \tau < 1$  put

$$p_{s,\tau}(\xi) = \det(g - s - \xi(P - \tau)), \quad q_{s,\tau}(\xi) = \det(g_d - s - \xi(P - \tau)) \quad (\xi \in \mathbb{C}).$$

**Lemma 3.3** (the pencil). *Let  $s \in \mathbb{R}$  and  $0 < \tau < 1$ .*

- (1)  $p_{s,\tau}$  and  $q_{s,\tau}$  have degree  $M$ , the same leading coefficient  $c_\tau = \det(\tau - P) = (\tau - 1)^{M-r} \tau^r \neq 0$ , and the same sum of roots.
- (2) Every root  $\xi$  of  $p_{s,\tau}$  or of  $q_{s,\tau}$  satisfies  $|\xi| \leq \|g - s\| / \min(\tau, 1 - \tau)$ .
- (3) Every non-real root  $\xi$  of  $p_{s,\tau}$  satisfies  $|\xi| \leq \|g - s\| / \sqrt{\tau(1 - \tau)}$ .
- (4) If  $s \leq \lambda_{\min}$  or  $s \geq \lambda_{\max}$ , all roots of  $p_{s,\tau}$  are real.
- (5) All roots of  $q_{s,\tau}$  are real.
- (6) The roots of  $p_{s,\tau}$ , as an unordered  $M$ -tuple, depend continuously on  $(s, \tau) \in \mathbb{R} \times (0, 1)$ .

Consequently  $\rho$  is continuous on  $\mathbb{R} \times (0, 1)$ , vanishes unless  $\lambda_{\min} < s < \lambda_{\max}$ , satisfies  $\rho(s, \tau) \leq \frac{M}{2\pi} \|g - s\| (\tau(1 - \tau))^{-1/2}$ , has support in  $K$ , and is integrable on  $\mathbb{R}^2$ .

*Proof.* (1)  $g - s - \xi(P - \tau) = (\tau - P)[\xi - (P - \tau)^{-1}(g - s)]$ , so  $p_{s,\tau}(\xi) = c_\tau \det(\xi - (P - \tau)^{-1}(g - s))$  and the roots are the eigenvalues of  $(P - \tau)^{-1}(g - s)$ , with sum  $\text{Tr}[(P - \tau)^{-1}(g - s)]$ ; likewise for  $q_{s,\tau}$  with  $g_d$ . Since  $(P - \tau)^{-1} = \frac{P}{1 - \tau} - \frac{B}{\tau}$  commutes with  $B$ ,  $\text{Tr}[(P - \tau)^{-1}(BgP + PgB)] = 0$ , and  $g - g_d = BgP + PgB$ .

(2) A root  $\xi$  comes with a unit vector  $v$  such that  $(g - s)v = \xi(P - \tau)v$ . As  $|(P - \tau)v|^2 = (1 - \tau)^2 |Pv|^2 + \tau^2 |Bv|^2 \geq \min(\tau, 1 - \tau)^2$ ,  $|\xi| \min(\tau, 1 - \tau) \leq \|g - s\|$ . For  $q_{s,\tau}$  use  $\|g_d - s\| \leq \|g - s\|$ .

(3) With  $v$  as in (2),  $\langle v, (g - s)v \rangle = \xi \langle v, (P - \tau)v \rangle$ , and both inner products are real. If  $\text{Im} \xi \neq 0$ , then  $\langle v, (P - \tau)v \rangle = 0$  and  $\langle v, (g - s)v \rangle = 0$ ; the first gives  $|Pv|^2 = \tau$ ,  $|Bv|^2 = 1 - \tau$ , hence  $|(P - \tau)v|^2 = \tau(1 - \tau)$ , and the bound follows as in (2).

(4) If  $s \leq \lambda_{\min}$  then  $g - s \geq 0$ , and a non-real root would give  $\langle v, (g - s)v \rangle = 0$ , hence  $(g - s)v = 0$ , hence  $\xi(P - \tau)v = 0$ , which is impossible. The case  $s \geq \lambda_{\max}$  is the same.

(5) Relative to  $\mathbb{C}^M = \text{ran } B \oplus \text{ran } P$ ,  $g_d - s - \xi(P - \tau) = (BgB - s + \xi\tau) \oplus (PgP - s - \xi(1 - \tau))$ ; it is singular if and only if  $\xi = (s - \beta)/\tau$  for an eigenvalue  $\beta$  of  $BgB|_{\text{ran } B}$  or  $\xi = (\mu - s)/(1 - \tau)$  for an eigenvalue  $\mu$  of  $PgP|_{\text{ran } P}$ .

(6) The coefficients of  $p_{s,\tau}$  are polynomials in  $(s, \tau)$ , and the leading coefficient  $c_\tau$  does not vanish on  $(0, 1)$ ; the roots of a polynomial of fixed degree with non-vanishing leading coefficient depend continuously on the coefficients. The final assertions follow from (3), (4), (6), and  $\int_0^1 (\tau(1 - \tau))^{-1/2} d\tau = \pi$ .  $\square$

### 3.3. A Jensen-type identity.

**Lemma 3.4.** *Let  $p(\xi) = c \prod_{i=1}^M (\xi - \xi_i)$  and  $q(\xi) = c \prod_{i=1}^M (\xi - y_i)$  with  $c \neq 0$ , all  $y_i$  real, and  $\sum_i \xi_i = \sum_i y_i$ . Then  $\log |p/q| \in L^1(\mathbb{R})$  and*

$$\int_{\mathbb{R}} \log \left| \frac{p(\xi)}{q(\xi)} \right| d\xi = \pi \sum_{i=1}^M |\text{Im} \xi_i|.$$

*Proof.* Write  $\xi_i = \alpha_i + i\beta_i$ . Then  $\log |p/q| = A + C$  with  $A(\xi) = \frac{1}{2} \sum_i \log(1 + \beta_i^2 / (\xi - \alpha_i)^2) \geq 0$  and  $C(\xi) = \sum_i [\log |\xi - \alpha_i| - \log |\xi - y_i|]$ . Each summand of  $A$  is integrable, with  $\int_{\mathbb{R}} \frac{1}{2} \log(1 + b^2/u^2) du = |b| \int_0^\infty \log(1 + v^{-2}) dv = \pi|b|$ , since  $v \log(1 + v^{-2}) + 2 \arctan v$  is an antiderivative of  $\log(1 + v^{-2})$  with limits 0 at 0 and  $\pi$  at  $\infty$ . The function  $C$  has only logarithmic singularities, and  $C(\xi) = \sum_i [\log |1 - \alpha_i/\xi| - \log |1 - y_i/\xi|] = O(\xi^{-2})$  as  $|\xi| \rightarrow \infty$  because  $\sum \alpha_i = \text{Re} \sum \xi_i = \sum y_i$ ; so  $C \in L^1(\mathbb{R})$ . For  $L > \max_i (|\alpha_i|, |y_i|)$  and real  $a$  with  $|a| < L$ ,  $m_L(a) := \int_{-L}^L \log |\xi - a| d\xi =$

$(L - a) \log(L - a) + (L + a) \log(L + a) - 2L$  is even in  $a$  with  $m_L''(a) = 2L/(L^2 - a^2)$ , so  $m_L(a) = m_L(0) + a^2/L + O(a^4/L^3)$ . Hence  $\int_{-L}^L C = \sum_i (\alpha_i^2 - y_i^2)/L + O(L^{-3}) \rightarrow 0$ , and  $\int_{\mathbb{R}} C = 0$ .  $\square$

**3.4. Real slices and limits.** For  $\xi \in \mathbb{R}$  let  $\lambda_1(\xi), \dots, \lambda_M(\xi)$  be the eigenvalues of  $g - \xi P$  and  $\lambda_1^0(\xi), \dots, \lambda_M^0(\xi)$  those of  $g_d - \xi P$ , that is, the eigenvalues of  $BgB|_{\text{ran } B}$  together with those of  $PgP|_{\text{ran } P}$  shifted by  $-\xi$ . Put

$$U_\xi(w) = \sum_{j=1}^M [(w - \lambda_j(\xi))_+ - (w - \lambda_j^0(\xi))_+], \quad n_\xi(w) = \sum_{j=1}^M [\mathbf{1}_{w > \lambda_j(\xi)} - \mathbf{1}_{w > \lambda_j^0(\xi)}].$$

**Lemma 3.5** (real slices). *Let  $\xi \in \mathbb{R}$ .*

- (1)  $U_\xi$  is continuous, piecewise linear and compactly supported, and  $U'_\xi = n_\xi$  almost everywhere;  $n_\xi$  is bounded with compact support.
- (2) For every  $a \in \mathbb{C} \setminus \{0\}$ ,  $\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = a^{-2} \mathcal{D}(a, a\xi)$ .
- (3) For  $w \notin \text{spec}(g - \xi P) \cup \text{spec}(g_d - \xi P)$ ,  $(\mathcal{H}n_\xi)(w) = \frac{1}{\pi} \log |\det(g - \xi P - w) / \det(g_d - \xi P - w)|$ .

*Proof.* (1)  $U_\xi$  vanishes below all eigenvalues, and above all of them it equals  $\sum_j (\lambda_j^0 - \lambda_j) = \text{Tr}(g_d - \xi P) - \text{Tr}(g - \xi P) = 0$ . (2) As distributions,  $U'_\xi = \sum_j (\delta_{\lambda_j(\xi)} - \delta_{\lambda_j^0(\xi)})$ , and  $U_\xi$  has compact support, so two integrations by parts give  $\int e^{aw} U_\xi = a^{-2} \sum_j (e^{a\lambda_j(\xi)} - e^{a\lambda_j^0(\xi)}) = a^{-2} [\text{Tr } e^{a(g - \xi P)} - \text{Tr}_B e^{aBgB} - e^{-a\xi} \text{Tr}_P e^{aPgP}] = a^{-2} \mathcal{D}(a, a\xi)$ . (3) Pair  $\lambda_j$  with  $\lambda_j^0$ : the function  $\mathbf{1}_{w > \lambda_j} - \mathbf{1}_{w > \lambda_j^0}$  is  $\pm$  the indicator of the interval between them, and  $\frac{1}{\pi} \int_{\alpha}^{\beta} \frac{du}{w - u} = \frac{1}{\pi} \log \left| \frac{w - \alpha}{w - \beta} \right|$ ; summing over  $j$  gives  $\frac{1}{\pi} \log |\prod_j (w - \lambda_j) / \prod_j (w - \lambda_j^0)|$ .  $\square$

**Lemma 3.6** (limits). *For every Hermitian matrix  $A$ ,*

$$\lim_{t \rightarrow +\infty} \text{Tr } e^{A - tP} = \text{Tr}_B e^{BAB} \quad \text{and} \quad \lim_{t \rightarrow -\infty} e^t \text{Tr } e^{A - tP} = \text{Tr}_P e^{PAP}.$$

*Proof.* Both sides of the first statement are multiplied by  $e^c$  if  $A$  is replaced by  $A + c$ , so we may assume  $A \geq \mathbf{1}$ . Let  $\mu_1 \geq \dots \geq \mu_M$  be the eigenvalues of  $H_t = A - tP$  and  $\beta_1 \geq \dots \geq \beta_r$  those of  $BAB|_{\text{ran } B}$ . The min-max principle with subspaces of  $\text{ran } B$ , on which  $H_t = A$ , gives  $\mu_k \geq \beta_k$  for  $k \leq r$ . For  $v = b + p$  ( $b \in \text{ran } B$ ,  $p \in \text{ran } P$ ) and  $t > \|A\|$ ,  $\langle v, H_t v \rangle \leq \langle b, Ab \rangle + 2\|A\|\|b\|\|p\| - (t - \|A\|)\|p\|^2 \leq \langle b, (A + \varepsilon_t)b \rangle$  with  $\varepsilon_t = \|A\|^2/(t - \|A\|)$ . The form  $v \mapsto \langle Bv, (A + \varepsilon_t)Bv \rangle$  has the eigenvalues  $\beta_k + \varepsilon_t > 0$  and 0, so  $\mu_k \leq \beta_k + \varepsilon_t$  for  $k \leq r$ . Finally  $\mu_{r+1} \leq \|A\| - t$  by the min-max principle with the subspace  $\text{ran } P$  of codimension  $r$ . Hence  $\text{Tr } e^{H_t} \rightarrow \sum_k e^{\beta_k}$ . The second statement is the first one with  $B$  and  $P$  exchanged, since  $e^t \text{Tr } e^{A - tP} = \text{Tr } e^{A - |t|B}$  for  $t < 0$ .  $\square$

**3.5. Proof of Theorem D.** The properties of  $\rho$  are Lemma 3.3. We prove (1.6) in four steps.

*Step 0:  $\mathcal{D}/a^2$  is entire.*  $\mathcal{D}$  is entire on  $\mathbb{C}^2$ . Since  $e^{-tP} = B + e^{-tP}$ ,  $\mathcal{D}(0, t) = \text{Tr } e^{-tP} - (M - r)e^{-t} - r = 0$  and  $\partial_a \mathcal{D}(0, t) = \text{Tr}(ge^{-tP}) - e^{-t} \text{Tr}(PgP) - \text{Tr}(BgB) = 0$ . Hence  $\mathcal{G}(a, t) := \mathcal{D}(a, t)/a^2 = \int_0^1 (1 - u) \partial_a^2 \mathcal{D}(ua, t) du$  is entire on  $\mathbb{C}^2$ .

*Step 1: a distribution with support in  $K$ .* For  $\zeta = (\zeta_1, \zeta_2) \in \mathbb{C}^2$  put  $\hat{\mathcal{G}}(\zeta) = \mathcal{G}(-i\zeta_1, i\zeta_2)$ , so that  $e^{-i(\zeta_1 s + \zeta_2 \tau)} = e^{as - t\tau}$  when  $(a, t) = (-i\zeta_1, i\zeta_2)$ . Let  $\eta = \text{Im } \zeta$  and  $H_K(\eta) = \max_{(s, \tau) \in K} (s\eta_1 + \tau\eta_2)$ , the supporting function of  $K$ . For  $X = ag - tP$  the Hermitian part of  $X$  is  $\eta_1 g + \eta_2 P$ , because  $\text{Re } a = \eta_1$  and  $\text{Re } t = -\eta_2$ . If  $u(\theta) = e^{\theta X} v$  then  $\frac{d}{d\theta} |u|^2 = 2\langle u, (\eta_1 g + \eta_2 P)u \rangle \leq 2\lambda_{\max}(\eta_1 g + \eta_2 P)|u|^2$ , so  $|\text{Tr } e^X| \leq Me^{\lambda_{\max}(\eta_1 g + \eta_2 P)} \leq Me^{H_K(\eta)}$ , the last step because  $\lambda_{\max}(\eta_1 g + \eta_2 P) = \max_{|v|=1} (\eta_1 \langle v, gv \rangle + \eta_2 \langle v, Pv \rangle)$  and  $(\langle v, gv \rangle, \langle v, Pv \rangle) \in K$ . Similarly  $|e^{-t} \text{Tr}_P e^{aPgP}| \leq (M - r)e^{H_K(\eta)}$  and  $|\text{Tr}_B e^{aBgB}| \leq re^{H_K(\eta)}$ , since the eigenvalues of the compressions lie in

$[\lambda_{\min}, \lambda_{\max}]$  and  $(\mu, 1), (\beta, 0) \in K$ . Hence  $|\mathcal{D}| \leq 2Me^{H_K(\eta)}$ , and  $|\hat{\mathcal{G}}(\zeta)| \leq 2Me^{H_K(\eta)}$  for  $|\zeta_1| \geq 1$ . For  $|\zeta_1| < 1$  the maximum principle for  $z \mapsto \hat{\mathcal{G}}(z, \zeta_2)$  on the disc  $|z| \leq 2$  gives  $|\hat{\mathcal{G}}(\zeta)| \leq \frac{M}{2} \max_{|z|=2} e^{H_K(\operatorname{Im} z, \eta_2)} \leq \frac{M}{2} e^{H_K(\eta) + 3c_K}$ , where  $c_K = \max(|\lambda_{\min}|, |\lambda_{\max}|)$  is a Lipschitz constant of  $H_K$  in  $\eta_1$  and  $|\operatorname{Im} z - \eta_1| \leq 3$ . So  $|\hat{\mathcal{G}}(\zeta)| \leq Ce^{H_K(\operatorname{Im} \zeta)}$  on  $\mathbb{C}^2$ . By the Paley–Wiener–Schwartz theorem [18, Theorem 7.3.1] there is a distribution  $\Theta$  with support in the convex compact set  $K$  whose Fourier–Laplace transform  $\Theta(e^{-i\langle \cdot, \zeta \rangle})$  is  $\hat{\mathcal{G}}(\zeta)$ ; equivalently,

$$(3.2) \quad \Theta(e^{as-t\tau}) = \mathcal{G}(a, t) = \frac{\mathcal{D}(a, t)}{a^2} \quad \text{for all } (a, t) \in \mathbb{C}^2.$$

On  $\mathbb{R}^2$ ,  $|\hat{\mathcal{G}}| \leq C$ , since  $H_K(0) = 0$ .

*Step 2:*  $\Theta = \rho$  on the open strip  $\mathbb{R} \times (0, 1)$ . Let  $\varphi \in C_c^\infty(\mathbb{R}^2)$  and  $\hat{\varphi}(k) = \int \varphi(z) e^{-ik \cdot z} dz$ . The Fourier transform of  $\Theta$  is the bounded function  $\hat{\mathcal{G}}$  on  $\mathbb{R}^2$  [18, Theorem 7.1.14], so

$$(3.3) \quad \Theta(\varphi) = (2\pi)^{-2} \int_{\mathbb{R}^2} \hat{\mathcal{G}}(k) \hat{\varphi}(-k) dk,$$

with an integrable integrand. The map  $(\sigma, \xi) \mapsto k = (\sigma, -\sigma\xi)$  is a diffeomorphism of  $(\mathbb{R} \setminus \{0\}) \times \mathbb{R}$  onto the complement of the null set  $\{k_1 = 0\}$ , with Jacobian  $|\sigma|$ ; by Fubini,

$$\Theta(\varphi) = (2\pi)^{-2} \int_{\mathbb{R}} d\xi \int_{\mathbb{R}} d\sigma |\sigma| \hat{\mathcal{G}}(\sigma, -\sigma\xi) \hat{\varphi}(-\sigma, \sigma\xi).$$

Fix  $\xi$ . (i) By Lemma 3.5(2) with  $a = -i\sigma$ ,  $\hat{\mathcal{G}}(\sigma, -\sigma\xi) = \mathcal{D}(a, a\xi)/a^2 = \hat{U}_\xi(\sigma)$ . (ii)  $\hat{\varphi}(-\sigma, \sigma\xi) = \iint \varphi(s, \tau) e^{i\sigma(s-\xi\tau)} ds d\tau = \hat{\Psi}_\xi(-\sigma)$ , where  $\Psi_\xi(w) = \int_{\mathbb{R}} \varphi(w + \xi\tau, \tau) d\tau$  belongs to  $C_c^\infty(\mathbb{R})$ . (iii) By Lemma 3.5(1),  $i\sigma\hat{U}_\xi = \hat{n}_\xi$ , so  $|\sigma|\hat{U}_\xi(\sigma) = -i \operatorname{sgn}(\sigma) \hat{n}_\xi(\sigma) = (\mathcal{H}n_\xi)^\wedge(\sigma)$ , with  $\mathcal{H}n_\xi \in L^2(\mathbb{R})$ . By Parseval's identity  $\int fh = (2\pi)^{-1} \int \hat{f}(\sigma) \hat{h}(-\sigma) d\sigma$  ( $f \in L^2$ ,  $h$  Schwartz),

$$\frac{1}{2\pi} \int_{\mathbb{R}} |\sigma| \hat{U}_\xi(\sigma) \hat{\Psi}_\xi(-\sigma) d\sigma = \int_{\mathbb{R}} (\mathcal{H}n_\xi)(w) \Psi_\xi(w) dw = \iint (\mathcal{H}n_\xi)(s - \xi\tau) \varphi(s, \tau) ds d\tau.$$

(iv) Since  $g - \xi P - (s - \xi\tau) = g - s - \xi(P - \tau)$ , Lemma 3.5(3) gives  $(\mathcal{H}n_\xi)(s - \xi\tau) = \frac{1}{\pi} \log |p_{s,\tau}(\xi)/q_{s,\tau}(\xi)|$  for  $\tau \notin \{0, 1\}$  and almost every  $s$ . Therefore

$$(3.4) \quad \Theta(\varphi) = \frac{1}{2\pi^2} \int_{\mathbb{R}} d\xi \iint \varphi(s, \tau) \log \left| \frac{p_{s,\tau}(\xi)}{q_{s,\tau}(\xi)} \right| ds d\tau.$$

Now assume  $\operatorname{supp} \varphi \subseteq [-S_0, S_0] \times [\epsilon, 1 - \epsilon]$  with  $0 < \epsilon < \frac{1}{2}$ . For  $(s, \tau)$  in this set, Lemma 3.3(2) puts all roots of  $p_{s,\tau}$  and  $q_{s,\tau}$  in the disc  $|\xi| \leq R := (\|g\| + S_0)/\epsilon$ , and by Lemma 3.3(1) the two polynomials have the same leading coefficient and the same sum of roots. For  $|\xi| \leq 2R$ ,  $|\log |p/q|| \leq \sum_i (|\log |\xi - \xi_i|| + |\log |\xi - y_i||)$ , whose integral over  $[-2R, 2R]$  is bounded uniformly. For  $|\xi| > 2R$ , using  $|\log |1 - z| + \operatorname{Re} z| \leq |z|^2$  for  $|z| \leq \frac{1}{2}$  and  $\operatorname{Re} \sum_i (\xi_i - y_i) = 0$ ,  $|\log |p/q||(\xi) = |\sum_i (\log |1 - \xi_i/\xi| - \log |1 - y_i/\xi|)| \leq 2MR^2/\xi^2$ . So the integrand of (3.4) is absolutely integrable on  $\mathbb{R} \times \operatorname{supp} \varphi$ , and Fubini and Lemma 3.4 (applicable by Lemma 3.3(1),(5)) give

$$\Theta(\varphi) = \iint \varphi(s, \tau) \left[ \frac{1}{2\pi^2} \int_{\mathbb{R}} \log \left| \frac{p_{s,\tau}(\xi)}{q_{s,\tau}(\xi)} \right| d\xi \right] ds d\tau = \iint \varphi(s, \tau) \rho(s, \tau) ds d\tau.$$

*Step 3: no singular part on the edges.* By Lemma 3.3,  $\rho$  defines a distribution with support in  $K$ , and  $\Theta - \rho$  vanishes on  $\mathbb{R} \times (0, 1)$  and outside  $K$ , so its support lies in the union of the disjoint segments  $L_0 = [\lambda_{\min}, \lambda_{\max}] \times \{0\}$  and  $L_1 = [\lambda_{\min}, \lambda_{\max}] \times \{1\}$ . Splitting  $\Theta - \rho$  with smooth cutoffs and applying the structure theorem for distributions supported in a hyperplane [18,



Theorem 2.3.5] to each piece, there are  $N_0 \geq 0$  and compactly supported distributions  $\alpha_k, \beta_k$  on  $\mathbb{R}$  with

$$\Theta - \rho = \sum_{k=0}^{N_0} [\alpha_k \otimes \delta^{(k)}(\tau) + \beta_k \otimes \delta^{(k)}(\tau - 1)].$$

Evaluating on  $e^{as-t\tau}$ , and using  $\langle \delta^{(k)}, f \rangle = (-1)^k f^{(k)}(0)$ ,

$$\mathcal{G}(a, t) - \iint e^{as-t\tau} \rho(s, \tau) ds d\tau = \sum_{k=0}^{N_0} t^k [A_k(a) + e^{-t} B_k(a)],$$

where  $A_k(a) = \alpha_k(e^{as})$  and  $B_k(a) = \beta_k(e^{as})$ . Fix a real  $a \neq 0$ . As  $t \rightarrow +\infty$ :  $\mathcal{G}(a, t) \rightarrow 0$  by Lemma 3.6 (with  $A = ag$ ), the integral tends to 0 by dominated convergence ( $\rho \in L^1$  with compact support,  $e^{-t\tau} \leq 1$ , and  $e^{-t\tau} \rightarrow 0$  for  $\tau > 0$ ), and  $t^k e^{-t} B_k(a) \rightarrow 0$ ; so the polynomial  $\sum_k A_k(a) t^k$  tends to 0, and every  $A_k(a) = 0$ . As  $t \rightarrow -\infty$ :  $e^t \mathcal{G}(a, t) = a^{-2} [e^t \text{Tr} e^{ag-tP} - \text{Tr}_P e^{aPgP} - e^t \text{Tr}_B e^{aBgB}] \rightarrow 0$  by Lemma 3.6, and  $e^t \iint e^{as-t\tau} \rho = \iint e^{as+t(1-\tau)} \rho \rightarrow 0$ ; so  $\sum_k B_k(a) t^k \rightarrow 0$  and every  $B_k(a) = 0$ . The functions  $A_k, B_k$  are entire (Fourier–Laplace transforms of compactly supported distributions) and vanish on  $\mathbb{R} \setminus \{0\}$ , hence identically, so  $\alpha_k = \beta_k = 0$ . Thus  $\Theta = \rho$ , and (3.2) is (1.6).

*Step 4: the  $\tau$ -marginals.* By Duhamel's formula,

$$\partial_a^2 \text{Tr} e^{ag-tP} \big|_{a=0} = \int_0^1 \text{Tr} (g e^{-\theta t P} g e^{-(1-\theta)t P}) d\theta,$$

and with  $e^{-\theta t P} = B + e^{-\theta t} P$  this equals  $\text{Tr}(BgB)^2 + e^{-t} \text{Tr}(PgP)^2 + 2\|BgP\|_F^2 (1 - e^{-t})/t$ . The first two terms are removed by the subtractions in  $\mathcal{D}$ , so

$$\mathcal{G}(0, t) = \frac{1}{2} \partial_a^2 \mathcal{D}(0, t) = \|BgP\|_F^2 \frac{1 - e^{-t}}{t} = \|BgP\|_F^2 \int_0^1 e^{-t\tau} d\tau.$$

By (3.2) at  $a = 0$ , where now  $\Theta = \rho$ , and uniqueness of Laplace transforms,  $(\int \rho(s, \tau) ds) d\tau = \|BgP\|_F^2 d\tau$  on  $[0, 1]$ ; the marginal is continuous in  $\tau \in (0, 1)$  (dominated convergence, Lemma 3.3), so the identity holds for every  $\tau$ . This completes the proof of Theorem D.  $\square$

**3.6. Proof of Theorem C.** Let  $\nu_k = x_k + iy_k$  ( $k = 1, \dots, M$ ) be the eigenvalues of  $X = B + ig$ . Since  $\text{Re}\langle v, Xv \rangle = \langle v, Bv \rangle \in [0, 1]$  for unit  $v$ ,  $x_k \in [0, 1]$ . Let  $\omega = \sum_k \omega_{\nu_k}$  (Lemma 3.2(5)) and  $\sigma = \sum_\mu \delta_\mu$ , the sum over the eigenvalues  $\mu$  of  $PgP|_{\text{ran } P}$ . By Lemma 3.2(5), the left side of (1.5) is  $\int (w - \lambda)_+ d(\omega - \sigma)(w)$ ; both measures are finite, and  $\omega$  is the sum of finitely many atoms and a measure with a density that is  $O(e^{-\pi|w|})$  as  $|w| \rightarrow \infty$  (Lemma 3.1(1)).

(a) *Laplace transforms.* For real  $a$  with  $0 < |a| < \pi$ , Lemma 3.2(5) gives

$$\hat{\omega}(a) := \int e^{aw} \omega(dw) = \sum_k e^{ay_k} \frac{\sin(a(1-x_k))}{\sin a} = \frac{\text{Tr} e^{ag+iaP} - \text{Tr} e^{ag-iaP}}{2i \sin a},$$

because  $e^{-ia(X-1)} = e^{ag+iaP}$  has the eigenvalues  $e^{ay_k} e^{ia(1-x_k)}$  and  $e^{ia(X^*-1)} = e^{ag-iaP}$  has the eigenvalues  $e^{ay_k} e^{-ia(1-x_k)}$ . Also  $\hat{\sigma}(a) = \text{Tr}_P e^{aPgP}$ . In  $\mathcal{D}(a, -ia) - \mathcal{D}(a, ia)$  the  $B$ -terms cancel, and

$$\mathcal{D}(a, -ia) - \mathcal{D}(a, ia) = \text{Tr} e^{ag+iaP} - \text{Tr} e^{ag-iaP} - (e^{ia} - e^{-ia}) \text{Tr}_P e^{aPgP} = 2i \sin a [\hat{\omega}(a) - \hat{\sigma}(a)].$$

By Theorem D at  $t = \mp ia$ , the left side is  $2ia^2 \iint e^{as} \sin(a\tau) \rho(s, \tau) ds d\tau$ . With Lemma 3.1(3) for  $x = 1 - \tau$ ,  $\sin(a\tau)/\sin a = \int e^{-au} K_{1-\tau}(u) du$ , and Tonelli's theorem (all integrands are

nonnegative for real  $a$ ),

$$(3.5) \quad \hat{\omega}(a) - \hat{\sigma}(a) = a^2 \int_{\mathbb{R}} e^{aw} V(w) dw, \quad V(w) := \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s-w) \rho(s, \tau) ds d\tau.$$

$V$  is finite, continuous and  $O(e^{-\pi|w|})$ :  $\int K_{1-\tau} = \tau \leq 1$ ,  $\sup_s \rho(s, \tau) \leq C(\tau(1-\tau))^{-1/2}$ ,  $\rho$  has compact support, and  $K_{1-\tau}(u) \leq 2 \sin(\pi\tau) e^{-\pi|u|}$  for  $|u| \geq 1$ .

(b) *Moments.* By (3.5),  $\hat{\omega} - \hat{\sigma} = O(a^2)$  as  $a \rightarrow 0$  (both sides extend holomorphically to  $|\operatorname{Re} a| < \pi$ ), so  $\omega$  and  $\sigma$  have the same mass and the same first moment. Put  $W(w) = \int (w-u)_+ d(\omega - \sigma)(u)$ . Then  $W$  is continuous; for  $w$  below all atoms of  $\omega$  and  $\sigma$ ,  $W(w) = \int_{u < w} (w-u) d\omega(u) = O(e^{-\pi|w|})$ , and for  $w$  above all atoms, by the moment identities,  $W(w) = \int_{u > w} (u-w) d\omega(u) = O(e^{-\pi w})$ ; the same holds for  $W'$ . As  $W'' = \omega - \sigma$  in the sense of distributions, two integrations by parts give  $\int e^{aw} W(w) dw = a^{-2}(\hat{\omega}(a) - \hat{\sigma}(a)) = \int e^{aw} V(w) dw$  for real  $0 < |a| < \pi$ . Both sides are holomorphic in  $|\operatorname{Re} a| < \pi$ , so they agree on  $i\mathbb{R}$ : the integrable continuous functions  $W$  and  $V$  have the same Fourier transform, and  $W = V$ .

(c) *Conclusion.* Since  $(u-\lambda)_+ = (\lambda-u)_+ + (u-\lambda)$  and  $\omega - \sigma$  has zero mass and zero first moment,  $\int (u-\lambda)_+ d(\omega - \sigma) = W(\lambda) = V(\lambda)$ , which is (1.5).

(d) *Equality.*  $K_{1-\tau}(s-\lambda) > 0$  for  $0 < \tau < 1$ , and  $\rho \geq 0$  is continuous on  $\mathbb{R} \times (0, 1)$ ; so  $V(\lambda) = 0$  for one  $\lambda$  forces  $\rho = 0$ , hence  $\mathcal{D} \equiv 0$  by Theorem D, hence  $\|BgP\|_F^2 = 0$  by Step 4 of Section 3.5, that is,  $BgP = 0 = PgB$  and  $[B, g] = BgP - PgB = 0$ . Conversely, if  $[B, g] = 0$  then  $g = g_d$ ,  $p_{s,\tau} = q_{s,\tau}$ ,  $\rho = 0$  and  $V \equiv 0$ .  $\square$

### 3.7. Remarks.

*Remark 3.7* ( $M = 2$ ). Let  $B = \operatorname{diag}(1, 0)$  and  $g = \begin{pmatrix} \alpha & c \\ \bar{c} & \beta \end{pmatrix}$ . Then  $p_{s,\tau}(\xi)$  has discriminant  $(m(\tau) - s)^2 - 4\tau(1-\tau)|c|^2$  with  $m(\tau) = (1-\tau)\alpha + \tau\beta$ , so

$$\rho(s, \tau) = \frac{(4\tau(1-\tau)|c|^2 - (s - m(\tau))^2)_+^{1/2}}{2\pi\tau(1-\tau)} :$$

each slice  $\tau = \text{const}$  is a semicircle law of radius  $2|c|\sqrt{\tau(1-\tau)}$  and mass  $|c|^2 = \|BgP\|_F^2$ , centred on the segment from  $\alpha$  (the eigenvalue of  $BgB$ , at  $\tau = 0$ ) to  $\beta$  (that of  $PgP$ , at  $\tau = 1$ ).

*Remark 3.8* (convex order). In the proof of Theorem C,  $\omega$  (the sweep of the eigenvalues of  $B + ig$  onto the left edge of  $\bar{S}$  by harmonic measure) and  $\sigma$  (the eigenvalue measure of  $PgP|_{\operatorname{ran} P}$ ) have equal masses  $M - r$  and equal means, and  $\int (\lambda - u)_+ d(\omega - \sigma) \geq 0$  for all  $\lambda$ . Hence Theorem C says exactly that  $\sigma$  is dominated by  $\omega$  in the convex order:  $\operatorname{Tr}_P f(PgP) \leq \int f d\omega$  for every convex  $f$  of at most linear growth, with equality for all such  $f$  if and only if  $[B, g] = 0$ . Applying this to  $(P, g)$ , whose matrix  $P + ig = \mathbf{1} - (B + ig)^*$  has the eigenvalues  $1 - \bar{v}$ , gives the mirror statement: the eigenvalue measure of  $BgB|_{\operatorname{ran} B}$  is dominated in convex order by the sweep of  $\operatorname{spec}(B + ig)$  onto the right edge. When  $B$  and  $g$  commute both statements are identities. In this sense Theorem C is a Schur–Horn type majorisation for the non-normal matrix  $B + ig$ ; first-order (distribution-function) versions are false whenever  $[B, g] \neq 0$ , because  $\omega$  has a density as soon as one eigenvalue lies in the open strip.

*Remark 3.9* (relation to Heinävaara’s tracial joint spectral measures). For Hermitian  $A, B$ , Heinävaara [17] constructs a positive measure  $\mu_{A,B}$  on  $\mathbb{R}^2$  such that  $\operatorname{tr} \mathcal{T}(f)(xA + yB) = \int f(ax + by) d\mu_{A,B}(a, b)$  for a large class of  $f$ , where  $\mathcal{T}(f)(x) = \int_0^1 \frac{1-t}{t} f(xt) dt$  (so  $\mathcal{T}$  maps  $x^k$  to  $x^k/(k(k+1))$ ), computes the absolutely continuous part of  $\mu_{A,B}$  explicitly as  $\frac{1}{2\pi} \sum_i |\operatorname{Im} \lambda_i(C_{a,b})|$  for an auxiliary pencil  $C_{a,b}$  ([17, Theorem 3.1]), and deduces Stahl’s theorem. That proof computes a Fourier transform in polar coordinates, one line through the origin at a time, and uses

an integral identity for  $\log |\det|$  of a pencil. Our proof of Theorem D follows the same route in an affine chart adapted to the projection  $P$ : the lines are the real slices of Lemma 3.5, the division by  $a^2$  plays the role of  $\mathcal{T}$ , the pinched subtractions remove the contributions of the edges  $\tau \in \{0, 1\}$ , and Lemma 3.4 is the corresponding  $\log |\det|$  identity. We use only the statement of Theorem D in the form given above, and we have given a complete proof of it. Appendix B gives a second proof, in which the Fourier inversion along lines is replaced by a Radon identity for the pencil (Theorem B.1), proved with a complex Burgers equation for its roots.

*Remark 3.10* (an independent proof for rank one, and an automorphism of order three). For rank  $B \leq 1$  there is a second proof of Theorem C (inequality part), by a different method; we sketch it, the details being in the accompanying notes [32]. Let  $B = vv^*$ , and let  $\nu_k = x_k + iy_k$  be the eigenvalues of  $B + ig$ , so that  $x_k \in [0, 1]$  and  $\sum_k x_k = 1$ . If all  $x_k > 0$ , the matrix determinant lemma gives  $\det(g - iB - t)/\det(g + iB - t) = \exp(-2\pi i F(t))$  for real  $t$ , where  $F(t) = \sum_k [\frac{1}{2} + \frac{1}{\pi} \arctan \frac{t - y_k}{x_k}]$  is the distribution function of the sum of the Cauchy laws with centres  $y_k$  and scales  $x_k$ , and the left side equals  $(1 - iG(t))/(1 + iG(t))$  with  $G(t) = \langle v, (g - t)^{-1}v \rangle$ ; the zeros of  $G$  are the eigenvalues of  $PgP|_{\text{ran } P}$ . Hence these eigenvalues are the integer quantiles  $\gamma_j = \min\{t : F(t) \geq j\}$ ,  $1 \leq j \leq M - 1$ , and conversely every configuration with  $x_k > 0$  and  $\sum x_k = 1$  arises. Theorem C for rank one thus becomes the inequality  $\sum_k h_0(\nu_k) \geq \sum_j (\gamma_j)_+$  for  $M$  points in the strip. It is proved by induction on  $M$ , for the more general constraint  $\sum_k x_k = s \leq 1$  with a correction term: a penalised minimiser is either reducible (a point on or near  $\partial S$ , handled by the induction hypothesis) or an interior critical point; at an interior critical point all  $\nu_k$  are zeros, counted with multiplicity, of a holomorphic function on  $S$  of the form  $\phi'_0 + \frac{i}{\pi} \sum_j \frac{c_j}{z - i\gamma_j} - m_0 + ia z + \varepsilon \pi \cos \pi z$  ( $c_j, \varepsilon > 0, a \geq 0$ ), and an argument-principle count shows that such a function has at most  $J + 1 \leq M - 1$  zeros in  $S$ , where  $J$  is the number of positive quantiles. Combined with a duality argument it also gives Theorem C whenever rank  $P \leq 3$ , hence for all  $M \leq 5$ .

The derivative  $\phi'_0(z) = -\frac{1}{\pi} \log(1 - e^{-i\pi z})$  has a curious structure. Let  $\chi = i\phi'_0$ . In the coordinate  $w = e^{-i\pi z}$ , which maps  $S$  onto the lower half-plane,  $\chi$  acts as the Möbius map  $w \mapsto 1/(1 - w)$ ; hence  $\chi$  is an automorphism of  $S$  of order three, with fixed point  $\frac{1}{3}$ , and  $z + \chi(z) + \chi(\chi(z)) = 1$  on  $S$  (the three values of  $w$  have product  $-1$  and arguments in  $(-\pi, 0)$ ). By the holomorphic functional calculus,  $X + \chi(X) + \chi(\chi(X)) = \mathbf{1}$  for every matrix  $X$  with spectrum in  $S$ . If  $B + ig$  has no eigenvalue on  $\partial S$ , the derivative of  $g \mapsto \sum_{\nu \in \text{spec}(B+ig)} h_0(\nu)$  in a Hermitian direction  $\delta g$  is  $\text{Tr}[\text{Re } \chi(B + ig) \delta g]$ , where  $\text{Re } Y = \frac{1}{2}(Y + Y^*)$ ; so the critical points of the defect in (1.5) (at  $\lambda = 0$ , where  $PgP|_{\text{ran } P}$  is invertible) are the solutions of  $\text{Re } \chi(B + ig) = Q$ , with  $Q$  the spectral projection of  $PgP$  for  $(0, \infty)$ .

#### 4. A NONCOMMUTATIVE CONTINUUM THEOREM

Let  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$  with  $dm = d\theta/2\pi$ , and let  $\mathbb{D}$  be the open unit disc. A *projection step field* on  $\mathbb{C}^M$  is given by points  $t_0 < t_1 < \dots < t_{n-1} < t_0 + 2\pi$  (indices modulo  $n$ ,  $t_n = t_0 + 2\pi$ ), the open arcs  $J_j = (t_j, t_{j+1})$ , and orthogonal projections  $B_j$  on  $\mathbb{C}^M$ ; it is the function  $B(\theta) = B_j$  for  $\theta \in J_j$  (values at the points  $t_j$  play no role). Its *conjugate function* is

$$(4.1) \quad g(\theta) = \text{p.v.} \int_{\mathbb{T}} \cot \frac{\theta - \varphi}{2} B(\varphi) dm(\varphi) = \frac{1}{\pi} \sum_{j=0}^{n-1} (B_j - B_{j-1}) \log \left| \sin \frac{\theta - t_j}{2} \right| \quad (\theta \notin \{t_j\}),$$

where the second expression follows from  $\frac{d}{d\varphi} \log \left| \sin \frac{\theta - \varphi}{2} \right| = -\frac{1}{2} \cot \frac{\theta - \varphi}{2}$  and summation by parts. Thus  $g$  is Hermitian, real-analytic on each arc  $J_j$ , and in  $L^p(\mathbb{T})$  for every  $p < \infty$ . We write  $P(\theta) = \mathbf{1} - B(\theta)$  and  $\bar{B} = \int_{\mathbb{T}} B dm$ .

**Lemma 4.1** (Herglotz function of a step field). *For  $z \in \mathbb{D}$  let  $\mathcal{F}(z) = \int_{\mathbb{T}} \frac{e^{i\varphi} + z}{e^{i\varphi} - z} B(\varphi) dm(\varphi)$ . Then:*

- (1)  $\mathcal{F}(z) = \bar{B} + \frac{i}{\pi} \sum_j (B_j - B_{j-1}) \operatorname{Log}(1 - ze^{-it_j})$  (principal logarithm);
- (2)  $\mathcal{F}$  is holomorphic on  $\mathbb{D}$  and  $\mathcal{F}(0) = \bar{B}$ ; its Hermitian part  $\operatorname{Re} \mathcal{F}(z) := \frac{1}{2}(\mathcal{F}(z) + \mathcal{F}(z)^*)$  is  $\int_{\mathbb{T}} P_z(\varphi) B(\varphi) dm(\varphi)$ , where  $P_z > 0$  is the Poisson kernel;
- (3)  $\mathcal{F}$  extends continuously to  $\bar{\mathbb{D}} \setminus \{e^{it_j}\}$ , with  $\mathcal{F}(e^{i\theta}) = B(\theta) + ig(\theta)$  for  $\theta \notin \{t_j\}$ ;
- (4) for  $\frac{1}{2} \leq r < 1$ ,  $\|\mathcal{F}(re^{i\theta})\| \leq C_0 + \frac{1}{\pi} \sum_j |\log |\sin \frac{\theta - t_j}{2}||$ , with  $C_0 = 1 + n(\log 2 + \frac{\pi}{2})/\pi$ .

*Proof.* (1) Since  $\frac{e^{i\varphi} + z}{e^{i\varphi} - z} = 1 + \frac{2}{i} \frac{d}{d\varphi} \operatorname{Log}(1 - ze^{-i\varphi})$  and  $\operatorname{Re}(1 - ze^{-i\varphi}) > 0$ ,  $\frac{1}{2\pi} \int_a^b \frac{e^{i\varphi} + z}{e^{i\varphi} - z} d\varphi = \frac{b-a}{2\pi} - \frac{i}{\pi} [\operatorname{Log}(1 - ze^{-ib}) - \operatorname{Log}(1 - ze^{-ia})]$ . Multiply by  $B_j$  for the arc  $J_j$ , sum over  $j$ , and regroup the logarithms by endpoints. (2) is clear. (3) The right side of (1) is continuous on  $\bar{\mathbb{D}} \setminus \{e^{it_j}\}$ . For  $u \in (0, 2\pi)$ ,  $1 - e^{iu} = 2 \sin \frac{u}{2} e^{i(u-\pi)/2}$ , so  $\operatorname{Log}(1 - e^{iu}) = \log |2 \sin \frac{u}{2}| + \frac{i}{2}(u - \pi)$ . The imaginary part of  $\mathcal{F}(e^{i\theta})$  is therefore  $\frac{1}{\pi} \sum_j (B_j - B_{j-1}) \log |2 \sin \frac{\theta - t_j}{2}| = g(\theta)$  (the  $\log 2$  terms cancel since  $\sum_j (B_j - B_{j-1}) = 0$ ). For the real part let  $\theta \in J_x$  and  $u_j = (\theta - t_j) \bmod 2\pi \in (0, 2\pi)$ , so  $u_j = \theta - t_j$  for  $j \leq x$  and  $u_j = \theta - t_j + 2\pi$  for  $j > x$  (with  $0 \leq j \leq n-1$ ). Then  $\operatorname{Re} \mathcal{F}(e^{i\theta}) = \bar{B} - \frac{1}{2\pi} \sum_j (B_j - B_{j-1}) u_j$ , and summation by parts gives  $\sum_j (B_j - B_{j-1}) u_j = -\sum_j (B_j - B_{j-1}) t_j + 2\pi \sum_{j>x} (B_j - B_{j-1}) = 2\pi(\bar{B} - B_{n-1}) + 2\pi(B_{n-1} - B_x)$ , so  $\operatorname{Re} \mathcal{F}(e^{i\theta}) = B_x$ . (4)  $\|B_j - B_{j-1}\| \leq 1$ , and for  $\frac{1}{2} \leq r < 1$ ,  $\sqrt{2} |\sin \frac{u}{2}| \leq |1 - re^{iu}| \leq 2$ , hence  $|\operatorname{Log}(1 - re^{iu})| \leq \log 2 + \frac{\pi}{2} + |\log |\sin \frac{u}{2}||$ .  $\square$

**Theorem 4.2** (continuum theorem). *Let  $B$  be a projection step field on  $\mathbb{C}^M$  with conjugate function  $g$ , and let  $A$  be a measurable field of Hermitian matrices with  $0 \leq A(\theta) \leq P(\theta)$ . Put  $\alpha = \int \tau(A) dm$ ,  $\beta = \int \tau(B) dm$ , and let  $\beta_1, \dots, \beta_M \in [0, 1]$  be the eigenvalues of  $\bar{B}$ . For  $\lambda \in \mathbb{R}$  and  $\theta \notin \{t_j\}$  put*

$$D_\lambda(\theta) = \sum_{\nu \in \operatorname{spec}(B(\theta) + ig(\theta))} h_\lambda(\nu) - \operatorname{Tr}[(P(\theta)(g(\theta) - \lambda)P(\theta))_+],$$

$$S_\lambda(\theta) = \tau[(P(\theta)(g(\theta) - \lambda)P(\theta))_+] - \tau[A(\theta)(g(\theta) - \lambda)].$$

Then  $D_\lambda \geq 0$ ,  $S_\lambda \geq 0$ , both are integrable, and

$$(4.2) \quad \lambda\alpha + h_\lambda(\beta) - \int_{\mathbb{T}} \tau(Ag) dm = \left[ h_\lambda(\beta) - \frac{1}{M} \sum_{i=1}^M h_\lambda(\beta_i) \right] + \frac{1}{M} \int_{\mathbb{T}} D_\lambda dm + \int_{\mathbb{T}} S_\lambda dm,$$

where the bracket is also nonnegative. In particular  $\int \tau(Ag) dm \leq \lambda\alpha + h_\lambda(\beta)$  for every  $\lambda$ , and if  $0 < \alpha, \beta$  and  $\alpha + \beta < 1$ , then  $\int \tau(Ag) dm \leq \Phi_c(\alpha, \beta)$ .

*Proof. Signs.*  $D_\lambda \geq 0$  is Theorem C for the pair  $(B(\theta), g(\theta))$  (note that  $\operatorname{Tr}[(P(g - \lambda)P)_+]$  is the same whether  $P(g - \lambda)P$  is regarded on  $\mathbb{C}^M$  or on  $\operatorname{ran} P$ ). For  $S_\lambda$ :  $0 \leq A \leq P$  gives  $BAB = 0$ , hence  $AB = 0$  and  $A = PAP$ ; with  $Y = P(g - \lambda)P$ ,  $\tau(A(g - \lambda)) = \tau(AY) \leq \tau(AY_+) \leq \tau(PY_+) = \tau(Y_+)$ . The bracket is nonnegative because  $x \mapsto h_\lambda(x)$  is concave on  $[0, 1]$  (Lemma 3.2(4)) and  $\frac{1}{M} \sum_i \beta_i = \tau(\bar{B}) = \beta$ .

*A harmonic function.* Let  $K_0 = \{v : B_j v = 0 \ \forall j\}$  and  $K_1 = \{v : B_j v = v \ \forall j\}$ ; these subspaces and  $K' = (K_0 \oplus K_1)^\perp$  reduce every  $B_j$ , and  $\mathcal{F}(z) = 0 \oplus \mathbf{1} \oplus \mathcal{F}'(z)$  accordingly. For a unit vector  $v \in K'$  and  $z \in \mathbb{D}$ ,  $\langle v, \operatorname{Re} \mathcal{F}(z)v \rangle = \sum_j \omega_j(z) \langle v, B_j v \rangle$  with harmonic measures  $\omega_j(z) > 0$ ,  $\sum_j \omega_j(z) = 1$ ; this is 0 only if  $v \in K_0$  and 1 only if  $v \in K_1$ , which is impossible. By compactness the numerical range of  $\operatorname{Re} \mathcal{F}'(z)$  is a compact subset of  $(0, 1)$ , so  $\operatorname{spec} \mathcal{F}'(z) \subseteq S$ . By the holomorphic functional calculus (a Riesz–Dunford integral over a contour in  $S$ , locally

constant in  $z$ ),  $z \mapsto \operatorname{Re} \operatorname{Tr} \phi_\lambda(\mathcal{F}'(z))$  is harmonic on  $\mathbb{D}$ . Hence

$$u_\lambda(z) := \sum_{\nu \in \operatorname{spec} \mathcal{F}(z)} h_\lambda(\nu) = \dim K_0 h_\lambda(0) + \dim K_1 h_\lambda(1) + \operatorname{Re} \operatorname{Tr} \phi_\lambda(\mathcal{F}'(z))$$

is harmonic on  $\mathbb{D}$ , and  $u_\lambda(0) = \sum_i h_\lambda(\beta_i)$  because  $\mathcal{F}(0) = \bar{B}$ .

*Boundary passage.* Every eigenvalue  $\nu$  of  $\mathcal{F}(z)$ ,  $z \in \mathbb{D} \setminus \{e^{it_j}\}$ , lies in  $\bar{S}$  and satisfies  $|\operatorname{Im} \nu| \leq \|\mathcal{F}(z)\|$ ; by Lemma 3.2(3),  $0 \leq u_\lambda(re^{i\theta}) \leq M(\|\mathcal{F}(re^{i\theta})\| + |\lambda| + 2)$ , which by Lemma 4.1(4) is bounded, for  $\frac{1}{2} \leq r < 1$ , by an integrable function of  $\theta$ . For  $\theta \notin \{t_j\}$ ,  $\mathcal{F}(re^{i\theta}) \rightarrow B(\theta) + ig(\theta)$  as  $r \rightarrow 1$  (Lemma 4.1(3)), the eigenvalues converge, and  $h_\lambda$  is continuous on  $\bar{S}$ ; so  $u_\lambda(re^{i\theta}) \rightarrow u_\lambda^*(\theta) := \sum_{\nu \in \operatorname{spec}(B(\theta) + ig(\theta))} h_\lambda(\nu)$ . The mean value property on  $|z| = r$  and dominated convergence give  $\int_{\mathbb{T}} u_\lambda^* dm = \sum_i h_\lambda(\beta_i)$ .

*Conclusion.* By definition,  $\tau(A(g - \lambda)) = \frac{1}{M} u_\lambda^* - \frac{1}{M} D_\lambda - S_\lambda$  off  $\{t_j\}$ , and every term is integrable ( $g \in L^p$ ,  $A$  bounded,  $u_\lambda^*$  dominated as above). Integrating,  $\int \tau(Ag) dm - \lambda\alpha = \frac{1}{M} \sum_i h_\lambda(\beta_i) - \frac{1}{M} \int D_\lambda - \int S_\lambda$ , which is (4.2). The last assertion follows from (4.2) and Lemma 4.4 below, whose proof uses only the identity (4.2) for  $M = 1$ .  $\square$

**Lemma 4.3** (two adjacent arcs). *Let  $\alpha, \beta \geq 0$  with  $\alpha + \beta \leq 1$ , and let  $g_\beta$  be the conjugate function of the scalar step field  $\mathbf{1}_{(-2\pi\beta, 0)}$ . Then  $\int_0^{2\pi\alpha} g_\beta dm = \Phi_c(\alpha, \beta)$ . Equivalently, for  $a, b \geq 0$  with  $a + b \leq N$ ,  $\int_0^a \int_{-b}^0 \cot \frac{\pi(\xi - \eta)}{N} d\eta d\xi = N^2 \Phi_c(a/N, b/N)$ .*

*Proof.* The function  $\Xi(u) = -\frac{N^2}{2\pi^2} \operatorname{Cl}_2(2\pi u/N)$  is continuous, odd and  $N$ -periodic, with  $\Xi''(u) = \cot \frac{\pi u}{N}$  for  $u \notin N\mathbb{Z}$  (since  $\operatorname{Cl}_2''(\theta) = -\frac{1}{2} \cot \frac{\theta}{2}$ ) and  $\Xi'$  locally integrable. Integrating twice,  $\int_0^a \int_{-b}^0 \Xi''(\xi - \eta) d\eta d\xi = \Xi(a + b) - \Xi(a) - \Xi(b) + \Xi(0)$ , which equals  $\frac{N^2}{2\pi^2} [\operatorname{Cl}_2(2\pi \frac{a}{N}) + \operatorname{Cl}_2(2\pi \frac{b}{N}) - \operatorname{Cl}_2(2\pi \frac{a+b}{N})] = N^2 \Phi_c(\frac{a}{N}, \frac{b}{N})$ , because  $-\operatorname{Cl}_2(\theta) = \operatorname{Cl}_2(2\pi - \theta)$ . The first form follows by the substitution  $\theta = 2\pi\xi/N$ ,  $\varphi = 2\pi\eta/N$ .  $\square$

**Lemma 4.4** (the classical value). *Let  $\alpha, \beta > 0$  with  $\alpha + \beta < 1$ . The function  $f(\lambda) = \lambda\alpha + h_\lambda(\beta)$  is strictly convex on  $\mathbb{R}$ , and*

$$\min_{\lambda \in \mathbb{R}} f(\lambda) = \Phi_c(\alpha, \beta), \quad \text{attained only at } \lambda_0 = \frac{1}{\pi} \log \frac{\sin \pi(\alpha + \beta)}{\sin \pi\alpha}.$$

*In particular  $\min_\lambda [\frac{\lambda}{4} + h_\lambda(\frac{1}{4})] = \Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$ , attained only at  $\lambda^* = \frac{\log 2}{2\pi}$ .*

*Proof.* Since  $h_\lambda(\beta) = h_0(\beta - i\lambda)$ , Lemma 3.2(4) gives  $f''(\lambda) = K_\beta(\lambda) > 0$ . Apply Theorem 4.2 with  $M = 1$  to  $B = \mathbf{1}_{(-2\pi\beta, 0)}$  and  $A = \mathbf{1}_{(0, 2\pi\alpha)}$ . The conjugate function  $g_\beta(\theta) = \frac{1}{\pi} \log |\sin \frac{\theta + 2\pi\beta}{2} / \sin \frac{\theta}{2}|$  has derivative  $\frac{1}{2\pi} [\cot \frac{\theta + 2\pi\beta}{2} - \cot \frac{\theta}{2}] < 0$  on  $(0, 2\pi(1 - \beta))$ , where it decreases from  $+\infty$  to  $-\infty$ . With  $\lambda_0 = g_\beta(2\pi\alpha)$ , which is the stated value,  $S_{\lambda_0} = 0$  off finitely many points: on  $(0, 2\pi\alpha)$  both terms equal  $g_\beta - \lambda_0 > 0$ , on  $(2\pi\alpha, 2\pi(1 - \beta))$  both vanish, and on the arc of  $B$  both vanish. For  $M = 1$ ,  $D_\lambda \equiv 0$  (if  $B(\theta) = 1$  both terms of  $D_\lambda$  vanish, if  $B(\theta) = 0$  both equal  $(g - \lambda)_+$  and the bracket in (4.2) vanishes). Hence  $f(\lambda) - \int Ag_\beta dm = \int S_\lambda \geq 0$  for all  $\lambda$ , with equality at  $\lambda_0$ , and  $\int Ag_\beta dm = \Phi_c(\alpha, \beta)$  by Lemma 4.3. For  $\alpha = \beta = \frac{1}{4}$ ,  $\lambda_0 = \frac{1}{\pi} \log \frac{1}{\sin(\pi/4)} = \frac{\log 2}{2\pi}$  and  $\Phi_c = \frac{1}{2\pi^2} [2 \operatorname{Cl}_2(\frac{\pi}{2}) + \operatorname{Cl}_2(\pi)] = G/\pi^2$ .  $\square$

The identity  $\min_\lambda f = \Phi_c$  is a Legendre-type identity,  $\inf_y [h_0(x + iy) - py] = \Phi_c(x, p)$ ; it can also be derived from the Bloch–Wigner evaluation of  $\operatorname{Im} \operatorname{Li}_2$  at the vertex of a triangle with angles  $\pi(\alpha, \beta, 1 - \alpha - \beta)$  [43]. At  $\alpha = \beta = \frac{1}{4}$  it is the classical value  $\operatorname{Im} \operatorname{Li}_2(\frac{1-i}{2}) = \frac{\pi}{8} \log 2 - G$  [27].

**Corollary 4.5** (equality). *Let  $B$  be a projection step field on  $\mathbb{C}^M$  with  $\int B \, dm = \frac{1}{4}\mathbf{1}$ , and let  $0 \leq A \leq \mathbf{1} - B$  be measurable with  $\int \tau(A) \, dm = \frac{1}{4}$ . Then*

$$\frac{G}{\pi^2} - \int_{\mathbb{T}} \tau(Ag) \, dm = \frac{1}{M} \int_{\mathbb{T}} D_{\lambda^*} \, dm + \int_{\mathbb{T}} S_{\lambda^*} \, dm \geq 0.$$

*If equality holds, then  $[B(\theta), g(\theta)] = 0$  for almost every  $\theta$ .*

*Proof.* All  $\beta_i$  equal  $\frac{1}{4}$ , so the bracket in (4.2) vanishes; take  $\lambda = \lambda^*$  and use Lemma 4.4. In the equality case  $D_{\lambda^*} = 0$  almost everywhere, and the equality clause of Theorem C gives  $[B(\theta), g(\theta)] = 0$ .  $\square$

*Remark 4.6* (scope). For  $M = 1$ , Theorem 4.2 contains the circle version of the bathtub bound obtained from the Stein–Weiss theorem [41, 25] (cf. [30, Proposition 4.6]); the proof above uses only the mean value property of  $u_{\lambda}$ . The theorem extends to measurable projection fields, by the standard boundary theory of Herglotz functions and the exponential integrability of conjugate functions of bounded functions [45]; we only need step fields.

*Remark 4.7* (non-commuting maximisers in the continuum). The bound  $\Phi_c(\alpha, \beta)$  is attained by many fields, including non-commuting ones. For instance, if  $\Theta$  is a rational  $M \times M$  inner function with  $\Theta(0) = 0$  (a finite Blaschke–Potapov product), the spectral projections  $B = \mathbf{1}_{(-2\pi\beta, 0)}(\Theta(e^{i\theta}))$  and  $A = \mathbf{1}_{(0, 2\pi\alpha)}(\Theta(e^{i\theta}))$  satisfy  $\int B \, dm = \beta\mathbf{1}$  and  $\int A \, dm = \alpha\mathbf{1}$  (because  $\int \Theta(e^{i\theta})^k \, dm = \Theta(0)^k = 0$  for  $k \geq 1$ ), the conjugate function of  $B$  is  $g_{\beta}(\Theta(e^{i\theta}))$ , and in the measurable version of Theorem 4.2 (Remark 4.6) all inequalities of (4.2) are equalities at  $\lambda_0$ , since  $A, B$  and  $g$  commute pointwise; the field  $B$  does not commute with itself at different points unless the factors of  $\Theta$  commute. Thus the continuum problem alone cannot detect the structure of the optimal discrete strategies. In the proof of Theorem B the needed rigidity comes instead from the cell structure (Proposition 7.3). We do not use this remark.

## 5. CELL EMBEDDINGS

**5.1. Cells and kernels.** Let  $\ell \in \Delta_d$ , extended to  $\mathbb{Z}$  by  $\ell_x = \ell_{x \bmod d}$ . Put  $L_0 = 0$  and  $L_{x+1} = L_x + \ell_x$ , so that  $L_{x+d} = L_x + d$  and  $L_{x+N} = L_x + N$ , and define the cells  $I_x = [L_x, L_{x+1}) \subseteq \mathbb{T}_N := \mathbb{R}/N\mathbb{Z}$ ,  $x \in \mathbb{Z}_N$ ; their lengths are  $\ell_x$ , and  $S_r(m) = L_{r+m} - L_r$ . We write  $\kappa(u) = \cot(\pi u/N)$  also for real  $u \notin N\mathbb{Z}$ . For  $x \neq y$  in  $\mathbb{Z}_N$  let

$$K^{\ell}(x, y) = \int_{I_x} \int_{I_y} \kappa(\xi - \eta) \, d\eta \, d\xi, \quad K^{\ell}(x, x) = 0,$$

and for a  $Q$ -configuration  $Q$  put  $Q^{\ell}(Q) = \sum_{x, y \in \mathbb{Z}_N} K^{\ell}(x, y) \tau(A_x B_y)$ .

**Lemma 5.1.** *For  $x \neq y$  the integral defining  $K^{\ell}(x, y)$  converges absolutely, and*

$$K^{\ell}(x, y) = \Xi(L_{x+1} - L_y) - \Xi(L_x - L_y) - \Xi(L_{x+1} - L_{y+1}) + \Xi(L_x - L_{y+1}).$$

*Moreover  $K^{\ell}(x + d, y + d) = K^{\ell}(x, y) = -K^{\ell}(y, x)$ , and  $K^{\ell}(x, y) = 0$  if  $\ell_x = 0$  or  $\ell_y = 0$ .*

*Proof.* The cells are disjoint;  $\kappa(\xi - \eta)$  is singular on  $\bar{I}_x \times \bar{I}_y$  only at a common endpoint, where  $|\kappa(\xi - \eta)| \leq C/|\xi - \eta|$  with  $\xi, \eta$  on opposite sides of it, and  $\int_0^1 \int_0^1 \frac{ds \, dt}{s+t} < \infty$ . The formula follows by integrating  $\Xi'' = \kappa$  twice, as in Lemma 4.3. The symmetries follow from  $I_{x+d} = I_x + d$  and the oddness of  $\kappa$ .  $\square$

**Proposition 5.2** (cell inequalities). *For every  $\ell \in \Delta_d$  and every  $Q$ -configuration  $Q$  on  $\mathbb{C}^M$ ,*

$$Q^{\ell}(Q) \leq \frac{N^2 G}{\pi^2},$$



with equality for every window. More precisely, let  $B$  and  $A$  be the step fields on  $\mathbb{T}$  equal to  $B_x$  and  $A_x$  on the arc  $J_x = (2\pi L_x/N, 2\pi L_{x+1}/N)$ , for every  $x$  with  $\ell_x > 0$ , and let  $g$  be the conjugate function of  $B$ ; then  $\int B \, dm = \int A \, dm = \frac{1}{4}\mathbf{1}$ ,  $Q^\ell(Q) = N^2 \int_{\mathbb{T}} \tau(Ag) \, dm$ , and Corollary 4.5 applies.

*Proof.* The arcs  $J_x$  with  $\ell_x > 0$  are disjoint, nonempty and cover  $\mathbb{T}$  up to finitely many points, so  $B$  is a projection step field;  $0 \leq A \leq \mathbf{1} - B$  because  $A_x B_x = 0$ ; and by (2.4),  $\int B \, dm = \frac{1}{N} \sum_x \ell_x B_x = \frac{d}{N} \mathbf{1} = \frac{1}{4} \mathbf{1}$ , and likewise  $\int A \, dm = \frac{1}{4} \mathbf{1}$ , so  $\int \tau(A) \, dm = \frac{1}{4}$ . For  $\theta \in J_x$ ,  $\tau(A(\theta)g(\theta)) = \sum_{y \neq x} \tau(A_x B_y) \int_{J_y} \cot \frac{\theta - \varphi}{2} \, dm(\varphi)$ , because  $\tau(A_x B_x) = 0$  (no principal value is needed). Integrating over  $J_x$ , summing over  $x$ , and substituting  $\theta = 2\pi\xi/N$ ,  $\varphi = 2\pi\eta/N$  gives  $\int \tau(Ag) \, dm = N^{-2} \sum_{x \neq y} K^\ell(x, y) \tau(A_x B_y) = N^{-2} Q^\ell(Q)$ , all integrals being absolutely convergent by Lemma 5.1. Corollary 4.5 gives the inequality. For the window  $\mathbf{1}_{[c, c+d)}$  ( $M = 1$ ),  $A = 1$  exactly on  $[L_c, L_c + d)$  and  $B = 1$  exactly on  $[L_c - d, L_c)$  (each window contains every residue once), two adjacent arcs of length  $d$ ; Lemma 4.3 gives  $Q^\ell = N^2 \Phi_c(\frac{1}{4}, \frac{1}{4}) = N^2 G/\pi^2$ .  $\square$

## 5.2. Rotation averages.

**Proposition 5.3** (averaged cell inequalities). *For every  $\ell \in \Delta_d$  and every  $Q$ -configuration,*

$$(5.1) \quad \frac{1}{d} \sum_{j=0}^{d-1} \left[ Q^\ell(\text{rot}_j Q) - \frac{N^2 G}{\pi^2} \right] = \langle u^\ell, \delta \rangle,$$

with  $u^\ell$  as in (1.7). Equivalently  $u_m^\ell = \tilde{K}(m) + \tilde{K}(2d - m)$  with  $\tilde{K}(m) = \frac{1}{d} \sum_{r=0}^{d-1} K^\ell(r + m, r)$ . Consequently  $\langle u^\ell, \delta \rangle \leq 0$ .

*Proof.* By definition  $Q^\ell(\text{rot}_j Q) = \sum_{x,y} K^\ell(x + j, y + j) \tau(A_x B_y)$ . Since  $K^\ell$  is invariant under  $(x, y) \mapsto (x + d, y + d)$ , the average over  $0 \leq j \leq d - 1$  equals the average over  $j \in \mathbb{Z}_N$ , which depends only on  $x - y$  and equals  $\tilde{K}(x - y)$ . Hence the left side of (5.1), plus  $N^2 G/\pi^2$ , is  $\sum_{m \in \mathbb{Z}_N} \tilde{K}(m) T(m)$ . The function  $\tilde{K}$  is odd on  $\mathbb{Z}_N$  (Lemma 5.1), so  $\tilde{K}(0) = \tilde{K}(2d) = 0$  and, by Lemma 2.5(1),

$$\sum_m \tilde{K}(m) T(m) = \sum_{m=1}^{2d-1} \tilde{K}(m) \mathcal{N}(m) = d \tilde{K}(d) + \sum_{m=1}^{d-1} [\tilde{K}(m) + \tilde{K}(2d - m)] \mathcal{N}(m).$$

For the window,  $\mathcal{N}(m) = m$  and every rotation is a window, so Proposition 5.2 gives  $N^2 G/\pi^2 = d \tilde{K}(d) + \sum_m [\tilde{K}(m) + \tilde{K}(2d - m)] m$ . Subtracting gives (5.1) with  $u_m^\ell = \tilde{K}(m) + \tilde{K}(2d - m)$ . It remains to identify this with (1.7). By Lemma 5.1,  $K^\ell(r + m, r) = \Xi(S_r(m + 1)) - \Xi(S_r(m)) - \Xi(S_{r+1}(m)) + \Xi(S_{r+1}(m - 1))$ , and  $S_{r+d} = S_r$ , so  $\tilde{K}(m) = \Delta_m^2 \Pi(m)$  with  $\Pi(k) = \frac{1}{d} \sum_r \Xi(S_r(k))$ . For  $0 \leq j \leq d$ ,  $S_r(2d - j) = d + S_r(d - j) = 2d - S_{r+d-j}(j)$ , so  $\Pi(2d - j) = \frac{1}{d} \sum_r \Xi(2d - S_r(j))$ , and  $\tilde{K}(2d - m) = \Delta_m^2 [\frac{1}{d} \sum_r \Xi(2d - S_r(m))]$  for  $1 \leq m \leq d - 1$ . Adding gives (1.7). The last assertion follows from Proposition 5.2.  $\square$

**Proposition 5.4** (the cone condition implies optimality). *Let  $d \geq 2$ , and suppose that  $v = \int u^\ell \mu(d\ell) + \sum_s w_s t_s$  for a finite positive Borel measure  $\mu$  on  $\Delta_d$  and  $w_s \geq 0$ . Then  $\langle v, \delta \rangle \leq 0$  for every  $Q$ -configuration on every  $\mathbb{C}^M$ ; hence  $F(V) \leq F_{\text{DKZ}}$  for every family  $V$  of order-four unitaries, and  $I_d \leq I_{\text{ME}}(d)$  for every projective strategy on a maximally entangled state.*

*Proof.* The map  $\ell \mapsto u^\ell$  is continuous on the compact set  $\Delta_d$  (as  $\Xi^\#$  is continuous on  $[0, 2d]$ ), so the integral is defined, and  $\langle v, \delta \rangle = \int \langle u^\ell, \delta \rangle \mu(d\ell) + \sum_s w_s \langle t_s, \delta \rangle \leq 0$  by Proposition 5.3 and Lemma 2.5(3). The rest is Lemma 2.5(2) and Theorem 2.1.  $\square$

*Remark 5.5* (uniform cells and junction terms). For uniform cells  $\ell \equiv 1$ ,  $K^1(x, y) = \bar{\kappa}(x - y)$  with  $\bar{\kappa}(m) = \int_{-1}^1 (1 - |s|) \kappa(m + s) ds$ , and  $u^1 = v + j$ , where  $j_m = |e_J(m)| + |e_J(2d - m)| \geq 0$  and  $e_J(m) = \kappa(m) - \bar{\kappa}(m)$  is the junction function of [30, Lemma 4.9]. So the uniform embedding gives  $\langle v, \delta \rangle \leq -\langle j, \delta \rangle$ , a bound with an error made of junction terms; this is how the commutative case is treated in [30], where those terms are estimated using the cyclic order of optimal colourings, an argument with no known operator analogue. The cone condition replaces this estimate: since  $u^1$  is not proportional to  $v$ , non-uniform cells are needed.

*Remark 5.6* (Poisson cells). Let  $\ell = d \cdot (Y_0, \dots, Y_{d-1})$  with  $(Y_r)$  uniformly distributed on the probability simplex (the Dirichlet law with all parameters 1; equivalently, the cell boundaries are  $d$  independent uniform points on a circle of length  $d$ ). Then  $S_r(m)/d$  has the Beta( $m, d - m$ ) law, and the Bernstein derivative formula gives  $\mathbb{E}u_m^\ell = \int_0^d (\Xi^\#)''(s) b_{m,d}(s/d) ds$  with  $b_{m,d}(x) = \binom{d}{m} x^m (1 - x)^{d-m}$ . As  $(\Xi^\#)''(s) = \frac{4d}{\pi s} + O(1)$  and  $\int_0^1 b_{m,d}(x) \frac{dx}{x} = \frac{1}{m}$ , the averaged cell vector reproduces *exactly* the singular part  $\frac{4d}{\pi m}$  of  $v_m$ : the discrete kernel is the continuum kernel averaged over Poisson-like cells. The residual  $\mathbb{E}u^\ell - v$  is smooth and of relative size  $O(1/d)$ ; the constructions of Section 6 correct it by modulating the Dirichlet parameters. (This remark explains the construction and is not used in the proofs; Lemma 6.3 below is the precise statement that is used.)

## 6. THE CONE CONDITION FOR EVERY $d$ (COMPUTER-ASSISTED)

This section proves Theorem E. The proof is computer-assisted: it depends on computations in rigorous interval arithmetic (the interval type of the Python library `mpmath` [21], with outward rounding, at 160 bits for  $d \leq 2000$  and at 120–240 bits for  $d \geq 2001$ ). Below we state the mathematical reductions, with proofs, and then exactly what is verified, by which program, and with which margins. Programs, certificates and logs are listed in Section 11. There are three regimes:  $2 \leq d \leq 200$  (Section 6.2),  $201 \leq d \leq 2000$  (Section 6.5) and  $d \geq 2001$  (Section 6.6).

**6.1. Potentials.** For a probability measure  $\mu$  on  $\Delta_d$  and  $0 \leq m \leq d$  let  $P_\mu(m) = \int \frac{1}{d} \sum_{r=0}^{d-1} \Xi^\#(S_r(m)) \mu(d\ell)$ . Since  $S_r(0) = 0$  and  $S_r(d) = d$ ,  $P_\mu(0) = \Xi^\#(0)$  and  $P_\mu(d) = \Xi^\#(d)$  for every  $\mu$ , and  $\int u^\ell \mu(d\ell) = \Delta^2 P_\mu$  on  $\{1, \dots, d-1\}$ . Let  $P_v$  be the solution of  $\Delta^2 P_v = v$  on  $\{1, \dots, d-1\}$  with  $P_v(0) = \Xi^\#(0)$  and  $P_v(d) = \Xi^\#(d)$ . For  $X: \{0, \dots, d\} \rightarrow \mathbb{R}$  write  $X_A(m) = \frac{1}{2}(X(m) - X(d - m))$  and  $X_S(m) = \frac{1}{2}(X(m) + X(d - m))$ .

**Lemma 6.1** (reduction to potentials). *Suppose that  $\mu$  satisfies (i)  $(P_\mu)_A = (P_v)_A$ , and (ii)  $W := (P_v - P_\mu)_S$  has  $\Delta^2 W(m) \geq 0$  for  $1 \leq m \leq d-1$ . Then (1.8) holds with this  $\mu$  and suitable  $w_s \geq 0$ .*

*Proof.*  $W$  is symmetric with  $W(0) = W(d) = 0$ , and by (i)  $P_\mu + W = P_v$ ; applying  $\Delta^2$  gives  $\int u^\ell d\mu + \Delta^2 W = v$ . The vector  $\Delta^2 W$  is symmetric under  $m \mapsto d - m$  and nonnegative, hence a nonnegative combination of the  $t_s$ .  $\square$

### 6.2. Regime I: $2 \leq d \leq 200$ .

**Proposition 6.2** (computer-assisted). *For every  $2 \leq d \leq 200$  there are  $K_d \leq 6.3d$  cell vectors  $\ell^{(k)} = n^{(k)}/16$  with integer entries  $n_r^{(k)} \geq 1$  and  $\sum_r n_r^{(k)} = 16d$ , and weights  $c_k > 0$  with  $d \cdot \min_k c_k \geq 0.0816$ , such that  $\sum_k c_k u^{\ell^{(k)}} = v$ . In particular Theorem E holds with  $\mu = \sum_k c_k \delta_{\ell^{(k)}}$ , which is supported in  $\Delta_d^+$ , and  $w = 0$ .*

*What is verified.* Since all window sums of  $\ell = n/16$  are multiples of  $1/16$ , only the values  $\Xi^\#(j/16)$ ,  $0 \leq j \leq 16d$ , are needed, and  $\Xi^\#(j/q) = -\frac{N^2}{2\pi^2} [\text{Cl}_2(\pi j/K_q) + \text{Cl}_2(\pi(K_q - j)/K_q)]$  with

$q = 16$ ,  $K_q = 2dq$ . Each value  $\text{Cl}_2(t)$ ,  $0 < t \leq \pi$ , is enclosed using

$$\text{Cl}_2(t) = t - t \log t + \sum_{n=1}^{70} \frac{|B_{2n}| t^{2n+1}}{2n(2n+1)!} + R_{70}(t), \quad 0 \leq R_{70}(t) \leq \frac{t \pi^2/6}{\frac{3}{4} \cdot 71 \cdot 143 \cdot 471},$$

with exact rational Bernoulli numbers; the tail bound follows from  $|B_{2n}| = 2(2n)!\zeta(2n)/(2\pi)^{2n}$ ,  $\zeta(2n) \leq \zeta(2)$  and  $(t/2\pi)^{2n} \leq 4^{-n}$ . The vectors  $u^{\ell(k)}$  are then enclosed through the window-sum form (1.7). The cell vectors and approximate weights  $\tilde{c}$  were found by floating-point linear programming (HiGHS [19]); this search is not part of the verification. The exact weights are  $c = \tilde{c} + U^\top y$ , where  $U$  is the matrix with columns  $u^{\ell(k)}$  and  $(UU^\top)y = v - U\tilde{c}$ ; then  $Uc = v$  exactly. With an approximate inverse  $R$  of  $UU^\top$ , the bound  $\|y\|_\infty \leq \|R(v - U\tilde{c})\|_\infty / (1 - \|\mathbf{1} - RUU^\top\|_\infty)$  holds as soon as  $\|\mathbf{1} - RUU^\top\|_\infty < 1$ ; all quantities are computed in interval arithmetic (160 bits), so the bound holds for the exact  $U$ . The program `verify_cone.py` (driver `run_verify.py`) certifies, for every  $2 \leq d \leq 200$ :  $\|\mathbf{1} - RUU^\top\|_\infty \leq 9.4 \cdot 10^{-34}$ ,  $\|y\|_\infty \leq 2.1 \cdot 10^{-11}$ , and  $\min_k c_k > 0$ . Every  $d$  was certified at the first attempt;  $3.5d \leq K_d \leq 6.3d$ ; the smallest certified weight is  $4.25 \cdot 10^{-4}$  (at  $d = 198$ ), and  $0.0816 \leq d \cdot \min_k c_k \leq 0.2572$ . The certificates contain the integer vectors  $n^{(k)}$  and the weights. A complete re-verification of all 199 stored certificates (program `lp_reverify.py`, which reruns `verify_cone.py`) passed and reproduced every certified bound. For  $2 \leq d \leq 20$  the same certificates are also re-evaluated by the Lean kernel in exact rational interval arithmetic (Section 11.5).

*Independent check.* The program `crosscheck_certs.py` recomputes  $\sum_k c_k u^{\ell(k)} - v$  without interval arithmetic, with `mpmath.clsin` at 50 digits and, for  $d \leq 10$ , with the cell-pair kernel of Lemma 5.1 instead of the window-sum form: the maximal deviation is between  $3 \cdot 10^{-15}$  and  $2.5 \cdot 10^{-13}$  for  $d \in \{3, 5, 8, 10, 20, 37\}$ , and at most  $1.5 \cdot 10^{-10}$  for  $d \in \{150, 200\}$  (weights stored in double precision).

**6.3. Dirichlet cells.** For  $0 < b < d$  let  $X_b$  have the Beta( $b, d-b$ ) law, and put  $\Xi(b) = \mathbb{E} \Xi^\sharp(dX_b)$ ,  $\Xi(0) = \Xi^\sharp(0)$ ,  $\Xi(d) = \Xi^\sharp(d)$ . If  $\beta_0, \dots, \beta_{d-1} > 0$  with  $\sum_r \beta_r = d$  and  $\ell/d$  has the Dirichlet law with parameters  $(\beta_r)$ , then by the aggregation property of the Dirichlet law  $S_r(m)/d$  has the Beta( $B_r(m), d - B_r(m)$ ) law,  $B_r(m) = \beta_r + \dots + \beta_{r+m-1}$ , so that

$$(6.1) \quad \mathbb{E} \left[ \frac{1}{d} \sum_r \Xi^\sharp(S_r(m)) \right] = \frac{1}{d} \sum_r \Xi(B_r(m)).$$

**Lemma 6.3** (the singular part is reproduced exactly). *Write  $\Xi^\sharp = \Xi_s + \Xi_r$  with  $\Xi_s(u) = \frac{4d}{\pi} u \log \frac{u}{d}$  and  $\Xi_r(u) = d^2 g_r(u/d)$ , where*

$$g_r(x) = -\frac{8}{\pi^2} \left[ \text{Cl}_2\left(\frac{\pi x}{2}\right) + \text{Cl}_2\left(\pi - \frac{\pi x}{2}\right) \right] - \frac{4}{\pi} x \log x$$

*extends to an odd analytic function on the disc  $|x| < 2$ . Let  $\Xi_s, \Xi_r$  be the corresponding parts of  $\Xi$ , and let  $P_{v,s}, P_{v,r}$  be the solutions of  $\Delta^2 P = \Xi_s''$ , respectively  $\Xi_r''$ , on  $\{1, \dots, d-1\}$  with the boundary values of  $\Xi_s$ , respectively  $\Xi_r$ , at 0 and  $d$  (so  $P_v = P_{v,s} + P_{v,r}$ ). Then for every integer  $0 \leq m \leq d$ ,*

$$\Xi_s(m) = P_{v,s}(m) = \frac{4d}{\pi} \left[ m\psi(m) + 1 - \frac{m}{d} - m\psi(d) \right],$$

*where  $\psi$  is the digamma function and  $m\psi(m)$  is read as  $-1$  at  $m = 0$ . Consequently  $P_v - \Xi = P_{v,r} - \Xi_r$  involves only the analytic function  $g_r$ .*

*Proof.* Near  $x = 0$ ,  $\text{Cl}_2(\theta) = \theta - \theta \log \theta + (\text{odd power series})$  shows that the logarithms cancel;  $g_r$  is analytic for  $|x| < 2$  because  $\text{Cl}_2$  is analytic on  $(0, 2\pi)$ . For  $X \sim \text{Beta}(b, d-b)$ ,  $\mathbb{E}[X \log X] = \frac{b}{d} [\psi(b+1) - \psi(d+1)]$ , so  $\Xi_s(b) = \frac{4d}{\pi} b [\psi(b+1) - \psi(d+1)]$ ; at integers use  $m\psi(m+1) = m\psi(m) + 1$

and  $\psi(d+1) = \psi(d) + \frac{1}{d}$ . The right side vanishes at  $m = 0$  and  $m = d$ , as do  $\Xi_s(0)$  and  $\Xi_s(d)$ , and  $\Delta^2[m\psi(m)] = \frac{1}{m}$  for  $m \geq 1$ , while  $\Xi_s''(m) = \frac{4d}{\pi m}$ ; so both sides solve the same discrete Dirichlet problem.  $\square$

**6.4. Gaussian modulation.** Call  $\Upsilon: \{0, \dots, d\} \rightarrow [0, \infty)$  a *variogram* if  $\Upsilon(m) = \Upsilon(d-m)$ ,  $\Upsilon(0) = \Upsilon(d) = 0$ , and, extending  $\Upsilon$   $d$ -periodically,

$$\hat{a}_k(\Upsilon) := -\frac{2}{d} \sum_{m=0}^{d-1} \Upsilon(m) \cos \frac{2\pi km}{d} \quad (1 \leq k < \frac{d}{2}), \quad \hat{a}_{d/2}(\Upsilon) := -\frac{1}{d} \sum_{m=0}^{d-1} (-1)^m \Upsilon(m) \quad (d \text{ even})$$

are all nonnegative. If all  $\hat{a}_k(\Upsilon) > 0$ , the function  $c(j) = \frac{1}{2}(\Upsilon(j+1) - 2\Upsilon(j) + \Upsilon(j-1))$  on  $\mathbb{Z}_d$  has discrete Fourier transform  $\hat{c}(k) = d \sin^2(\pi k/d) \hat{a}_k(\Upsilon)$  for  $1 \leq k < d/2$  (and  $2d \hat{a}_{d/2}$  at  $k = d/2$ ), and  $\hat{c}(0) = 0$ ; so there is a centred Gaussian vector  $(\eta_r)_{r \in \mathbb{Z}_d}$ , invariant in law under rotations, with covariance  $c$ , with  $\sum_r \eta_r = 0$ , with  $\text{Var}(\eta_r + \dots + \eta_{r+m-1}) = \Upsilon(m)$ , and with a positive density on the hyperplane  $\{\sum_r \eta_r = 0\}$ . Let  $E = \{\max_r |\eta_r| < \frac{3}{4}\}$  and let  $\mu_\Upsilon$  be the law of the random cell vector  $\ell$  which, given  $\eta$ , is  $d$  times a Dirichlet vector with parameters  $(1 + \eta_r)$  on  $E$  and with all parameters 1 on the complement  $E^c$ .

For  $1 \leq m \leq d-1$  let  $L_m = \frac{3}{4} \min(m, d-m)$ , and for  $t \geq 0$  and  $\zeta \sim N(0, 1)$  let

$$H(m, t) = \mathbb{E}[\bar{\Xi}(m + \sqrt{t}\zeta); |\sqrt{t}\zeta| < L_m] + \bar{\Xi}(m) \mathbb{P}(|\sqrt{t}\zeta| \geq L_m).$$

For  $1 \leq m < d/2$  put  $\text{Hc}_m(t) = [H(m, t) - \bar{\Xi}(m)] - [H(d-m, t) - \bar{\Xi}(d-m)]$  and  $\tau_m = \text{Dl}(m) - \text{Dl}(d-m)$  with  $\text{Dl} = P_v - \bar{\Xi}$ . The *decoupled equations* are

$$(6.2) \quad \text{Hc}_m(\Upsilon(m)) = \tau_m \quad (1 \leq m < d/2).$$

**Lemma 6.4** (decoupling). *Let  $\Upsilon$  be a variogram with all  $\hat{a}_k(\Upsilon) > 0$ , and put*

$$\varepsilon_t = \frac{3d}{8} \sup_{0 < b < d} |\bar{\Xi}'(b)| \cdot 2d \exp\left(-\frac{9}{32\Upsilon(1)}\right),$$

$$e_m(\Upsilon) = [P_{\mu_\Upsilon}(m) - \bar{\Xi}(m)] - [P_{\mu_\Upsilon}(d-m) - \bar{\Xi}(d-m)] - \text{Hc}_m(\Upsilon(m)).$$

Then  $|P_{\mu_\Upsilon}(m) - H(m, \Upsilon(m))| \leq \varepsilon_t$  for  $1 \leq m \leq d-1$ , and  $|e_m(\Upsilon)| \leq 2\varepsilon_t$  for  $1 \leq m < d/2$ .

*Proof.* By (6.1), rotation invariance of  $(\eta, E)$ , and  $B_r(m) = m + Z_r(m)$  with  $Z_r(m) = \eta_r + \dots + \eta_{r+m-1}$ ,  $P_{\mu_\Upsilon}(m) = \mathbb{E}[\bar{\Xi}(m + Z); E] + \mathbb{P}(E^c)\bar{\Xi}(m)$  with  $Z = Z_0(m) \sim N(0, \Upsilon(m))$ . On  $E$ ,  $|Z| < \frac{3}{4}m$ , and since  $\sum_r \eta_r = 0$  also  $|Z| < \frac{3}{4}(d-m)$ ; so  $\{|Z| \geq L_m\} \subseteq E^c$ , and  $P_{\mu_\Upsilon}(m) - H(m, \Upsilon(m)) = -\mathbb{E}[(\bar{\Xi}(m + Z) - \bar{\Xi}(m))\mathbf{1}_{|Z| < L_m} \mathbf{1}_{E^c}]$ , which is bounded by  $L_m \sup |\bar{\Xi}'| \mathbb{P}(E^c)$ . Finally  $\mathbb{P}(E^c) \leq \sum_r \mathbb{P}(|\eta_r| \geq \frac{3}{4}) \leq 2d \exp(-\frac{9}{32\Upsilon(1)})$  since  $\text{Var} \eta_r = \Upsilon(1)$ , and  $L_m \leq \frac{3d}{8}$ . The bound on  $e_m$  follows since  $\Upsilon(d-m) = \Upsilon(m)$ .  $\square$

**Lemma 6.5** (from the decoupled to the exact equations). *Let  $\text{lo}_m < \text{hi}_m$  for  $1 \leq m < d/2$ , and let  $\mathcal{Q}$  be the set of variograms with  $\Upsilon(m) \in [\text{lo}_m, \text{hi}_m]$  for  $1 \leq m < d/2$  (and a fixed value  $\Upsilon(d/2)$  if  $d$  is even). Suppose that*

- (B1)  $\hat{a}_k(\Upsilon) > 0$  for all  $k$  and all  $\Upsilon \in \mathcal{Q}$ ;
- (B2)  $\varepsilon_t \leq \bar{\varepsilon}$  for all  $\Upsilon \in \mathcal{Q}$ ;
- (B3)  $\text{Hc}_m(\text{lo}_m) < \tau_m - 2\bar{\varepsilon}$  and  $\text{Hc}_m(\text{hi}_m) > \tau_m + 2\bar{\varepsilon}$  for all  $m$ .

Then some  $\Upsilon^\# \in \mathcal{Q}$  satisfies  $(P_{\mu_{\Upsilon^\#}})_A = (P_v)_A$ , that is, condition (i) of Lemma 6.1.

*Proof.* Condition (i) for  $\mu_\Upsilon$  is the system  $F_m(\Upsilon) := \text{Hc}_m(\Upsilon(m)) + e_m(\Upsilon) - \tau_m = 0$ ,  $1 \leq m < d/2$ . By (B1) the Gaussian law depends continuously on  $\Upsilon \in \mathcal{Q}$  and has a positive density on the hyperplane  $\{\sum \eta_r = 0\}$ , so the boundary of  $E$  is a null set and  $\Upsilon \mapsto P_{\mu_\Upsilon}$  is continuous; hence  $F$  is continuous on the box  $\mathcal{Q}$ . By Lemma 6.4, (B2) and (B3),  $F_m < 0$  on the face  $\Upsilon(m) = \text{lo}_m$

and  $F_m > 0$  on the face  $\Upsilon(m) = \text{hi}_m$ . The Poincaré–Miranda theorem [24] gives a zero of  $F$  in  $\mathcal{Q}$ .  $\square$

Condition (B3) is obtained from a sign change and a derivative bound: if  $\text{Hc}_m(a_m) < \tau_m < \text{Hc}_m(b_m)$ ,  $\text{lo}_m \leq a_m - \varrho$ ,  $\text{hi}_m \geq b_m + \varrho$ , and  $\text{Hc}'_m \geq c_* > 0$  on  $[\text{lo}_m, \text{hi}_m]$  with  $c_*\varrho \geq 2\bar{\varepsilon}$ , then  $\text{Hc}_m(\text{lo}_m) \leq \text{Hc}_m(a_m) - c_*\varrho < \tau_m - 2\bar{\varepsilon}$ , and similarly at  $\text{hi}_m$ , so (B3) holds. For condition (ii) of Lemma 6.1 at  $\Upsilon^\#$ , note that  $W = (P_v - P_{\mu_{\Upsilon^\#}})_S$  and that  $|P_{\mu_{\Upsilon}}(m) - H(m, \Upsilon(m))| \leq \bar{\varepsilon}$  by Lemma 6.4, while both functions take the same values at  $m = 0$  and  $m = d$ . So it suffices that  $\Delta^2(P_v - H(\cdot, \Upsilon))_S > 4\bar{\varepsilon}$  on  $\{1, \dots, d-1\}$  for every  $\Upsilon \in \mathcal{Q}$ . Since moreover  $(P_v - P_{\mu_{\Upsilon^\#}})_A = 0$ , one can instead use the one-sided slack  $X_\Upsilon(m) := P_v(m) - H(m, \Upsilon(m))$  ( $1 \leq m \leq d/2$ ), extended by  $X_\Upsilon(d-m) = X_\Upsilon(m)$  and  $X_\Upsilon(0) = 0$ : then  $W = X_{\Upsilon^\#} + e$  with  $|e| \leq \bar{\varepsilon}$ , and it suffices that  $\Delta^2 X_\Upsilon > 4\bar{\varepsilon}$  for every  $\Upsilon \in \mathcal{Q}$ .

### 6.5. Regime II: $201 \leq d \leq 2000$ .

**Proposition 6.6** (computer-assisted). *For every  $201 \leq d \leq 2000$  the hypotheses of Lemma 6.5 and condition (ii) of Lemma 6.1 hold for explicit intervals  $[\text{lo}_m, \text{hi}_m]$ . Hence Theorem E holds with  $\mu = \mu_{\Upsilon^\#}$ , a probability measure with  $\mu(\Delta_d^+) = 1$ .*

*What is verified.* For each  $201 \leq d \leq 2000$ , one run of the program `verify_cert.py` certifies all hypotheses of Lemma 6.5 and condition (ii) of Lemma 6.1. Every quantity is computed in one process, at one precision (160-bit interval arithmetic throughout) and by one version of the code, whose SHA-256 hash is stored in every certificate (Section 11). The run computes:

- (1) enclosures of  $\Xi^{(2j)}(m)$  for  $j \leq 7$  and all integers  $1 \leq m \leq d-1$  (and bounds for  $\Xi^{(16)}$  on discs, used in (3)): the singular part through  $\Xi_s(b) = \frac{4d}{\pi}b[\psi(b+1) - \psi(d+1)]$  and the polygamma values  $\psi^{(n)}(m+1) = (-1)^{n+1}n![\zeta(n+1) - H_m^{(n+1)}]$ , where  $\zeta(s)$ ,  $2 \leq s \leq 17$ , is enclosed by the Euler–Maclaurin formula in exact rational arithmetic with its remainder bound; the regular part through the series  $\Xi_r(b) = d^2 \sum_{i \text{ odd}} g_i \frac{(b)_i}{(d)_i}$  ( $g_i$  the Taylor coefficients of  $g_r$ ,  $(b)_i$  the rising factorial,  $\mathbb{E}X_b^i = (b)_i/(d)_i$ ), whose Taylor coefficients at the integers are computed by exact truncated products for  $i \leq 141$ , with a tail bound from  $0 \leq \frac{d^k}{db^k} \frac{(b)_i}{(d)_i} \leq k! \binom{i}{k} / (d)_k$  on  $[0, d]$ ;
- (2) the targets  $\tau_m$ , through Lemma 6.3 and the discrete potential  $P_{v,r}$  (prefix sums), with  $g_r(1)$  enclosed including the tail of its series;
- (3) root boxes  $[a_m, b_m]$  for (6.2): the Gaussian average is expanded exactly to order 14,  $H(m, t) - \Xi(m) = \sum_{j=1}^7 t^j \Xi^{(2j)}(m) / (2^j j!) + R(m, t)$ , where  $R$  contains the order-16 Lagrange remainder (bounded through  $\sup |\Xi^{(16)}|$  on  $|z| \leq \frac{1}{2} \min(m, d-m)$  and a Gaussian tail estimate on the rest of the window, using  $|\psi^{(n)}(z)| \leq (n-1)!z^{-n} + n!z^{-n-1}$ ) and the truncation  $|\sqrt{t}\zeta| < L_m$  (truncated Gaussian moments); an untrusted search proposes  $[a_m, b_m]$ , and a certified sign change of  $\text{Hc}_m - \tau_m$  at its ends is checked;
- (4) the tail quantities, in logarithms:  $\sup |\Xi'| \leq \frac{4d}{\pi}(H_d + 1) + S_1 d$  ( $H_d$  the harmonic number,  $S_1 = \sum_{i \text{ odd}} i|g_i| \leq 1.8884$ ),  $\mathbb{P}(E^c)$  and  $\bar{\varepsilon}$  from Lemma 6.4, and a radius  $\varrho = 2^e$ ,  $e \in \mathbb{Z}$ , with  $\varrho \geq 8\bar{\varepsilon}/c_{1,\min}$ , where  $c_{1,\min} = \min_m \frac{1}{2}(\Xi''(m) - \Xi''(d-m))$ ;
- (5) the box  $\mathcal{Q} = \prod_m [\text{lo}_m, \text{hi}_m]$  with  $\text{lo}_m \leq a_m - \varrho$  and  $\text{hi}_m \geq b_m + \varrho$  (outward rounding), and, on the whole of  $\mathcal{Q}$ : condition (B1); condition (B2), using  $\Upsilon(1) \leq \text{hi}_1$ ; the bound  $c_D := \min_m \inf\{\text{Hc}'_m(t) : t \in [\text{lo}_m, \text{hi}_m]\} > 0$  with  $c_D \varrho > 2\bar{\varepsilon}$ , which gives (B3) by the remark after Lemma 6.5 (the derivative of the order-14 expansion and of the remainder is bounded with the heat equation  $\partial_t \phi_t = \frac{1}{2} \phi_t''$  for the Gaussian density and two integrations by parts on  $[-L_m, L_m]$ ); condition (B3) once more, directly, by interval evaluation of

$Hc_m - \tau_m$  at  $\text{lo}_m$  and  $\text{hi}_m$ ; and  $\Delta^2(P_v - H(\cdot, \Upsilon))_S > 4\bar{\varepsilon}$  on  $\{1, \dots, d-1\}$  for all  $\Upsilon \in \mathcal{Q}$ , which gives condition (ii) by the same remark.

All factorials, binomial coefficients, Bernoulli numbers and rational constants enter as exact integers or rationals with outward rounding, the results of the elementary interval functions of `mpmath` are widened by a further 16 units in the last place, every decision is a comparison of interval endpoints, and every certified bound is stored as the exact decimal expansion of the corresponding endpoint. The program `audit_cert_g.py` re-checks every condition from these stored values, together with the code hash: all 1800 certificates pass. The smallest certified values over  $201 \leq d \leq 2000$  are:  $d^2 \min_k \hat{a}_k \geq 0.06319$  for even  $d$  (at  $d = 2000$ ; the minimum over  $k$  is at  $k = \lfloor d/2 \rfloor$ ) and  $d^2 \min_k \hat{a}_k \geq 0.12638$  for odd  $d$ ;  $d \cdot \min_m \Delta^2(P_v - H)_S \geq 0.660$ ;  $c_D/(c_{1,\min}/4) \geq 4.0009$  (a ratio above 1 is needed) and  $c_D \varrho/(2\bar{\varepsilon}) \geq 10^{0.61}$ ; the face values of (B3) exceed  $2\bar{\varepsilon}$  by a factor of at least  $10^{16.7}$ ;  $\mathbb{P}(E^c) \leq 4.2 \cdot 10^{-37}$ ,  $\bar{\varepsilon} \leq 10^{-31.16}$  and  $\varrho \leq 2^{-94}$ . (Earlier runs, which used separate programs for the Gaussian and the fixed-point conditions and 100-bit arithmetic for  $d \leq 600$ , gave lower apparent margins, 0.0285 and 0.0798; they are superseded by the certificates described here.)

**6.6. Regime III:**  $d \geq 2001$ . For large  $d$  the variogram coefficients of the decoupled solution are of size  $d^{-2}$  at high frequencies and are controlled by a local criterion. Let  $\varepsilon = 1/d$ ,  $A_0 = \frac{\pi}{2} - 1$ ,

$$B(b) = b[2\psi'(b+1) + b\psi''(b+1)], \quad f(b) = b^2 \left( \frac{1}{B(b)} - 1 \right) \quad (b > 0), \quad f(0) = 0,$$

$a(m) = \frac{1}{6} - f(m)$  for  $m \geq 0$ , and  $F_A(m) = A_0\varepsilon[f(m) + f(d-m) - f(d)]$  for  $0 \leq m \leq d$ . For a symmetric function  $X$  on  $\mathbb{Z}_d$  let  $\Delta^4 X$  be its centred periodic fourth difference.

**Lemma 6.7** (positivity criterion). (1) (computer-assisted) *The sequence  $(a(m))_{m \geq 0}$  is positive and convex, and  $m^2 a(m) \leq \frac{13}{180}$  for  $m \geq 1$ . Consequently, for all  $\theta$ ,*

$$\mathcal{A}(\theta) := a(0) + 2 \sum_{m \geq 1} a(m) \cos m\theta \geq \Delta^2 a(1) = 2f(1) - f(2) \geq 0.105487,$$

where  $f(1) = (\pi^2/3 - 2\zeta(3))^{-1} - 1$  and  $f(2) = 4(2\pi^2/3 + 4 - 8\zeta(3))^{-1} - 4$ .

(2) *Let  $d \geq 5$ , let  $\Upsilon_e$  be symmetric on  $\mathbb{Z}_d$  with  $\Upsilon_e(0) = 0$ , and let  $D_B = \Delta^4(\Upsilon_e - F_A)$ . If  $D_B(1) \geq -\gamma\varepsilon^2$  and  $D_B(m) \geq \beta'\varepsilon^3$  for  $2 \leq m \leq d/2$ , with  $\gamma \geq 0$  and  $\beta' > 0$ , then for  $1 \leq k \leq d/2$*

$$\hat{a}_k(\Upsilon_e) \geq w_k \frac{\varepsilon^2}{8} \left[ P_d - \frac{4\gamma^2}{\beta'd(1-4/d)} \right], \quad P_d = 16A_0 \left( \inf_{\theta} \mathcal{A}(\theta) - \frac{0.1446}{d-1} \right),$$

where  $w_k = 1$  for  $k < d/2$  and  $w_{d/2} = \frac{1}{2}$ .

(3) *If  $|\Upsilon(m) - \Upsilon_e(m)| \leq \delta_{\max}$  for all  $m$ , then  $\hat{a}_k(\Upsilon) \geq \hat{a}_k(\Upsilon_e) - 2w_k \delta_{\max}$ .*

*Proof.* (1) The positivity, convexity and the bound  $m^2 a(m) \leq \frac{13}{180}$  are verified in interval arithmetic for  $m \leq 3000$ , and for  $m > 3000$  from the asymptotic expansion  $f(b) = \frac{1}{6} - \frac{13}{180}b^{-2} + \frac{683}{7560}b^{-4} + E(b)$ ,  $|E(b)| \leq 0.409b^{-6}$ , obtained from the Binet expansion of the polygamma functions with its explicit remainder (program `t35_polya.py`). Since  $a$  is convex with  $a(m) \rightarrow 0$ , Pólya's argument writes  $\mathcal{A} = \sum_{L \geq 1} \Delta^2 a(L) L \mathcal{K}_L$  with the nonnegative Fejér kernels  $\mathcal{K}_L(\theta) = L^{-2} (\sin \frac{L\theta}{2} / \sin \frac{\theta}{2})^2$ , and  $L \mathcal{K}_L \equiv 1$  for  $L = 1$ . The values of  $f(1), f(2)$  follow from  $\psi'(2) = \frac{\pi^2}{6} - 1$ ,  $\psi''(2) = 2 - 2\zeta(3)$ ,  $\psi'(3) = \frac{\pi^2}{6} - \frac{5}{4}$ ,  $\psi''(3) = \frac{9}{4} - 2\zeta(3)$ .

(2) For symmetric  $X$  on  $\mathbb{Z}_d$  with  $X(0) = 0$ ,  $\theta = 2\pi k/d$  and  $s = \sin \frac{\theta}{2}$ , summation by parts gives  $8ds^4 \hat{a}_k(X) = w_k \sum_{m=1}^{d-1} \Delta^4 X(m) (1 - \cos m\theta)$ , because the symbol of  $\Delta^4$  is  $16s^4$  and  $\sum_m \Delta^4 X(m) = 0$ . For  $X = F_A$ , using  $F_A(m) = A_0\varepsilon[\frac{1}{6} + a(d) - a(m) - a(d-m)]$  (valid



also at  $m = 0$ ),  $\sum_m \Delta^4 F_A(m)(1 - \cos m\theta) = 16A_0\varepsilon s^4[a(0) + 2\sum_{m=1}^{d-1} a(m)\cos m\theta + a(d)] \geq 16A_0\varepsilon s^4[\mathcal{A}(\theta) - 2\sum_{m \geq d} a(m)] \geq P_d\varepsilon s^4$ , since  $2\sum_{m \geq d} a(m) \leq \frac{26}{180(d-1)} < \frac{0.1446}{d-1}$ . For the remainder,  $D_B(d-m) = D_B(m)$ ,  $1 - \cos \theta = 2s^2$ , and  $\sum_{m=2}^{d-2} (1 - \cos m\theta) = d - 2 + 2\cos \theta \geq d - 4$ , so  $\sum_m D_B(m)(1 - \cos m\theta) \geq -4\gamma\varepsilon^2 s^2 + \beta'\varepsilon^2(1 - 4\varepsilon)$ . Dividing by  $8ds^4$  and putting  $y = s^{-2} \geq 1$ ,  $\hat{a}_k(\Upsilon_e) \geq w_k \frac{\varepsilon^2}{8} [P_d - 4\gamma\varepsilon y + \beta'\varepsilon(1 - 4\varepsilon)y^2]$ , and the minimum over  $y \in \mathbb{R}$  of the bracket is the stated bound.

(3)  $|\hat{a}_k(\delta)| \leq \frac{2}{d} \sum_m |\delta(m)| \leq 2\delta_{\max}$ , and half of this for  $k = d/2$ .  $\square$

For example,  $\gamma = 3$ ,  $\beta' = \frac{1}{4}$  and  $\delta_{\max} = \varepsilon^2/20$  in Lemma 6.7(2),(3) give  $\hat{a}_k(\Upsilon) \geq w_k \varepsilon^2 [\frac{1}{8}(0.962733 - \frac{36}{0.25 \cdot 2001 \cdot (1-4/2001)}) - \frac{1}{10}] \geq 0.011 w_k \varepsilon^2$  for every  $d \geq 2001$ ; the bounds certified below are much better.

**Proposition 6.8** (computer-assisted, uniform in  $d$ ). *Let  $\Upsilon_e$  be the explicit model of Appendix A. For every  $d \geq 2001$ , with  $\varepsilon = 1/d$ , let  $\bar{\varepsilon}_d$  be the quantity  $\varepsilon_t$  of Lemma 6.4 evaluated with  $\Upsilon(1) \leq 0.6451\varepsilon$  and  $\sup |\Xi'| \leq \frac{4d}{\pi}(H_d + 1) + 1.8884d$ , let  $c_d = 0.1935\varepsilon$  and  $\varrho_d = 2\bar{\varepsilon}_d/c_d$ . Then:*

- (1) *for each  $1 \leq m < d/2$  there are  $a_m < b_m$  within  $6.6468 \cdot 10^{-4}\varepsilon^2$  of  $\Upsilon_e(m)$  with  $\text{Hc}_m(a_m) < \tau_m < \text{Hc}_m(b_m)$ , and  $\text{Hc}'_m \geq c_d$  on  $[a_m - \varrho_d, b_m + \varrho_d]$ ;*
- (2)  *$D_B[\Upsilon_e](1) \geq -2.765192\varepsilon^2$  and  $D_B[\Upsilon_e](m) \geq 6.022431\varepsilon^3$  for  $2 \leq m \leq d/2$ ;*
- (3) *for every symmetric  $\Upsilon$  with  $|\Upsilon(m) - \Upsilon_e(m)| \leq 6.6468 \cdot 10^{-4}\varepsilon^2 + \varrho_d$  for  $1 \leq m \leq d/2$ , the one-sided slack  $X_\Upsilon$  of the remark after Lemma 6.5 satisfies  $\Delta^2 X_\Upsilon(m) \geq 0.620154\varepsilon - 4\varrho_d d^2$  for  $1 \leq m \leq d-1$ ;*
- (4) *every such  $\Upsilon$  has  $\Upsilon(1) \leq 0.6451\varepsilon$ ; consequently, for every  $d \geq 2001$  and in natural logarithms,  $\log \mathbb{P}(E^c) \leq -864$ ,  $\log \bar{\varepsilon}_d \leq -847$  and  $\log \varrho_d \leq -837$  (each bound decreases in  $d$ , so  $d = 2001$  is the worst case); in particular  $2\varrho_d$ ,  $4(\bar{\varepsilon}_d + \varrho_d d^2)$  and  $\varrho_d$  are smaller than  $0.1186944\varepsilon^2$ ,  $0.620154\varepsilon$  and  $10^{-6}\varepsilon$ , respectively, by factors larger than  $e^{800}$ .*

*What is verified.* The model  $\Upsilon_e$  is real-analytic in  $(x, \varepsilon) = (m/d, 1/d)$ , and every statement is proved by interval Taylor models over boxes in  $(x, \varepsilon)$  or  $(w, \varepsilon)$ ,  $w = 1/m$ , with  $\varepsilon \in [0, 1/2001]$ , so that each box covers all  $d \geq 2001$  at once (Appendix A); the box endpoints lie on exact decimal grids and the  $\varepsilon$ -ranges are stored exactly. Item (1) is the *pointwise lemma* (program `a10_pointwise.py`, 159 boxes; it also certifies that the model root is unique) together with the derivative bound of `a12_fp_alld.py`: a certified lower bound  $0.15204\varepsilon \geq c_d$ . Item (2) is assembled from `a06_zone0.py` (outer zone  $x \in [0.011, 0.501]$ : a fourth-derivative density  $\geq 7.419$ ), `a07_zoneI.py` (inner zone  $x \leq 0.013$ ,  $m \geq 3$ :  $\geq 6.022431$ ), `a08_A2.py` (the leading inner term  $A_2(m) = \Delta^4 \varphi_2(m)$  is  $\geq 0$  for all  $m \geq 2$ , and  $A_2(1) = -2.7559456\dots$ ) and `a09_smallm.py` ( $m = 1, 2$ :  $D_B(1) \geq -2.765192\varepsilon^2$ ,  $D_B(2) \geq 206.884\varepsilon^3$ ); the contribution of the term  $A_0\varepsilon f(d-m)$  of  $F_A$  is at most  $2.03 \cdot 10^{-11}\varepsilon^3$  (`a13_constants.py`). Item (3) is `a11_slack.py` (slack densities of the model  $\geq 0.651054$  in the outer zone,  $\geq 0.8260$  in the inner zone,  $\geq 0.9100$  at  $m = 1$ ) together with a perturbation bound of  $0.030900\varepsilon$ : with  $G(b, t) = H(b, t) - \Xi(b)$  and  $G_4$  its expansion to order  $t^4$ , the program `a13_constants.py` certifies  $|G - G_4| \leq 138.103\varepsilon^3$  (orders 5–7 and the remainder) and  $|\partial_t G_4| \leq 11.5703d$ , and the Euler–Maclaurin defect of the model target is bounded analytically. Item (4) follows from the scan of `a10_pointwise.py` ( $u_e \leq 0.645042$  for  $m = 1$ ), Lemma 6.4 and item (1); every margin is monotone in  $d$ , so  $d = 2001$  is the worst case. The program `audit_alld.py` checks the coverage of all boxes exactly, with rational arithmetic, and evaluates every margin in interval arithmetic from the logs. A final rigour pass (documented with the certificates) completed the written proofs of the model identities used in Appendix A, among them the first-order identity of the inner zone, which was also checked symbolically and

numerically to  $6 \cdot 10^{-25}$ , and corrected several code-level enclosure defects; one of them had left a sliver of size about  $10^{-20}$  of the parameter range uncovered at  $d = 2001$ . No conclusion changed.

*Proof of Theorem E for  $d \geq 2001$ .* Let  $\text{lo}_m = a_m - \varrho_d$  and  $\text{hi}_m = b_m + \varrho_d$ . Since  $c_d \varrho_d = 2\bar{\varepsilon}_d$ , the remark after Lemma 6.5 and Proposition 6.8(1),(4) give (B2) and (B3) with  $\bar{\varepsilon} = \bar{\varepsilon}_d$ . For  $\Upsilon$  in the resulting box,  $|\Upsilon(m) - \Upsilon_e(m)| \leq 6.6468 \cdot 10^{-4} \varepsilon^2 + \varrho_d$ , so Lemma 6.7, with  $\gamma = 2.765192$ ,  $\beta' = 6.022431$ ,  $P_d \geq 0.962733$  (from  $\inf \mathcal{A} \geq 0.1054878997577$ ) and  $d \geq 2001$ , gives

$$\begin{aligned} \hat{a}_k(\Upsilon) &\geq w_k \left[ \frac{\varepsilon^2}{8} \left( 0.962733 - \frac{4\gamma^2}{\beta' \cdot 2001 \cdot (1 - 4/2001)} \right) - 2(6.6468 \cdot 10^{-4} \varepsilon^2 + \varrho_d) \right] \\ &\geq w_k (0.1186944 \varepsilon^2 - 2\varrho_d) > 0, \end{aligned}$$

which is (B1). Condition (ii) of Lemma 6.1 at the point  $\Upsilon^\#$  provided by Lemma 6.5 follows from the remark after Lemma 6.5 in its one-sided form, since by Proposition 6.8(3),(4)  $\Delta^2 X_\Upsilon \geq 0.620154 \varepsilon - 4\varrho_d d^2 > 4\bar{\varepsilon}_d$  on the box. Lemmas 6.5 and 6.1 give Theorem E with  $\mu = \mu_{\Upsilon^\#}$ , and  $\mu(\Delta_d^+) = 1$  because, given  $\eta$ ,  $\ell/d$  is a Dirichlet vector with all parameters  $\geq \frac{1}{4}$ .  $\square$

**6.7. Summary.** Propositions 6.2, 6.6 and 6.8 prove Theorem E for every  $d \geq 2$ . In all three regimes  $\mu(\Delta_d \setminus \Delta_d^+) = 0$ : in Regime I every cell vector has  $n_r^{(k)} \geq 1$ , and in Regimes II and III the cell vector is almost surely in the open simplex. This property is used in the proof of Theorem B.

All values are certified, and a single program checks all three regimes. It verifies the 199 certificates of Regime I (their exact structure, and a matching verification record of `verify_cone.py`), runs `audit_cert_g.py` on the 1800 certificates of Regime II (1800 of 1800 pass, all with the same code hash), checks every box and every margin of Regime III, and prints ALL-D CERTIFIED (every  $d \geq 2$ ). This program, `audit_alld.py`, and its output are included with the certificates.

## 7. PROOFS OF THEOREMS A AND B

**7.1. Proof of Theorem A.** By Theorem E and Proposition 5.4,  $I_d \leq I_{\text{ME}}(d)$  for every projective strategy on every maximally entangled state, and the DKZ strategy attains  $I_{\text{ME}}(d)$  (Section 2.2).  $\square$

*Remark 7.1 (a deficit identity).* The proof gives more than an inequality. For a family  $V$  with  $Q$ -configuration  $Q$ , Lemma 2.5, Theorem E, Proposition 5.3 and Corollary 4.5 give

$$\begin{aligned} F_{\text{DKZ}} - F(V) &= \frac{N^2}{2} \int_{\Delta_d} \frac{1}{d} \sum_{j=0}^{d-1} \left[ \frac{1}{M} \int_{\mathbb{T}} D_{\lambda^*}^{(\ell,j)} dm + \int_{\mathbb{T}} S_{\lambda^*}^{(\ell,j)} dm \right] \mu(d\ell) \\ &\quad + \frac{1}{2} \sum_s w_s [d - \mathcal{N}(s) - \mathcal{N}(d-s)], \end{aligned}$$

where  $D_{\lambda^*}^{(\ell,j)}$  and  $S_{\lambda^*}^{(\ell,j)}$  are the (nonnegative) strip and bathtub defects of Theorem 4.2 for the cell embedding of  $\text{rot}_j Q$  with cells  $\ell$ . Thus the CGLMP deficit of any projective strategy on a maximally entangled state is an explicit positive combination of integrated defects of Theorem C, of bathtub slacks, and of pair slacks.

**7.2. Proof of Theorem B.** We need one lemma about cotangent sums.

**Lemma 7.2** (residue lemma). *Let  $t_0, \dots, t_{n-1} \in \mathbb{R}$  be pairwise distinct modulo  $2\pi$  and  $J_0, \dots, J_{n-1} \in M_M(\mathbb{C})$ . If  $\sum_j J_j \cot \frac{\theta - t_j}{2} = 0$  for all  $\theta$  in a nonempty open interval  $I \subseteq \mathbb{R} \setminus \bigcup_j (t_j + 2\pi\mathbb{Z})$ , then  $J_0 = \dots = J_{n-1} = 0$ .*

*Proof.* Entrywise,  $R(z) = \sum_j J_j \cot \frac{z-t_j}{2}$  is holomorphic on the connected open set  $\Omega = \mathbb{C} \setminus \bigcup_j (t_j + 2\pi\mathbb{Z})$  and vanishes on  $I$ , so  $R = 0$  on  $\Omega$ . Near  $z = t_j$  the terms with index  $j' \neq j$  are holomorphic, and  $\cot \frac{z-t_j}{2} = \frac{2}{z-t_j} + O(z-t_j)$ ; so the residue of  $R$  at  $t_j$  is  $2J_j$ , which must vanish.  $\square$

**Proposition 7.3** (pointwise commutation forces commutation). *Let  $\ell \in \Delta_d^+$ , let  $Q$  be a  $Q$ -configuration, and let  $B$  and  $g$  be the step field and its conjugate function from Proposition 5.2. If  $[B(\theta), g(\theta)] = 0$  for almost every  $\theta$ , then  $Q_x Q_y = Q_y Q_x$  for all  $x, y \in \mathbb{Z}_N$ .*

*Proof.* Since all cells have positive length, the points  $t_x = 2\pi L_x/N$  ( $x \in \mathbb{Z}_N$ ) are pairwise distinct modulo  $2\pi$  and the arcs  $J_x = (t_x, t_{x+1})$  are nonempty. Fix  $x$ . On  $J_x$ ,  $B(\theta) = B_x$  and  $g$  is real-analytic by (4.1), so  $\theta \mapsto [B_x, g(\theta)]$  is continuous on  $J_x$  and vanishes almost everywhere there, hence identically; so does its derivative,

$$0 = \frac{d}{d\theta} [B_x, g(\theta)] = \frac{1}{2\pi} \sum_{j \in \mathbb{Z}_N} [B_x, B_j - B_{j-1}] \cot \frac{\theta - t_j}{2} \quad (\theta \in J_x).$$

By Lemma 7.2,  $[B_x, B_j] = [B_x, B_{j-1}]$  for every  $j$ ; so  $j \mapsto [B_x, B_j]$  is constant on  $\mathbb{Z}_N$ , with value  $[B_x, B_x] = 0$ . As  $x$  runs over  $\mathbb{Z}_N$ ,  $B_x = Q_{x+d}$  runs over all the  $Q$ 's.  $\square$

*Proof of Theorem B.* Let a projective strategy  $R$  on  $\Phi_D$  attain  $I_{\text{ME}}(d)$ , let  $V$  be its reduced family (Theorem 2.1) and  $Q$  the  $Q$ -configuration of  $V$ . Then  $F(V) = F_{\text{DKZ}}$ , so  $\langle v, \delta \rangle = 0$  by Lemma 2.5.

*Step 1: a tight cell inequality with all cells positive.* By Theorem E,  $0 = \langle v, \delta \rangle = \int \langle u^\ell, \delta \rangle \mu(d\ell) + \sum_s w_s \langle t_s, \delta \rangle$ , and every term is nonpositive (Proposition 5.3 and Lemma 2.5(3)). Hence the continuous nonpositive function  $\ell \mapsto \langle u^\ell, \delta \rangle$  vanishes  $\mu$ -almost everywhere. As  $\mu(\Delta_d^+) = \mu(\Delta_d) > 0$ , there is a cell vector  $\ell^* \in \Delta_d^+$  with  $\langle u^{\ell^*}, \delta \rangle = 0$ . By Proposition 5.3, the  $d$  numbers  $Q^{\ell^*}(\text{rot}_j Q) - N^2 G/\pi^2$  are nonpositive with mean 0; so  $Q^{\ell^*}(Q) = N^2 G/\pi^2$ .

*Step 2: commutation almost everywhere.* For the step fields of Proposition 5.2 built from  $\ell^*$ ,  $\int \tau(Ag) dm = N^{-2} Q^{\ell^*}(Q) = G/\pi^2$ ; by Corollary 4.5,  $[B(\theta), g(\theta)] = 0$  for almost every  $\theta$ .

*Step 3: commutation.* By Proposition 7.3 all  $Q_x$  commute. Hence the unitaries  $E_k = \sum_a i^a Q_{k-da}$  commute, and so do the  $V_k = E_k^*$ .

*Step 4: the commutative case.* By Theorem 2.2(2),(3),  $d \mid D$  and there is a unitary  $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$  with  $u R_i u^* = R_i^{\text{DKZ}} \otimes \mathbf{1}$  for  $i = 1, \dots, 4$ .

*Step 5: measurements, state and uniqueness.* From  $u U_x u^* = U_x^{\text{DKZ}} \otimes \mathbf{1}$  we get  $u P_a^x u^* = P_a^{x, \text{DKZ}} \otimes \mathbf{1}$  (spectral projections). From  $u (U'_y)^\top u^* = (U_y^{\text{DKZ}})^\top \otimes \mathbf{1}$ , by transposition,  $\bar{u} U'_y \bar{u}^* = U_y^{\text{DKZ}} \otimes \mathbf{1}$ , so  $\bar{u} Q_b^y \bar{u}^* = Q_b^{y, \text{DKZ}} \otimes \mathbf{1}$ . For any unitary  $u$ ,  $(u \otimes \bar{u}) \sum_i |ii\rangle = \sum_{j,k} (u u^*)_{jk} |jk\rangle = \sum_j |jj\rangle$ , so  $(u \otimes \bar{u}) \Phi_D = \Phi_d \otimes \Phi_{D/d}$  in the product basis. If  $u_1, u_2$  both have these properties, then  $u_2 u_1^*$  commutes with all  $R_i^{\text{DKZ}} \otimes \mathbf{1}$ ; the  $R_i^{\text{DKZ}} = G R_i^{(0)} G^*$  generate  $M_d(\mathbb{C})$  (the algebra generated by the  $R_i^{(0)}$  contains  $X$  and  $R_1^{(0)}(R_2^{(0)})^{-1} = z^{-1}(\mathbf{1} + (i-1)|0\rangle\langle 0|)$ , hence all matrix units [30, proof of Corollary C]), so  $u_2 u_1^* \in \mathbf{1}_d \otimes U(D/d)$ . Finally, for fixed  $D$  the projective strategies on  $\Phi_D$  form a compact set on which  $I_d$  is continuous, so the maximum is attained; if it were  $I_{\text{ME}}(d)$ , then  $d \mid D$ .  $\square$

*Remark 7.4* (why all cells must be positive). If  $\ell_r = 0$  for some residue  $r$ , then  $K^\ell(x, y) = 0$  whenever  $x \equiv r$  or  $y \equiv r \pmod{d}$ , so  $Q^\ell$  does not depend on the measurement of site  $r$ , and a tight cell inequality cannot force that measurement to commute with the others. The uniform cell vector lies in  $\Delta_d^+$ , but  $u^1$  is not proportional to  $v$  (Remark 5.5), so  $\langle v, \delta \rangle = 0$  does not make the uniform cell inequality tight. Step 1 needs a cone representation charging  $\Delta_d^+$ , which Theorem E provides.

## 8. THE CONTINUUM LIMIT

**Corollary 8.1.** *As  $d \rightarrow \infty$ ,*

$$I_{\text{ME}}(d) = \frac{32G}{\pi^2} - \frac{c_1}{d} + O(d^{-2}), \quad c_1 = 2 + \frac{4}{\pi} - \frac{32G}{\pi^2} = 0.30342\dots,$$

*so the maximum of  $I_d$  over projective strategies on maximally entangled states tends to  $32G/\pi^2 = 2.96981\dots$*

*Proof.* By Lemma 2.5(2),  $F_{\text{DKZ}} = \sum_{m=1}^{d-1} m \csc x_m$  with  $x_m = \frac{\pi m}{2d}$ . Write  $m \csc x_m = \frac{2d}{\pi}(1 + f(x_m))$  with  $f(x) = x \csc x - 1$ , which is smooth on  $[0, \frac{\pi}{2}]$  with  $f(0) = 0$ ,  $f(\frac{\pi}{2}) = \frac{\pi}{2} - 1$  and  $\int_0^{\pi/2} f = 2G - \frac{\pi}{2}$  (because  $\int_0^{\pi/2} x \csc x \, dx = 2G$ ). The trapezoidal rule gives  $\sum_{m=1}^{d-1} f(x_m) = \frac{2d}{\pi}(2G - \frac{\pi}{2}) - \frac{1}{2}(\frac{\pi}{2} - 1) + O(\frac{1}{d})$ , hence  $F_{\text{DKZ}} = \frac{8G}{\pi^2}d^2 - (\frac{1}{2} + \frac{1}{\pi})d + O(1)$  and  $I_{\text{ME}}(d) = \frac{4F_{\text{DKZ}}}{d(d-1)}$  has the stated expansion.  $\square$

In the language of Sections 4 and 5,  $N^{-2}(2F_{\text{DKZ}} + d) \rightarrow \Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$ : the discrete value converges to the value of the continuum problem, which by Theorem 4.2 is the same for matrix-valued and for scalar configurations. The continuum theorem alone does not give the exact discrete bound: the discrete and continuum problems differ by terms of order  $d$  (Remark 5.5), and the exact bound requires the cell embeddings and Theorem E.

## 9. THE NOISE AND KULLBACK–LEIBLER CLAUSES OF PROBLEM 27B

Besides the clause proved in Theorems A and B, Part B of Problem 27 [36] asks to show that the DKZ measurements realise the highest resistance of the violation to noise, and the best discrimination against local realism in the sense of the Kullback–Leibler divergence [42]. In this section we give partial answers to both clauses: the noise clause holds for the violation of the CGLMP inequality, and fails for the violation of local realism when measurements may have outcomes that never occur, while for complete von Neumann measurements it remains open for  $d \geq 4$ ; the Kullback–Leibler clause fails for every  $d \geq 4$ . Throughout, the settings are  $x, y \in \{1, 2\}$  and the outcomes  $a, b \in \mathbb{Z}_d$ , as in (1.1), and  $\text{DKZ}_d$  denotes the DKZ strategy on  $\Phi_d$ . The statements and proofs, the certificates, the verification programs and the re-verification reports of this section are in the directory `clauses/` of [32]; its file `LEDGER.md` records the wording of Problem 27 on the archived page [36] and the status of each clause.

**9.1. Critical visibilities.** A *behaviour* is a family  $p = (p(\cdot, \cdot | x, y))_{x, y}$  of probability distributions on  $\mathbb{Z}_d \times \mathbb{Z}_d$ . For  $\lambda = (a_1, a_2, b_1, b_2) \in \mathbb{Z}_d^4$  let  $\delta_\lambda(a, b | x, y) = [a = a_x][b = b_y]$ . The local polytope is  $\mathcal{L}_d = \text{conv}\{\delta_\lambda : \lambda \in \mathbb{Z}_d^4\}$ ; a behaviour is *local* if it lies in  $\mathcal{L}_d$ , that is, if it is the behaviour of a random deterministic strategy (a *local model*). Relabellings of the outcomes of each measurement, of the settings and of the parties, and local post-processing of the outcomes, map  $\mathcal{L}_d$  into itself. Let  $p_0(a, b | x, y) = d^{-2}$  be the behaviour of completely random outcomes; it is local. For a behaviour  $p$  and  $0 \leq v \leq 1$  put  $p_v = vp + (1 - v)p_0$  and

$$v_c(p) = \max\{v \in [0, 1] : p_v \in \mathcal{L}_d\}.$$

As  $\mathcal{L}_d$  is closed and convex and contains  $p_0$ ,  $p_v$  is local for  $v \leq v_c(p)$  and nonlocal for  $v > v_c(p)$ ; a *smaller* critical visibility means a *higher* resistance to noise. Relabellings map  $\mathcal{L}_d$  onto itself and fix  $p_0$ , so they do not change  $v_c$ . For a strategy we write  $v_c$  of its behaviour. In the paper behind the problem, Gill measures noise resistance by mixing the behaviour with “completely random, uniform outcomes” [15, p. 140] as long as local realism is still violated, that is, by  $v_c$ . For strategies on maximally entangled states the noise clause can be read in three ways.

- (a) *CGLMP reading*: noise  $p_0$ , nonlocality detected by the CGLMP inequality  $I_d \leq 2$ .
- (b) *Gill’s literal reading*: noise  $p_0$ , nonlocality against all Bell inequalities, that is,  $v_c$ .

- (c) *White-noise reading*: noise on the state,  $v|\Phi_D\rangle\langle\Phi_D| + (1-v)\mathbf{1}/D^2$ , and nonlocality against all Bell inequalities. The noise then contributes the product behaviour  $\text{Tr } P_a^x \text{Tr } Q_b^y / D^2$ , which equals  $p_0$  if all outcomes of all measurements have trace  $D/d$ , for example for complete von Neumann (rank-one) measurements on  $\Phi_d$ ; for these, readings (b) and (c) coincide.

Formal Conjectures issue #3444 [33] formulates the clause with white noise on  $\Phi_d$  and complete von Neumann measurements. Reading (a) holds for every  $d$  (Corollary 9.1). Reading (b) fails for every  $d \geq 4$  when measurements may have outcomes that never occur (Theorem 9.2), and for complete von Neumann measurements on  $\Phi_d$  readings (b) and (c) are open for  $d \geq 4$  (Section 9.4).

## 9.2. The CGLMP reading.

**Corollary 9.1.** *Let  $p$  be the behaviour of a projective strategy on  $\Phi_D$ . Then  $I_d(p_v) = vI_d(p)$ , so the CGLMP inequality detects  $p_v$  (that is,  $I_d(p_v) > 2$ ) if and only if  $v > 2/I_d(p)$ . Consequently the CGLMP threshold  $2/I_d(p)$  (defined when  $I_d(p) > 2$ ) satisfies  $2/I_d(p) \geq 2/I_{\text{ME}}(d)$ , with equality if and only if the strategy is, up to a local unitary  $u \otimes \bar{u}$ , the DKZ strategy tensored with the identity (in particular  $d \mid D$ ). The same holds for white noise added to the state,  $v|\Phi_D\rangle\langle\Phi_D| + (1-v)\mathbf{1}/D^2$ , whenever every outcome of every measurement has  $\text{Tr } P_a^x = \text{Tr } Q_b^y = D/d$  (for example rank-one measurements with  $D = d$ ), because then the noise term produces  $p_0$ .*

*Proof.*  $I_d$  is linear in the behaviour, and  $I_d(p_0) = 0$  because each of the eight probabilities in each bracket of (1.1) equals  $1/d$  under  $p_0$ . Apply Theorems A and B. Under the trace condition,  $\text{Tr}((P_a^x \otimes Q_b^y)\mathbf{1}/D^2) = (D/d)^2/D^2 = d^{-2}$ .  $\square$

Corollary 9.1 is formally verified in Lean with the scope of Theorems A and B: without assumptions for  $2 \leq d \leq 20$ , and assuming the cone condition (Theorem E) for  $d \geq 21$  (Section 11.5). For measurements with outcomes of unequal traces, white noise on the state produces a local behaviour whose CGLMP value need not vanish, and Corollary 9.1 says nothing about it. The corollary concerns the CGLMP inequality only; for all Bell inequalities the corresponding statement is false for every  $d \geq 4$  when measurements may have outcomes that never occur (Theorem 9.2).

**9.3. Gill's literal reading.** Let  $\sigma_z, \sigma_x$  be Pauli matrices, let

$$A_1 = \sigma_z, \quad A_2 = \sigma_x, \quad B_1 = \frac{1}{\sqrt{2}}(\sigma_z + \sigma_x), \quad B_2 = \frac{1}{\sqrt{2}}(\sigma_z - \sigma_x)$$

be the optimal CHSH observables, and let  $\Pi_{\pm}^x = \frac{1}{2}(\mathbf{1} \pm A_x)$  and  $\Pi_{\pm}^y = \frac{1}{2}(\mathbf{1} \pm B_y)$ , real symmetric projections of rank one on  $\mathbb{C}^2$ . For  $d \geq 3$  define a projective strategy  $C_d$  on  $\Phi_d$  as follows.

- *d even.* Identify  $\mathbb{C}^d = \mathbb{C}^2 \otimes \mathbb{C}^{d/2}$  by  $|i\rangle|j\rangle \mapsto |i\frac{d}{2} + j\rangle$ , so that  $\Phi_d = \Phi_2 \otimes \Phi_{d/2}$ , and let  $P_0^x = \Pi_+^x \otimes \mathbf{1}$ ,  $P_1^x = \Pi_-^x \otimes \mathbf{1}$ ,  $Q_0^y = \Pi_+^y \otimes \mathbf{1}$  and  $Q_1^y = \Pi_-^y \otimes \mathbf{1}$ .
- *d odd.* Decompose  $\mathbb{C}^d = \bigoplus_{k=1}^{(d-1)/2} \text{span}\{|2k-2\rangle, |2k-1\rangle\} \oplus \mathbb{C}|d-1\rangle$  and let  $P_0^x = (\bigoplus_k \Pi_+^x) \oplus \mathbf{1}$ ,  $P_1^x = (\bigoplus_k \Pi_-^x) \oplus 0$ , and  $Q_0^y, Q_1^y$  likewise with  $\Pi_{\pm}^y$ .

In both cases  $P_a^x = Q_b^y = 0$  for  $a, b \geq 2$ : the outcomes  $2, \dots, d-1$  never occur. Put

$$v_{\text{even}}(d) = \frac{4(d-1)}{(\sqrt{2}-1)d^2 + 4(d-1)}, \quad v_{\text{odd}}(d) = \frac{4}{(\sqrt{2}-1)d + 4}.$$

**Theorem 9.2** (Gill's noise clause; computer-assisted for  $4 \leq d \leq 9$ ). (1) *For every  $d \geq 3$ ,*

*$v_c(C_d) = v_{\text{even}}(d)$  if  $d$  is even and  $v_c(C_d) = v_{\text{odd}}(d)$  if  $d$  is odd.*

- (2) *For every  $d \geq 2$ ,  $\frac{1}{2} \leq v_c(\text{DKZ}_d) \leq 2/I_{\text{ME}}(d)$ . For  $4 \leq d \leq 9$ ,  $v_c(\text{DKZ}_d) \geq r(d)$ , where  $r(4), \dots, r(9) = 0.69, 0.687, 0.684, 0.683, 0.682, 0.681$ .*



- (3) For every  $d \geq 4$ ,  $v_c(C_d) < v_c(\text{DKZ}_d)$ . Hence, on  $\Phi_d$  and on every  $\Phi_{kd}$ , among projective measurements with possibly unused outcomes, the DKZ measurements do not realise the highest resistance of the violation of local realism to uniform outcome noise.

The only computer-assisted step is Proposition 9.9 below, which is used for  $4 \leq d \leq 9$ . We first compute the behaviour of  $C_d$  and a Bell inequality that detects it.

**Lemma 9.3.** *Let  $s_{xy} = -1$  if  $x = y = 2$  and  $s_{xy} = 1$  otherwise, let  $\text{CH}(a, b|x, y) = \frac{1}{4}(1 + (-1)^{a+b}s_{xy}/\sqrt{2})$  for  $a, b \in \{0, 1\}$  and  $\text{CH}(a, b|x, y) = 0$  otherwise, and let  $\delta_{00}(a, b|x, y) = [a = b = 0]$ . The behaviour of  $C_d$  is  $\text{CH}$  if  $d$  is even and  $\frac{d-1}{d}\text{CH} + \frac{1}{d}\delta_{00}$  if  $d$  is odd.*

*Proof.* All projections are real symmetric, so the behaviour is  $\text{Tr}(P_a^x Q_b^y)/d$ . For  $e, f \in \{\pm 1\}$ ,  $\frac{1}{2} \text{Tr}(\frac{1+eA_x}{2} \cdot \frac{1+fB_y}{2}) = \frac{1}{4}(1 + \frac{ef}{2} \text{Tr}(A_x B_y))$  and  $\frac{1}{2} \text{Tr}(A_x B_y) = s_{xy}/\sqrt{2}$ . For even  $d$ ,  $\text{Tr}((\Pi \otimes \mathbf{1})(\Pi' \otimes \mathbf{1}))/d = \frac{1}{2} \text{Tr}(\Pi\Pi')$ . For odd  $d$  the trace splits into the  $(d-1)/2$  blocks, each contributing  $\text{Tr}(\Pi\Pi')/d$ , and the last dimension, which contributes  $[a = b = 0]/d$ .  $\square$

Let  $\varsigma(0) = 1$  and  $\varsigma(a) = -1$  for  $a \neq 0$ , let  $E_{xy}(p) = \sum_{a,b} \varsigma(a)\varsigma(b)p(a, b|x, y)$ , and let  $\mathcal{E}(p) = E_{11}(p) + E_{12}(p) + E_{21}(p) - E_{22}(p)$ : the CHSH expression after coarse-graining every measurement into the outcome 0 and the rest.

**Lemma 9.4** (a coarse-grained CHSH witness).  $\mathcal{E} \leq 2$  on  $\mathcal{L}_d$ ;  $\mathcal{E}(p_0) = 2(d-2)^2/d^2$ ,  $\mathcal{E}(\text{CH}) = 2\sqrt{2}$  and  $\mathcal{E}(\delta_{00}) = 2$ . Consequently the behaviour  $p_v$  of  $C_d$  violates  $\mathcal{E} \leq 2$  if and only if  $v > v_{\text{even}}(d)$  ( $d$  even), respectively  $v > v_{\text{odd}}(d)$  ( $d$  odd).

*Proof.* On  $\delta_\lambda$ ,  $\mathcal{E} = \varsigma(a_1)(\varsigma(b_1) + \varsigma(b_2)) + \varsigma(a_2)(\varsigma(b_1) - \varsigma(b_2)) = \pm 2$ , and  $\mathcal{E}$  is linear. Under  $p_0$  the two coarse-grained outcomes are independent with mean  $(2-d)/d$ , so  $E_{xy}(p_0) = (d-2)^2/d^2$ ; under  $\text{CH}$ ,  $E_{xy} = s_{xy}/\sqrt{2}$ , and under  $\delta_{00}$ ,  $E_{xy} = 1$ . As  $\mathcal{E}(p_v) = v\mathcal{E}(p) + (1-v)\mathcal{E}(p_0)$  and  $\mathcal{E}(p) > 2 > \mathcal{E}(p_0)$ , the inequality is violated if and only if  $v > (2 - \mathcal{E}(p_0))/(\mathcal{E}(p) - \mathcal{E}(p_0))$ . Here  $2 - \mathcal{E}(p_0) = 8(d-1)/d^2$ , and  $\mathcal{E}(p) - \mathcal{E}(p_0)$  equals  $(2\sqrt{2}d^2 - 2(d-2)^2)/d^2$  for  $p = \text{CH}$  and  $2(d-1)((\sqrt{2}-1)d+4)/d^2$  for  $p = \frac{d-1}{d}\text{CH} + \frac{1}{d}\delta_{00}$ ; the quotients are  $v_{\text{even}}(d)$  and  $v_{\text{odd}}(d)$ .  $\square$

The witness is optimal: at the threshold the behaviour is local. We give an explicit local model.

**Proposition 9.5** (a local model at the threshold). *Let  $d \geq 3$ ,  $s = 1/d$  and  $T = 4s(1-s)/(\sqrt{2} - (1-2s)^2)$ . Then  $T = v_{\text{even}}(d)$ , and  $T\text{CH} + (1-T)p_0$  is local.*

*Proof.* Multiplying numerator and denominator of  $T$  by  $d^2$  gives  $v_{\text{even}}(d)$ . Put  $c_1 = (1-T)(1-2s)$ ,  $c_2 = (1-T)(1-2s)^2$  and  $c_0 = 1 - 4c_1 + 3c_2$ . Since  $1-T = (\sqrt{2}-1)/(\sqrt{2} - (1-2s)^2)$ , one has  $c_1 - c_2 = 2s(1-T)(1-2s) \geq 0$  and

$$c_0 = \frac{4s[(1-s) - (\sqrt{2}-1)(1-3s)]}{\sqrt{2} - (1-2s)^2} > 0,$$

because  $(\sqrt{2}-1)(1-3s) < 1-3s \leq 1-s$ . Let  $\Lambda = (\Lambda_{A_1}, \Lambda_{A_2}, \Lambda_{B_1}, \Lambda_{B_2})$  be a random variable with values in  $\{0, 1, *\}^4$  and the following law:

- (1) with probability  $c_2$ ,  $\Lambda = (*, *, *, *)$ ;
- (2) for each  $r \in \{0, 1\}$ , with probability  $\frac{1}{2}(c_1 - c_2)$  each,  $\Lambda = (*, r, r, 1-r)$ ,  $(r, *, r, r)$ ,  $(r, 1-r, *, r)$  or  $(r, r, r, *)$ ;
- (3) with probability  $\frac{1}{8}c_0$  each,  $\Lambda$  is one of the eight  $\lambda \in \{0, 1\}^4$  with  $\sum_{x,y} s_{xy}(-1)^{\lambda_{A_x} + \lambda_{B_y}} = 2$ .

The probabilities sum to  $c_2 + 4(c_1 - c_2) + c_0 = 1$ . Alice outputs  $\Lambda_{A_x}$  if it lies in  $\{0, 1\}$ , and otherwise an independent uniformly random element of  $\{2, \dots, d-1\}$ ; Bob does the same with  $\Lambda_{B_y}$ . This is a local model; we show that its behaviour  $q$  is  $T\text{CH} + (1-T)p_0$ . Fix  $(x, y)$ , and recall



$1 - 2s = (d - 2)/d$ . If  $a, b \geq 2$ , only (1) contributes, and  $q(a, b|x, y) = c_2/(d - 2)^2 = (1 - T)/d^2$ . If  $a \geq 2 > b$ , only the pattern of (2) with  $*$  at  $A_x$  and the value  $b$  at  $B_y$  contributes, and  $q(a, b|x, y) = \frac{1}{2}(c_1 - c_2)/(d - 2) = (1 - T)/d^2$ ; the case  $b \geq 2 > a$  is symmetric. Let  $a, b \in \{0, 1\}$ . In each pattern of (2) whose  $*$  is not at  $A_x$  or  $B_y$ , one has  $(-1)^{\Lambda_{A_x} + \Lambda_{B_y}} = s_{xy}$ , and the two values of  $r$  give the two pairs  $(a, b)$  with  $(-1)^{a+b} = s_{xy}$ . Each strategy in (3) satisfies  $(-1)^{\lambda_{A_x} + \lambda_{B_y}} = s_{xy}$  for exactly three of the four pairs  $(x, y)$ , and the set of these eight strategies is invariant under  $\lambda \mapsto 1 - \lambda$ ; so for each  $(x, y)$ , six of them give a pair  $(a, b)$  with  $(-1)^{a+b} = s_{xy}$  (three for each such pair) and two give one of the other two pairs (one each). Hence, for  $a, b \in \{0, 1\}$ ,

$$q(a, b|x, y) = \frac{1}{4}(1 - 2c_1 + c_2) + \frac{1}{8}(-1)^{a+b}s_{xy}(1 - c_2),$$

$$TCH(a, b|x, y) + (1 - T)d^{-2} = \frac{1}{4}(T + 4(1 - T)s^2) + \frac{T}{4\sqrt{2}}(-1)^{a+b}s_{xy}.$$

The constant terms agree, because  $1 - 2c_1 + c_2 = 1 - (1 - T)(1 - 2s)(1 + 2s) = T + 4(1 - T)s^2$ , and so do the other terms, because  $1 - c_2 = \sqrt{2}T$  is equivalent to the definition of  $T$ .  $\square$

*Proof of Theorem 9.2(1).* By Lemmas 9.3 and 9.4,  $v_c(C_d) \leq v_{\text{even}}(d)$  for even  $d$  and  $v_c(C_d) \leq v_{\text{odd}}(d)$  for odd  $d$ . For even  $d$ , Proposition 9.5 gives equality. For odd  $d$  let  $s = 1/d$ ; then

$$vp_{C_d} + (1 - v)p_0 = vs\delta_{00} + (1 - vs)(v'\text{CH} + (1 - v')p_0), \quad v' = \frac{v(1 - s)}{1 - vs},$$

and for  $v = v_{\text{odd}}(d) = 4s/(\sqrt{2} - 1 + 4s)$  one finds  $1 - vs = (\sqrt{2} - (1 - 2s)^2)/(\sqrt{2} - 1 + 4s)$  and hence  $v' = T$ . So the behaviour at  $v = v_{\text{odd}}(d)$  is a mixture of the deterministic behaviour  $\delta_{00}$  and the local behaviour of Proposition 9.5.  $\square$

Thus the coarse-grained CHSH inequality  $\mathcal{E} \leq 2$  is an optimal witness for  $C_d$ . For the DKZ strategy we need lower bounds on  $v_c$ , that is, local models. The first one is universal.

**Lemma 9.6** (the  $\frac{1}{2}$ -lemma). *Let  $p$  be a behaviour all of whose marginals are uniform:  $\sum_b p(a, b|x, y) = \sum_b p(b, a|x, y) = 1/d$  for all  $a, x, y$ . Then  $p_v$  is local for  $0 \leq v \leq \frac{1}{2}$ . The constant  $\frac{1}{2}$  is optimal: the behaviour  $\text{PR}_d(a, b|x, y) = \frac{1}{d}[b - a \equiv [x = 2][y = 1] \pmod{d}]$ , a  $d$ -outcome version of the box of Popescu and Rohrlich [37], has uniform marginals and  $v_c(\text{PR}_d) = \frac{1}{2}$ .*

*Proof.* The probability distributions on  $\mathbb{Z}_d^4$  given by

$$P_1(a_1, a_2, b_1, b_2) = p(a_1, b_1|1, 1)p(a_2, b_2|2, 2), \quad P_2(a_1, a_2, b_1, b_2) = p(a_1, b_2|1, 2)p(a_2, b_1|2, 1)$$

are local models. Under  $P_1$  the pairs of outcomes for the settings (1, 1) and (2, 2) have the laws  $p(\cdot, \cdot|1, 1)$  and  $p(\cdot, \cdot|2, 2)$ , while for (1, 2) and (2, 1) the two outcomes are independent and uniformly distributed, with law  $p_0(\cdot, \cdot|x, y)$ ; under  $P_2$  the roles are exchanged. Hence  $\frac{1}{2}(P_1 + P_2)$  is a local model of  $\frac{1}{2}(p + p_0)$ , and for  $v \leq \frac{1}{2}$ ,  $p_v = 2v \cdot \frac{1}{2}(p + p_0) + (1 - 2v)p_0$  is local. For  $\text{PR}_d$  each of the four terms with  $k = 0$  in the first bracket of (1.1) equals 1 and all other probabilities in (1.1) vanish, so  $I_d(\text{PR}_d) = 4$ ; as  $I_d(p_0) = 0$ ,  $I_d$  detects  $(\text{PR}_d)_v$  for every  $v > \frac{1}{2}$ .  $\square$

The DKZ strategy has uniform marginals, so  $v_c(\text{DKZ}_d) \geq \frac{1}{2}$ . With the bases of Section 2.2,  $\langle \Phi_d | (|a\rangle_{A,x} \otimes |b\rangle_{B,y}) = d^{-3/2} \sum_j e^{2\pi i j(b-a-\alpha_x-\beta_y)/d}$ , and  $|\sum_{j=0}^{d-1} e^{2\pi i j\theta/d}|^2 = \sin^2(\pi\theta)/\sin^2(\pi\theta/d)$  with  $\sin^2(\pi(\alpha_x + \beta_y)) = \frac{1}{2}$  give the closed form

$$(9.1) \quad p_{\text{DKZ}}(a, b|x, y) = \frac{1}{d}h_{xy}(b - a), \quad h_{xy}(m) = \frac{1}{2d^2 \sin^2(\pi(m - \delta_{xy})/d)},$$

where  $\delta_{xy} = \alpha_x + \beta_y$ , that is,  $\delta_{11} = \delta_{22} = \frac{1}{4}$ ,  $\delta_{12} = -\frac{1}{4}$  and  $\delta_{21} = \frac{3}{4}$ ; each  $h_{xy}$  is a probability distribution on  $\mathbb{Z}_d$ .

**Lemma 9.7** (covariant local models). *Let  $\mu$  be a probability distribution on  $M_d = \{m \in \mathbb{Z}_d^{\{1,2\}^2} : m_{11} - m_{12} - m_{21} + m_{22} \equiv 0\}$ , with one-dimensional marginals  $\mu_{xy}$ . Then  $p(a, b|x, y) = \frac{1}{d}\mu_{xy}(b-a)$  is local.*

*Proof.* Draw  $m \sim \mu$  and, independently,  $c$  uniformly from  $\mathbb{Z}_d$ , and output  $A_1 = c$ ,  $A_2 = c + m_{11} - m_{21}$ ,  $B_1 = c + m_{11}$  and  $B_2 = c + m_{12}$ . Then  $B_y - A_x = m_{xy}$  for all  $x, y$  (for  $(2, 2)$  by the constraint), and each  $A_x$  is uniformly distributed and independent of  $m$ ; so  $\mathbb{P}(A_x = a, B_y = b) = \frac{1}{d}\mu_{xy}(b-a)$ .  $\square$

**Lemma 9.8** (absorption). *Let  $p$  have uniform marginals, let  $Q$  be a local behaviour with uniform marginals, and let  $0 < v_0 < v_1 \leq 1$ . If  $|Q(a, b|x, y) - p_{v_1}(a, b|x, y)| \leq (v_1 - v_0)/(2d^2v_0)$  for all  $a, b, x, y$ , then  $p_{v_0}$  is local.*

*Proof.* Let  $E = p_{v_1} - Q$ , whose marginals vanish, and  $\gamma = v_0/(v_1 - v_0)$ . Then

$$p_{v_0} = \frac{v_0}{v_1}p_{v_1} + \left(1 - \frac{v_0}{v_1}\right)p_0 = \frac{v_0}{v_1}Q + \left(1 - \frac{v_0}{v_1}\right) \cdot \frac{1}{2}(p_0 + S), \quad S = p_0 + 2\gamma E.$$

The hypothesis gives  $S \geq 0$ ; so  $S$  is a behaviour with uniform marginals, and  $\frac{1}{2}(p_0 + S)$  is local by Lemma 9.6.  $\square$

For  $p = p_{\text{DKZ}}$  and  $Q = \frac{1}{d}\mu_{xy}(b-a)$  as in Lemma 9.7, the condition of Lemma 9.8 reads  $|\mu_{xy}(m) - v_1 h_{xy}(m) - (1 - v_1)/d| \leq (v_1 - v_0)/(2dv_0)$  for all  $x, y, m$ .

**Proposition 9.9** (computer-assisted). *For every  $3 \leq d \leq 20$  there are rationals  $v_0(d) < v_1(d)$  and a probability distribution  $\mu$  on  $M_d$  with rational weights (common denominator  $10^{16}$ ) whose marginals satisfy this condition. Hence  $v_c(\text{DKZ}_d) \geq v_0(d)$ , where  $2/I_{\text{ME}}(d) - 1.02 \cdot 10^{-7} < v_0(d) < 2/I_{\text{ME}}(d)$ ; in particular  $v_c(\text{DKZ}_d) \geq r(d)$  for  $4 \leq d \leq 9$ . For example,  $v_0(4) = 0.690549638999$  and  $v_0(9) = 0.681080610999$ .*

*Proof.* The weights are checked to be nonnegative with sum 1 in integer arithmetic, and the marginals  $\mu_{xy}(m)$  are computed exactly. The numbers  $h_{xy}(m)$  are enclosed in rational intervals:  $\pi$  is enclosed by consecutive partial sums of Machin's formula  $\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}$  (alternating series, enclosure width below  $10^{-50}$ );  $\sin^2(\pi s)$  for rational  $s$  is reduced to  $\sin(\pi r)$  with  $r \in [0, \frac{1}{2}]$ , which lies between  $\sin(r\pi_-)$  and  $\sin(r\pi_+)$  for an enclosure  $[\pi_-, \pi_+]$  of  $\pi$ ; and  $\sin z$  for rational  $0 \leq z \leq 2$  is enclosed by consecutive partial sums of its Taylor series, which alternates with decreasing terms because  $z^2 < 6$ . The largest deviation  $|\mu_{xy}(m) - v_1 h_{xy}(m) - (1 - v_1)/d|$  over the enclosures is then compared exactly with  $(v_1 - v_0)/(2dv_0)$ ; the margins are about  $10^{-13}$  and the enclosure widths about  $10^{-15}$ . The distributions  $\mu$  were found by a linear program over the covariant local models ( $d^3$  variables), solved in floating point with HiGHS [19] slightly below its optimum and then rounded; this computation plays no role in the verification. The certificates (`certificates/dkz_d3.json`, ..., `dkz_d20.json`) and the checking program `dkz_cert.py` are in `clauses/noise-literal/` of [32].  $\square$

*Proof of Theorem 9.2(2) and (3).* (2) The lower bound  $\frac{1}{2}$  is Lemma 9.6. Since  $I_d(p_{\text{DKZ}}) = I_{\text{ME}}(d)$  (Section 2.2) and  $I_d(p_0) = 0$ ,  $I_d$  detects  $(p_{\text{DKZ}})_v$  for  $v > 2/I_{\text{ME}}(d)$ , so  $v_c(\text{DKZ}_d) \leq 2/I_{\text{ME}}(d)$ . The bounds  $r(d)$  follow from Proposition 9.9, since  $v_0(d) \geq r(d)$  for  $4 \leq d \leq 9$ .

(3) For  $d \geq 10$  we show  $v_c(C_d) < \frac{1}{2}$ . For even  $d$ ,  $v_{\text{even}}(d) < \frac{1}{2}$  if and only if  $(\sqrt{2}-1)d^2 - 4d + 4 > 0$ ; this holds at  $d = 9$  ( $81\sqrt{2} > 113$ ) and hence for all  $d \geq 9$ , the derivative  $2(\sqrt{2}-1)d - 4$  being positive for  $d \geq 5$ . For odd  $d$ ,  $v_{\text{odd}}(d) < \frac{1}{2}$  if and only if  $d > 4(\sqrt{2}+1) = 9.657\dots$ . For  $4 \leq d \leq 9$  we have  $v_c(C_d) < r(d)$ , by exact comparison in  $\mathbb{Q}(\sqrt{2})$ ; numerically  $v_c(C_d) = 0.6442, 0.6589, 0.5729, 0.5798, 0.5137, 0.5176$  for  $d = 4, \dots, 9$ , against  $r(d) = 0.69, \dots, 0.681$ . (For  $d = 8, 9$  the value exceeds  $\frac{1}{2}$ , which is why certificates are needed up to  $d = 9$ .) Finally,  $C_d \otimes \mathbf{1}_k$  and  $\text{DKZ}_d \otimes \mathbf{1}_k$  on  $\Phi_{kd} = \Phi_d \otimes \Phi_k$  have the behaviours of  $C_d$  and  $\text{DKZ}_d$ .  $\square$

*Remark 9.10* (exact values for the DKZ strategy). For  $3 \leq d \leq 20$  the lower bounds of Proposition 9.9 are within  $1.02 \cdot 10^{-7}$  of the upper bound  $2/I_{\text{ME}}(d)$ : up to this accuracy the CGLMP inequality is an optimal witness for  $\text{DKZ}_d$  against all Bell inequalities. For  $d = 3, 4, 5, 6$  the equality  $v_c(\text{DKZ}_d) = 2/I_{\text{ME}}(d)$  holds exactly. The link laws of the covariant local behaviours of Lemma 9.7 form the polytope spanned by the vectors  $(e_{m_{11}}, e_{m_{12}}, e_{m_{21}}, e_{m_{22}})$ ,  $m \in M_d$ ; its complete facet lists (66, 216, 1020 and 2462 facets for  $d = 3, 4, 5, 6$ ) were computed by an exact double description, and in exact arithmetic in  $\mathbb{Q}(\cos \frac{\pi}{4d})$  the link laws of  $(p_{\text{DKZ}})_v$  at  $v = 2/I_{\text{ME}}(d)$  satisfy all of them (computer-assisted; not needed for Theorem 9.2).

**Proposition 9.11** ( $d = 2$  and  $d = 3$ ). (1) For every quantum behaviour  $p$  with  $d = 2$  (any state, any measurements),  $v_c(p) \geq 1/\sqrt{2} = v_c(\text{DKZ}_2)$ .  
 (2) (computer-assisted) For every projective strategy on  $\Phi_3$ ,  $v_c \geq 2/I_{\text{ME}}(3) = 3\sqrt{3} - \frac{9}{2} = 0.696152\dots$ , with equality if and only if the strategy is  $\text{DKZ}_3$  up to relabellings and a local unitary  $u \otimes \bar{u}$ .  
 (3) The strategy on  $\Phi_2$  with the three-outcome measurements  $\{\Pi_+^x, \Pi_-^x, 0\}$  and  $\{\Pi_+^y, \Pi_-^y, 0\}$  has  $v_c = 8/(9\sqrt{2} - 1) = 0.682132\dots < v_c(\text{DKZ}_3)$ . The same behaviour arises on every  $\Phi_D$  with  $D$  even, in particular on  $\Phi_6$ , which also carries  $\text{DKZ}_3 \otimes \mathbf{1}_2$ .

So the clause holds for  $d = 2$ ; for  $d = 3$  it holds on  $\Phi_3$  and fails if every maximally entangled state is allowed.

*Proof.* (1) The local polytope  $\mathcal{L}_2$  is described by positivity and the eight CHSH inequalities [14]. Every CHSH expression vanishes on  $p_0$  and is at most  $2\sqrt{2}$  on quantum behaviours [8], so  $p_v$  satisfies all of them for  $v \leq 1/\sqrt{2}$ ; and  $I_2(\text{DKZ}_2) = 2\sqrt{2}$  gives  $v_c(\text{DKZ}_2) \leq 1/\sqrt{2}$ .

(2) The polytope  $\mathcal{L}_3$  has exactly 1116 facets: 36 positivity facets, 648 liftings of CHSH inequalities (every measurement coarse-grained into one outcome against the other two) and the 432 images of the CGLMP inequality under relabellings. This is the classification of Collins and Gisin [9]; we recomputed the list by an exact double description, verifying every facet. Let  $p$  be the behaviour and  $v \leq 2/I_{\text{ME}}(3)$ . Positivity holds. A facet of CGLMP type is a positive multiple of  $2 - I_3(g \cdot p)$  for a relabelling  $g$ ;  $g$  maps projective strategies on  $\Phi_3$  to projective strategies on  $\Phi_3$  (for the exchange of the parties use  $\text{Tr}(X^T Y) = \text{Tr}(Y^T X)$ ) and fixes  $p_0$ , so Theorem A gives  $I_3(g \cdot p_v) = v I_3(g \cdot p) \leq v I_{\text{ME}}(3) \leq 2$ . For a lifted CHSH facet let  $\tilde{A}_x, \tilde{B}_y$  be the Hermitian involutions of the coarse-grained measurements. Up to relabelling, the CHSH value of  $p$  is  $\frac{1}{3}[\text{Tr}(\tilde{A}_1^T(\tilde{B}_1 + \tilde{B}_2)) + \text{Tr}(\tilde{A}_2^T(\tilde{B}_1 - \tilde{B}_2))]$ , which by Hölder's inequality is at most  $\frac{1}{3}(\|\tilde{B}_1 + \tilde{B}_2\|_1 + \|\tilde{B}_1 - \tilde{B}_2\|_1)$ . By Jordan's lemma  $\mathbb{C}^3$  splits into subspaces of dimension at most two invariant under  $\tilde{B}_1$  and  $\tilde{B}_2$ ; a two-dimensional one contributes at most  $4\sqrt{2}$  and a one-dimensional one at most 2, so the CHSH value of  $p$  is at most  $(4\sqrt{2} + 2)/3$ , while that of  $p_0$  is  $\pm \frac{2}{9}$ . Hence no lifted CHSH facet is violated for  $v \leq 4/(3\sqrt{2} + 1) = 0.762974\dots$ , which exceeds  $2/I_{\text{ME}}(3)$ . So  $v_c \geq 2/I_{\text{ME}}(3)$ , with equality for  $\text{DKZ}_3$  by Theorem 9.2(2). If  $v_c = 2/I_{\text{ME}}(3)$ , then for every  $v > 2/I_{\text{ME}}(3)$  some facet of CGLMP type is violated by  $p_v$ ; as there are finitely many, one relabelling  $g$  works for a sequence  $v \downarrow 2/I_{\text{ME}}(3)$ , so  $I_3(g \cdot p) \geq I_{\text{ME}}(3)$ , and Theorem B applies to the relabelled strategy.

(3) As in Lemma 9.3, the behaviour is CH, now with  $d = 3$ . Lemma 9.4 and Proposition 9.5 for  $d = 3$  give  $v_c = v_{\text{even}}(3) = 8/(9\sqrt{2} - 1)$ , which is smaller than  $3\sqrt{3} - \frac{9}{2}$ . On  $\Phi_D$  with  $D$  even, the measurements  $\Pi_{\pm}^x \otimes \mathbf{1}_{D/2}, \Pi_{\pm}^y \otimes \mathbf{1}_{D/2}$  give the same behaviour.  $\square$

*Remark 9.12* (unused outcomes). The strategy  $C_d$  never produces the outcomes  $2, \dots, d - 1$ , while the noise  $p_0$  gives each of them probability  $1/d$ ; this is what lets a two-outcome CHSH structure beat DKZ, and  $v_c(C_d) \sim 4/((\sqrt{2} - 1)d) \rightarrow 0$ . On  $\Phi_d$ , projective measurements cannot avoid this: a PVM on  $\mathbb{C}^d$  all of whose  $d$  outcomes occur consists of  $d$  nonzero orthogonal projections, so it is rank-one, and for rank-one measurements on  $\Phi_d$  the noise  $p_0$  is white noise

on the state (reading (c)). A projective counterexample on  $\Phi_d$  that uses every outcome would therefore settle reading (c). With POVMs every outcome can be used. For  $0 < \varepsilon \leq \frac{1}{100}$  the effects  $M_a^x = (1 - \varepsilon)P_a^x + \frac{\varepsilon}{d}\mathbf{1}$  and  $N_b^y = (1 - \varepsilon)Q_b^y + \frac{\varepsilon}{d}\mathbf{1}$ , with  $(P, Q) = C_d$ , are invertible, every pair of outcomes has probability at least  $\varepsilon^2/d^2$ , and still  $v_c(M, N) < v_c(\text{DKZ}_d)$  for every  $d \geq 4$ . Indeed, the behaviour of  $(M, N)$  is obtained from that of  $C_d$  by the local channel that keeps each outcome with probability  $1 - \varepsilon$  and otherwise replaces it by a uniformly random one; this channel fixes  $p_0$  and preserves locality, so  $v_c(M, N) \geq v_c(C_d)$ , and the witness  $\mathcal{E}$  gives  $v_c(M, N) \leq w_\varepsilon(d) = (2 - 2e^2)/(\mathcal{E}_\varepsilon - 2e^2)$ , where  $e = (2 - d)/d$ ,  $\mathcal{E}_\varepsilon = 2\sqrt{2}(1 - \varepsilon)^2 + 2\varepsilon^2 e^2$  for even  $d$ , and  $\mathcal{E}_\varepsilon = (1 - \varepsilon)^2(2\sqrt{2}\frac{d-1}{d} + \frac{2}{d}) + 4\varepsilon(1 - \varepsilon)\frac{\varepsilon}{d} + 2\varepsilon^2 e^2$  for odd  $d$ . Exact computations show  $w_\varepsilon(d) \leq w_{1/100}(d) < v_0(d)$  for  $4 \leq d \leq 10$  (for example  $w_{1/100}(4) = 0.6602$  and  $w_{1/100}(9) = 0.5380$ ) and  $w_{1/100}(d) < \frac{1}{2}$  for  $d \geq 11$ . Projective measurements on  $\Phi_D$  with  $D > d$  can also use every outcome and still beat DKZ: one adds  $d - 2$  extra basis vectors carrying the outcomes  $2, \dots, d - 1$ ; for  $(d, D) = (4, 12), (5, 11), (6, 12), (7, 13), (8, 12), (9, 13)$  the resulting critical visibility is below  $v_0(d)$ , and for  $(4, 12)$  the strategy  $\text{DKZ}_4 \otimes \mathbf{1}_3$  lives on the same state.

*Remark 9.13* (prior work). The mechanism behind  $C_d$  is known. Acín, Durt, Gisin and Latorre [1, eq. (14)] found that the CHSH strategy on a maximally entangled state of Schmidt rank two embedded in  $\mathbb{C}^d \otimes \mathbb{C}^d$ , with two-outcome measurements and white noise, has critical visibility  $v_{\text{even}}(d)$ , which tends to 0, and concluded that resistance to noise is not a good measure of nonlocality; for even  $d$  the strategy  $C_d$  produces the same coarse-grained statistics on  $\Phi_d$  itself. In the two-outcome setting the exactness of this threshold is elementary; in the  $d$ -outcome scenario it is Proposition 9.5. Gruca, Laskowski and Żukowski [16] found numerically, for  $d \leq 5$ , that states of Schmidt rank two are the most noise-robust. Baek, Ryu and Lee [3] observed by linear programming that merging outcomes increases the white-noise robustness  $R_w = (1 - v_c)/v_c$  of the DKZ behaviour on  $\Phi_4$  and  $\Phi_{16}$ , from 0.448 to 0.552 for  $d = 4$ , and along the chain 0.4755, 0.6216, 1.0243, 1.7673 for  $d = 16$ ; these values agree with  $(I_{\text{ME}}(4) - 2)/2 = 0.44812$ ,  $(1 - v_{\text{even}}(4))/v_{\text{even}}(4) = 0.55229$  and  $(1 - v_{\text{even}}(16))/v_{\text{even}}(16) = 1.76731$ , so merged-outcome strategies beating DKZ on  $\Phi_4$  and  $\Phi_{16}$  under uniform noise were already visible there; they did not relate this to Problem 27. The thresholds  $2/I_{\text{ME}}(d)$  of the DKZ measurements are the numerical values of [22, 12, 10, 16], and Chen et al. [7] gave an analytic result for  $d = 3$ ; we did not check whether it contains a local model, so the novelty of our lower bound for  $d = 3$  is uncertain. As far as we found, the following are new: an exact theorem for every  $d \geq 4$  on  $\Phi_d$  itself, for both parities, the exact critical visibility of  $C_d$  in the  $d$ -outcome scenario, rigorous lower bounds for  $v_c(\text{DKZ}_d)$ , and the consequence for the noise clause of Problem 27 when unused outcomes are allowed.

**9.4. The white-noise reading.** In reading (c), the noise behaviour of a projective strategy on  $\Phi_D$  is  $n(a, b|x, y) = \text{Tr } P_a^x \text{Tr } Q_b^y / D^2$ , the product of the marginals of its behaviour  $p$ , and the question is whether  $vp + (1 - v)n$  is local for all  $v \leq 2/I_{\text{ME}}(d)$ , for every projective strategy on every  $\Phi_D$ , with equality of the critical visibility only for DKZ (up to  $u \otimes \bar{u}$  and an inert identity). For strategies with  $n = p_0$  this is reading (b), and on  $\Phi_d$  these are exactly the strategies that use every outcome (Remark 9.12), which  $C_d$  does not; for them the answer is positive for  $d = 2$  and, on  $\Phi_3$ , for  $d = 3$  (Proposition 9.11). For  $d \geq 4$  the question is open. Numerical searches have found no strategy more robust than DKZ for  $d \leq 8$  (for  $d \leq 5$  see also [16]; our searches are summarised in `clauses/LEDGER.md` of [32]). A proof has to control all facets of  $\mathcal{L}_d$ , not only the CGLMP inequality: already for  $d = 4$  the polytope of link laws of covariant local behaviours (Lemma 9.7), Masanes' polytope of correlation functions [29], has facets that are not images of the CGLMP inequality.

**9.5. The Kullback–Leibler clause.** Van Dam, Gill and Grünwald [42] measure the strength of a nonlocality proof with behaviour  $q$  by its Kullback–Leibler divergence from the local behaviours. For a probability distribution  $\sigma$  on the setting pairs  $\{1, 2\}^2$  let

$$U_q(\sigma) = \inf_{p \in \mathcal{L}_d} \sum_{x,y} \sigma_{xy} D(q(\cdot, \cdot | x, y) \| p(\cdot, \cdot | x, y)), \quad D(r \| s) = \sum_i r_i \log_2 \frac{r_i}{s_i},$$

and let  $S^{\text{UNI}}(q) = U_q(\frac{1}{4})$  (uniform settings),  $S^{\text{UC}}(q)$  the supremum of  $U_q(\sigma)$  over product distributions  $\sigma = \sigma_A \otimes \sigma_B$ , and  $S^{\text{COR}}(q) = \sup_{\sigma} U_q(\sigma)$ ; so  $S^{\text{UNI}} \leq S^{\text{UC}} \leq S^{\text{COR}}$ . All values are in bits.

The competitor is built from two copies of the CHSH strategy. Write  $d = 4k + \rho$  with  $k \geq 1$  and  $\rho \in \{0, 1, 2, 3\}$ , and split the computational basis of  $\mathbb{C}^d$ , and the set of outcomes, into consecutive groups of sizes  $4, \dots, 4$  ( $k$  times) and  $\rho$ . On a group of size four,  $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$ , Alice measures in the product basis  $e_{a_1}^x \otimes e_{a_2}^x$ , where  $e_0^x, e_1^x$  are the eigenvectors of  $A_x$  for the eigenvalues  $1, -1$ , with outcome  $2a_1 + a_2$  within the group, and Bob likewise with  $B_y$ : two copies of the CHSH strategy with shared settings. On the last group both parties measure the single basis vector ( $\rho = 1$ ), the eigenbases of  $A_x$  and  $B_y$  ( $\rho = 2$ ), or the DKZ bases for three outcomes ( $\rho = 3$ ). The resulting strategy  $B_d$  on  $\Phi_d$  consists of four orthonormal bases of  $\mathbb{C}^d$ , so every outcome occurs.

**Theorem 9.14** (the Kullback–Leibler clause; computer-assisted). *For every  $d \geq 4$  and  $X \in \{\text{UNI}, \text{UC}, \text{COR}\}$ ,*

$$S^X(\text{DKZ}_d) \leq 0.0687802606 < 0.0703203697 \leq S^{\text{UNI}}(B_d) \leq S^X(B_d).$$

*If  $4 \mid d$ , then  $S^{\text{UNI}}(B_d) \geq 0.0879004622$ . The same holds on  $\Phi_{kd}$  for  $\text{DKZ}_d \otimes \mathbf{1}_k$  and  $B_d \otimes \mathbf{1}_k$ . Hence for every  $d \geq 4$  the DKZ measurements do not give the best discrimination in any of the three senses; the margin is at least 0.00154 bits, and at least 0.0191 bits if  $4 \mid d$ .*

*Sketch of proof. Test factors.* Let  $r \geq 0$  be a function of  $(a, b, x, y)$ , positive where  $q$  is, with  $\sum_{x,y} \sigma_{xy} r(a_x, b_y | x, y) \leq 1$  for every  $\lambda$ . For  $p \in \mathcal{L}_d$ , Jensen’s inequality gives  $\sum \sigma_{xy} q \log(q/p) \geq \sum \sigma_{xy} q \log r - \log \sum \sigma_{xy} r p$ , and the last sum is at most 1 because it is linear in  $p$  and at most 1 on every  $\delta_\lambda$ . Hence  $U_q(\sigma) \geq \sum_{x,y} \sigma_{xy} \sum_{a,b} q \log r$ .

*The competitor.* The measurements of  $B_d$  are block diagonal with disjoint outcome groups, so its behaviour is the mixture  $\sum_j \frac{m_j}{d} q_j$  of the behaviours  $q_j$  of its blocks (of sizes  $m_j$ ), each supported on the outcomes of its group. For uniform  $\sigma$ , test factors of the blocks add: if each  $r_j$  is a test factor of  $q_j$  in its own scenario with all entries at most 2, then  $r = r_j$  on the cells of block  $j$ , and  $r = 0$  elsewhere, is a test factor of the mixture. Indeed, for a deterministic  $\lambda$  only the setting pairs whose two outcomes lie in the same group contribute; if these groups coincide, the contribution is at most that of a deterministic strategy of one block, and otherwise the contributing pairs are  $\{(1, 1), (2, 2)\}$  or  $\{(1, 2), (2, 1)\}$  and contribute at most  $\frac{1}{4}(2 + 2) = 1$ . Exact rational test factors for the four kinds of blocks, with all entries at most 1.24801 and feasibility checked by exhaustive exact enumeration, certify the values 0 (one basis vector), 0.046273845411 (CHSH), 0.057783024051 ( $\text{DKZ}_3$ ) and  $L_4 = 0.087900462237$  (two copies of CHSH). Hence  $S^{\text{UNI}}(B_d) \geq (4kL_4 + \rho L_\rho)/d$ , where  $L_\rho$  is the value of the last block; this bound is nondecreasing in  $k$ , and its minimum over  $d \geq 4$  is  $\frac{4}{5}L_4 \geq 0.0703203697$ , attained at  $d = 5$ .

*The DKZ strategy.* The partial fraction expansion  $\sum_{n \in \mathbb{Z}} (z + n)^{-2} = \pi^2 / \sin^2(\pi z)$  shows that  $h_{xy}$  in (9.1) is the image under  $\mathbb{Z} \rightarrow \mathbb{Z}_d$  of the probability distribution  $h_{xy}^\infty(m) = 1/(2\pi^2(m - \delta_{xy})^2)$  on  $\mathbb{Z}$ , which does not depend on  $d$ . Let  $\mu$  be a probability distribution of a triple  $(t, n_1, n_2) \in \mathbb{Z}^3$ , and let  $\mu_{11}, \mu_{12}, \mu_{21}, \mu_{22}$  be the distributions of  $n_1, n_2, n_1 - t, n_2 - t$ . With a uniformly random  $c \in \mathbb{Z}_d$ , the outputs  $A_1 = c, A_2 = c + t, B_1 = c + n_1, B_2 = c + n_2$  modulo



$d$  form a local model with behaviour  $\frac{1}{d}\bar{\mu}_{xy}(b-a)$ , where  $\bar{\mu}_{xy}$  is the image of  $\mu_{xy}$  in  $\mathbb{Z}_d$ . Its divergence from  $p_{\text{DKZ}}$  on the pair  $(x, y)$  is  $D(h_{xy} \| \bar{\mu}_{xy})$ , which is at most  $D(h_{xy}^\infty \| \mu_{xy})$  by the data-processing inequality. Hence

$$S^{\text{UNI}}(\text{DKZ}_d) \leq T(\mu) := \frac{1}{4} \sum_{x,y} D(h_{xy}^\infty \| \mu_{xy}) \quad \text{for every } d \geq 2.$$

An explicit distribution  $\mu^*$ , with rational weights in a finite window and a tail proportional to  $1/(t^2 - \frac{1}{4})$  (of total mass exactly 1 by telescoping), gives  $T(\mu^*) \leq 0.068780260568$ : the sum over  $|m| \leq 3000$  is evaluated in interval arithmetic, and for  $|m| > 3000$  every term is nonpositive because there  $\mu_{xy}^*(m) \geq h_{xy}^\infty(m)$ . Finally, the relabellings  $q \mapsto q'$  with  $q'(a, b|x, y) = q(-a, [y = 1] - b|3 - x, y)$  and  $q'(a, b|x, y) = q(-a - [x = 2], -b|x, 3 - y)$  fix  $p_{\text{DKZ}}$  and generate a group acting transitively on the setting pairs; as  $\sigma \mapsto U_q(\sigma)$  is concave and invariant under this group, uniform settings are optimal, and  $S^{\text{UNI}} = S^{\text{UC}} = S^{\text{COR}}$  for  $\text{DKZ}_d$ .  $\square$

The certificates and programs are in the directory `clauses/kl-divergence/` of [32]. There, the program `blocks_cert.py` checks the block test factors, which are stored in `blocks_testfactors.json`; the program `zcert.py` checks the distribution  $\mu^*$ , which is stored in `zmodel_cert_A10.json`; and the program `verify_main.py` re-checks the theorem independently from the stored exact data.

**Corollary 9.15.** *For  $X \in \{\text{UNI}, \text{UC}, \text{COR}\}$ ,  $d \geq 2$  and  $k \geq 1$ ,  $S^X(\text{DKZ}_d) \leq S^X(\text{DKZ}_{kd})$ , and*

$$0.0687464 \leq \sup_d S^X(\text{DKZ}_d) \leq 0.0687711.$$

*Sketch of proof.* The infimum defining  $U_q(\frac{1}{4})$  may be restricted to local models invariant under common shifts of all outcomes (averaging over the shifts does not increase the divergence, by convexity), and these are the models of the proof of Theorem 9.14 with  $\mathbb{Z}_{kd}$  in place of  $\mathbb{Z}$ ; reducing modulo  $d$  gives the monotonicity. The lower bound is the value at  $d = 128$  of an exact rational test factor invariant under common shifts, and the upper bound follows from a two-sided rigorous enclosure of the tail of  $T(\mu^*)$ .  $\square$

**Proposition 9.16** ( $d = 3$ ; computer-assisted). *There is a neighbourhood of  $\text{DKZ}_3$  in the set of projective strategies on  $\Phi_3$  on which each of the three strengths is at most its value at  $\text{DKZ}_3$ ,  $S(\text{DKZ}_3) = 0.05778302549332865$  bits, with equality only for strategies obtained from  $\text{DKZ}_3$  by changing the phases of the basis vectors and by a local unitary  $u \otimes \bar{u}$ .*

*Sketch of proof.* For  $d = 3$ , (1.1) reads  $I_3 = 4 - \sum_{x,y} \mathbb{E}[g_{xy}]$ , where the level  $g_{xy} \in \{0, 1, 2\}$  of a pair of outcomes  $(a, b)$  is the residue modulo 3 of  $a - b$ ,  $b - a$ ,  $b - a - 1$ ,  $a - b$  for  $(x, y) = (1, 1), (1, 2), (2, 1), (2, 2)$ ; so the CGLMP inequality is  $\sum_{x,y} \mathbb{E}[g_{xy}] \geq 2$ . With  $\beta = [(16 + 24\sqrt{3}) - \sqrt{3280 - 960\sqrt{3}}]/54$ , the test factor  $r = 1 + \beta(\frac{1}{2} - g_{xy})$  is feasible, and  $p_{\text{DKZ}}/r$  is a mixture with positive weights of the 30 deterministic strategies on the CGLMP facet; so  $r$  is an optimal test factor and gives the value. Every probability of a CGLMP level is stationary at  $\text{DKZ}_3$  (an exact identity in  $\mathbb{Q}(\sqrt{3})$ ), so  $\text{DKZ}_3$  is a critical point. Moving the local model inside the CGLMP facet so that the first- and second-order terms of the four divergences are equal, an interval Cholesky factorisation shows that the averaged Hessian is at most  $-\frac{1}{200}$  times the identity on a 16-dimensional slice transversal to the orbit of the phases and local unitaries. The rank of the measurements is locally constant, which gives the statement among all projective strategies.  $\square$

Whether  $\text{DKZ}_3$  is a global maximiser is open; 330 numerical searches found no strategy with a larger strength. As  $B_d$  consists of rank-one projective measurements, Theorem 9.14 also refutes the clause if POVMs or other ranks are allowed. The Kullback–Leibler clause was first shown to



fail for  $d = 4$  by Zhang [44], with an exact certificate, and in [31] we gave exact certificates for  $4 \leq d \leq 9$ ; numerically,  $\text{DKZ}_3$  appears to be optimal [42, 2]. As far as we found, the additivity of test factors over blocks with disjoint outcomes and the description of every  $\text{DKZ}_d$  as the reduction modulo  $d$  of one distribution on  $\mathbb{Z}$ , which gives a bound uniform in  $d$ , are new.

## 10. OPEN QUESTIONS

(1) *POVMs*. Theorems A and B concern projective measurements; the reduction of Section 2 uses projections in an essential way. For  $3 \leq d \leq 8$  POVMs do not increase the maximum, on maximally entangled states of every local dimension, and every optimal POVM strategy is projective, hence DKZ by Theorem B: we verified this with exact sum-of-squares certificates over  $\mathbb{Q}(\zeta_{4d})$  in the tracial moment relaxation with POVM constraints only (level 1+AB with doubly-localizing constraints), checked independently (repository folder `clauses/povm/` [32]). For  $d \geq 9$  the question is open; the numerical evidence of [26] for  $d = 3$  pointed the same way. (2) *All states*. The maximal value of  $I_d$  over all states is known exactly only for  $d \leq 8$  ([20, 31]). (3) *Robustness of rigidity*. The deficit identity of Remark 7.1 suggests a quantitative version of Theorem B; it would require a lower bound for the defect in Theorem C in terms of  $\| [B, g] \|_F^2$ . (4) *An analytic proof of Theorem E*. The mechanism (Remark 5.6) and the continuum limit of the Gaussian modulation are explicit, and the large- $d$  argument of Section 6.6 is an expansion with certified remainders; a proof without interval arithmetic for all  $d$  seems within reach but is not attempted here. Such a proof, once formalised, would also complete the Lean proof of Theorems A and B for  $d \geq 21$ . (5) *Noise on the state*. Is the critical visibility of every projective strategy on a maximally entangled state, for white noise on the state and against all Bell inequalities, at least  $2/I_{\text{ME}}(d)$ , with equality only for DKZ (reading (c) of Section 9.1)? This is open for every  $d \geq 4$ , already for rank-one measurements on  $\Phi_d$ ; for complete von Neumann measurements on  $\Phi_d$  it is the noise clause of Problem 27B in the formulation of [33]. (6) *Kullback–Leibler for small  $d$* . Whether  $\text{DKZ}_3$  maximises the statistical strength globally, which is the Kullback–Leibler clause of Problem 27B for  $d = 3$ , is open (Proposition 9.16 gives strict local optimality); the case  $d = 2$  is not treated here, and the exact value of  $\sup_d S(\text{DKZ}_d)$  (Corollary 9.15) is not known.

## 11. VERIFICATION AND REPRODUCIBILITY

**11.1. Computer-assisted components.** In the proof of Theorems A and B, only Theorem E (and the convexity statement of Lemma 6.7(1)) depend on computer calculations; the computations for Section 9 are described in Section 11.2. All verifications for Theorem E use interval arithmetic with outward rounding (`mpmath` [21]); floating-point computations are used only to find candidates (linear-programming columns, approximate roots) and play no role in the verification. Table 2 lists the programs; they are in the accompanying repository [32] (directory `cone-certificates/`), together with their outputs and with written accounts of every bound they use. The program `verify_cert.py` of Regime II has the SHA-256 hash

ff56740d15d40a07c99deb6433fb1b0f40b5a2d5b356b3ec9da0270ff1fab6a7,

which is stored in each of its 1800 certificates and checked by `audit_cert_g.py`. The single command `python audit_alld.py` re-checks all three regimes and prints ALL-D CERTIFIED (every  $d \geq 2$ ).

**11.2. The computations of Section 9.** The results of Section 9 use computer calculations at the following points, all in exact rational or algebraic arithmetic or with rigorous rational enclosures; floating-point computations were used only to find certificates. (i) Proposition 9.9: `dkz_cert.py` checks the certificates `certificates/dkz_d3.json`, ..., `dkz_d20.json`. (ii) Proposition 9.11(2): the facet list of  $\mathcal{L}_3$  is computed by `facets223.py` (exact double description,

| Statement                                    | Programs                                                                                                                                                                                                                                                                                                                      | Output and margins                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prop. 6.2 ( $d \leq 200$ )                   | <code>verify_cone.py</code> ,<br><code>run_verify.py</code> ,<br><code>lp_reverify.py</code> ; check<br><code>crosscheck_certs.py</code>                                                                                                                                                                                      | 199 certificates <code>cert_d{d}.json</code> , all<br>re-verified; $d \min_k c_k \geq 0.0816$                                                                                                                                                                                                                                                                |
| Prop. 6.6<br>( $201 \leq d \leq 2000$ )      | <code>verify_cert.py</code> (tests<br><code>test_verify_cert.py</code> ); audit<br><code>audit_cert_g.py</code>                                                                                                                                                                                                               | 1800 certificates <code>cert_g_d{d}.json</code> , all<br>pass; $d^2 \min \hat{a}_k \geq 0.06319$ (even $d$ ),<br>$\geq 0.12638$ (odd $d$ );<br>$d \min \Delta^2(P_v - H)_S \geq 0.660$ ; derivative<br>ratio $\geq 4.0009$                                                                                                                                   |
| Prop. 6.8 ( $d \geq 2001$ )                  | <code>a06_zone0.py</code> , <code>a07_zoneI.py</code> ,<br><code>a08_A2.py</code> , <code>a09_smallm.py</code> ,<br><code>a10_pointwise.py</code> ,<br><code>a11_slack.py</code> ,<br><code>a12_fp_alld.py</code> ,<br><code>a13_constants.py</code> ,<br><code>a14_identities.py</code> (libraries<br><code>a02–a05</code> ) | $Q_O \geq 7.419$ , $\Lambda(R_2) \geq 6.022$ ,<br>$D_B(1) \geq -2.765192\epsilon^2$ ,<br>$ \Upsilon_* - \Upsilon_\epsilon  \leq 6.6468 \cdot 10^{-4}\epsilon^2$ ,<br>$\hat{a}_k \geq 0.1186944 w_k \epsilon^2$ , $\Delta^2 W \geq 0.620154 \epsilon$ ,<br>$ \mathbf{G} - \mathbf{G}_4  \leq 138.103 \epsilon^3$ , $ \partial_t \mathbf{G}_4  \leq 11.5703 d$ |
| Lemma 6.7(1)<br>Theorem E, all<br>$d \geq 2$ | <code>t35_polya.py</code><br><code>audit_alld.py</code>                                                                                                                                                                                                                                                                       | $\inf \mathcal{A} \geq 0.1054878997577$<br>ALL-D CERTIFIED                                                                                                                                                                                                                                                                                                   |

TABLE 2. Programs for the computer-assisted part of the proof of Theorems A and B.

two insertion orders, every facet re-verified) and stored in `facets223.txt`; `phi3.py` checks the remaining exact identities. (iii) Remark 9.10: `dkz_exact.py`. (iv) Remark 9.12: `povm.py`. These programs are in `clauses/noise-literal/` of [32], and the single command `python verify_theorem.py` there re-checks (i)–(iv) and the exact identities of Section 9.3 (17 items, about three minutes). (v) Theorem 9.14: `zcert.py` (the bound on  $T(\mu^*)$ ) and `blocks_cert.py` (the block test factors), with the re-check `verify_main.py`; Corollary 9.15: `dkz_lower.py` and the programs of the re-verification; Proposition 9.16: `d3_cert.py` (exact arithmetic and an interval Cholesky factorisation). These programs are in `clauses/kl-divergence/`.

**11.3. Re-verification.** Each analytic proof was re-derived step by step in a separate verification pass, and the numerical checks were rewritten from scratch, without importing any of our programs; the reports and scripts are in the repository [32] (directories `proofs/` and `clauses/`). These are internal checks, not peer review. There were four such re-verifications for the proof of Theorems A–E, and three for Section 9.

- (1) Theorems D and C, by the proofs of Section 3: verdict correct. The re-implementation (adaptive quadrature between exactly located branch points of  $\rho$ , and  $\mathcal{D}$  by a 30-digit matrix exponential) reproduces (1.6) at 82 complex points  $(a, t)$  on 13 instances ( $M = 3, \dots, 6$ , all ranks for  $M \leq 5$ ), with worst relative error  $6.8 \cdot 10^{-13}$ , and the defect formula (1.5) at 51 pairs (instance,  $\lambda$ ) on 9 instances, including spectra up to  $\pm 150$ , clustered spectra and  $M = 6$ , to  $1.3 \cdot 10^{-12}$  (absolute). The marginal identity  $\int \rho(s, \tau) ds = \|BgP\|_F^2$  holds to  $10^{-13}$ , and the back-projection integral in Step 2 equals  $\rho$  to  $10^{-20}$  (40-digit arithmetic, determinants). Fixed-order quadrature rules between branch points are not sufficient:  $\rho$  has complex singularities close to the real  $s$ -axis.
- (2) The distribution-free proof of Appendix B, including the general case of Remark B.8: verdict correct. The identities involved, among them the Radon identity of Theorem B.1 on 22 instances with  $M \leq 8$ , were confirmed to 34–46 digits.

- (3) The reduction chain (Section 2), the continuum theorem (Section 4), the cell inequalities (Section 5), the logic of the cone condition (Section 6) and the rigidity argument (Section 7): verdict correct.
- (4) The rank-one argument of Remark 3.10: verdict correct (no mathematical error; some expository gaps, none of which changes a statement). The check includes the zero count on 1800 extreme instances and the inequality at  $M = 3, \dots, 8$  with entries up to 300, re-evaluated at 50 digits.
- (5) Theorem 9.2, first re-verification: verdict correct, no error. It re-derived every step, recomputed the facets of  $\mathcal{L}_3$  by a separate double description in different coordinates, certified the threshold of  $C_d$  separately, re-verified all eighteen certificates of Proposition 9.9 in 200-bit interval arithmetic, confirmed the exact values of Remark 9.10 for  $d = 3, 4, 5$  with explicit local models, and ran eleven tamper tests on our verification program, all of which were detected.
- (6) Theorem 9.2, second re-verification: verdict correct, no gap. It rebuilt the result from scratch, with its own certificates for the DKZ strategy, and found the explicit local model of Proposition 9.5.
- (7) Theorem 9.14, Corollary 9.15 and Proposition 9.16: verdict correct, no error. The re-verification code reproduces  $T(\mu^*) \leq 0.068780260568$  digit for digit, re-certifies the block values with its own test factors, and re-proves the local optimality at  $d = 3$  with a different chart, slice and right inverse (averaged Hessian at most  $-\frac{1}{1000}$  times the identity on the slice); the sharper bounds of Corollary 9.15 come from this verification.

11.4. **Further numerical checks.** The following checks are numerical; they are not part of any proof, and were made to guard against errors in formulas, signs and constants.

- Sections 2 and 5: on random non-commuting families ( $d = 3, \dots, 12$ ,  $M = 2, 3, 4$ ) the identities of Lemmas 2.4 and 2.5 and Proposition 5.3 hold to  $2 \cdot 10^{-12}$ ; with the certificates of Proposition 6.2 at  $d = 25$  and  $d = 40$ , the identity  $\langle v, \delta \rangle = \sum_k c_k \langle u^{\ell(k)}, \delta \rangle$  holds to  $1.6 \cdot 10^{-11}$  on random non-commuting configurations.
- Section 4: the closed form of Lemma 4.1 agrees with the Herglotz integral to  $2 \cdot 10^{-16}$ ; the deficit identity (4.2) holds term by term on cell-embedded fields ( $d = 4$ ,  $M = 2, 3$ ) to  $10^{-8}$ ; the values in Lemma 4.4 at  $\alpha = \beta = \frac{1}{4}$  agree to 40 digits.
- Consistency: Theorems A and B were proved for  $3 \leq d \leq 20$  by exact sum-of-squares certificates in [31], by a method sharing no step with the present one; and gradient ascents of  $F$  over non-commuting families ( $d = 3, \dots, 6$ ,  $M = 2, 3$ ) that reach  $F_{\text{DKZ}}$  end at commuting families, as Theorem B predicts.

11.5. **Formalisation.** The argument of this paper has been formalised in the Lean 4 theorem prover [34] with the mathematical library Mathlib [35], in the versions

Lean 4.33.1,      Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474,

in 98 files of about 33,400 lines (directory `lean/` of [32]). No file uses `sorry`, `admit`, an `axiom` declaration or `native_decide`, and `#print axioms` reports, for every main theorem, only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. Table 3 lists the main theorems (file `Main.lean` and the files it imports, and the file `CglmpNoise.lean`, which imports `Main.lean` and formalises Corollary 9.1). These statements use the internal model of the proof: projective measurements on  $\Phi_D$  for every local dimension  $D$ , the probabilities  $\text{Tr}(A^\top B)/D$ , and the expression  $I_d$  of (1.1).

A second development is a single file in the format of the Formal Conjectures repository (directory `formal-conjectures-27/` of [32]). It states Part B from scratch in the formulation of issue #3444 [33]: the state  $\Phi_d$  of  $\mathbb{C}^d \otimes \mathbb{C}^d$ , complete von Neumann  $d$ -outcome measurements,

the Born rule  $\text{Tr}[\rho(A \otimes B)]$ , the CGLMP functional of the problem page (local bound  $d - 1$ ), the DKZ measurements and white noise. It checks these definitions on special cases (the local bound, no violation in the computational basis, Tsirelson’s bound for  $d = 2$ , and for  $d = 3$  a non-maximally entangled state that beats  $\Phi_3$  [1]), proves that they reduce to the internal model, and proves optimality, uniqueness and the CGLMP noise clause in that formulation, with no hypotheses for  $2 \leq d \leq 20$  and with `Hyp_ConeCertPos_large` for every  $d$ . It builds inside that repository with its continuous-integration flag `--wfail`, and its main theorems depend only on the three standard axioms.

| Lean theorem                                                                                            | Statement                                                                                                                                            | Assumptions                                                                 |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <code>OQP27.maxEntClause_le_twenty</code>                                                               | for every $2 \leq d \leq 20$ : Theorems A and B, and the DKZ strategy attains $I_{\text{ME}}(d)$                                                     | none                                                                        |
| <code>OQP27.maxEntClause_all</code>                                                                     | Theorems A and B for every $d \geq 2$                                                                                                                | $\text{CONE}_d$ for $d \geq 21$ only ( <code>Hyp_ConeCertPos_large</code> ) |
| <code>OQP27.stripInequality,</code><br><code>OQP27.stripEquality</code>                                 | Theorem C for every matrix size $M$ , and its equality case                                                                                          | none                                                                        |
| <code>OQP27.StripL3b.theorem1_bmv2</code>                                                               | Theorem D, for all $(a, t) \in \mathbb{C}^2$                                                                                                         | none                                                                        |
| <code>OQP27.StripL3b.hyp_RI</code>                                                                      | the Radon identity, Theorem B.1                                                                                                                      | none                                                                        |
| <code>OQP27.Red.hyp_reduction</code>                                                                    | the exact reduction: strategy $\rightarrow$ clock model $\rightarrow$ Q-configuration $\rightarrow$ linear form (Theorem 2.1, Lemma 2.5)             | none                                                                        |
| <code>OQP27.Cell.continuumCell</code>                                                                   | the continuum theorem for step fields: Theorem C implies every cell inequality (Proposition 5.3)                                                     | none                                                                        |
| <code>OQP27.ClassB.</code><br><code>classicalTheoremB_all</code>                                        | the classical (commuting) theorem for every $d$ (Theorem 2.2), including the discrete Hilbert-transform rearrangement ( $N = 4d$ , $ A  =  B  = d$ ) | none                                                                        |
| <code>OQP27.coneCertPos_le_twenty</code>                                                                | Theorem E for $2 \leq d \leq 20$ : the certificates of Proposition 6.2 re-evaluated by the Lean kernel in exact rational interval arithmetic         | none                                                                        |
| <code>OQP27.ConeCertificate.</code><br><code>fullCheck_sound</code>                                     | soundness of the certificate checker                                                                                                                 | none                                                                        |
| <code>OQP27.dkz_cglmp</code>                                                                            | the DKZ strategy attains $I_{\text{ME}}(d)$ , for every $d$                                                                                          | none                                                                        |
| <code>OQP27.cglmpOf_noisy,</code><br><code>OQP27.noisy_violation_iff</code>                             | $I_d(vp + (1 - v)p_0) = vI_d(p)$ for every behaviour $p$ ; the CGLMP threshold of a strategy is $2/I_d$                                              | none                                                                        |
| <code>OQP27.noisy_violation_imp_le_twenty,</code><br><code>OQP27.noisy_threshold_rigid_le_twenty</code> | Corollary 9.1 for $2 \leq d \leq 20$                                                                                                                 | none                                                                        |
| <code>OQP27.noisy_violation_imp,</code><br><code>OQP27.noisy_threshold_rigid</code>                     | Corollary 9.1 for every $d \geq 2$                                                                                                                   | $\text{CONE}_d$ for $d \geq 21$ only                                        |

TABLE 3. The main theorems of the Lean formalisation. The last three rows are in the file `CglmpNoise.lean`.

Thus Theorems A and B, in the form stated in Lean, are machine-checked with no hypotheses for  $2 \leq d \leq 20$ . For  $d \geq 21$  the formal proof is complete except for the single input  $\text{CONE}_d$  (`Hyp_`

`ConeCertPos_large`), which is established by the interval-arithmetic computations of Section 6, outside Lean. Theorems C and D are formalised in full, for every matrix size and with no assumption: the identity (1.6) on  $\mathbb{C}^2$ , with  $\rho \geq 0$  measurable, supported in  $K$  and bounded by  $C(\tau(1-\tau))^{-1/2}$ , and the identity (1.5), that is, the strip inequality with its exact defect, together with its equality case. The formal proof of Theorem D follows the distribution-free route of Appendix B: the Radon identity, then Tonelli's theorem and the identity theorem. In the uniqueness step (b) of the proof of Theorem C, the formal proof uses the Fourier inversion theorem of Mathlib in place of Lemma B.7. The continuum theorem is formalised for step fields, with the explicit Herglotz functions of Lemma 4.1. The file `CgImpNoise.lean` defines the CGLMP expression of an arbitrary behaviour and the uniform behaviour  $p_0$ ; besides the theorems of Table 3, `OQP27.dkz_noisy_violation_iff` states that the noisy DKZ behaviour violates the CGLMP inequality exactly when  $vI_{\text{ME}}(d) > 2$ . These theorems, too, depend only on the three standard axioms.

A few auxiliary statements of this paper are proved on paper only, because the formal proof does not need them: the continuity of  $\rho$  (the formal proof uses its measurability, and a bound in Frobenius norm instead of the operator norm), the marginal identity  $\int \rho(s, \tau) ds = \|BgP\|_F^2$  of Theorem D, and the closed formula (1.3) for  $h_\lambda$  (in Lean,  $h_\lambda$  is defined as the Poisson integral of Lemma 3.2(1)). Likewise, the formal rigidity statement is the main assertion of Theorem B:  $d$  divides  $D$ , and a unitary  $u$  maps the measurements to the DKZ measurements tensored with the identity. The uniqueness of  $u$  and the strict inequality for  $d \nmid D$ , which follow from it by the short arguments at the end of Section 7, are not part of the formal statement.

*Remark 11.1* (the finite form of the cone condition). In Lean the cone condition is stated in finite form:  $v = \sum_j \alpha_j u^{\ell_j}$  for finitely many  $\ell_j \in \Delta_d$  and  $\alpha_j \geq 0$ , with  $\alpha_j > 0$  for at least one  $\ell_j \in \Delta_d^+$ . For  $21 \leq d \leq 200$  this is Proposition 6.2. For  $d \geq 201$  it follows from Theorem E. Indeed, the measure  $\mu$  of Propositions 6.6 and 6.8 is a probability measure carried by  $\Delta_d^+$  and  $\ell \mapsto u^\ell$  is continuous, so the barycentre  $\int u^\ell \mu(d\ell)$  lies in the convex hull of  $\{u^\ell : \ell \in \Delta_d^+\}$  (in finite dimension the barycentre of a probability measure carried by a set lies in its convex hull), and by Carathéodory's theorem it is a convex combination of at most  $d$  such vectors. Finally, each  $t_s$  is a positive multiple of  $u^\ell$  for the cell vector  $\ell$  with  $\ell_0 = a$ ,  $\ell_s = d - a$  ( $0 < a < d$ ) and all other cells empty. Indeed, by Lemma 5.1,  $K^\ell(x, y) = 0$  unless  $x$  and  $y$  are congruent to 0 or  $s$  modulo  $d$ , so  $u^\ell$  is supported on  $\{s, d - s\}$ ; its two entries are equal, by the reflections  $\xi \mapsto d + a - \xi$  and  $\xi \mapsto 2d + a - \xi$  of  $\mathbb{T}_N$ , and positive, because they are sums of integrals of  $\kappa(\xi - \eta)$  over cells with  $\xi - \eta \in (0, 2d)$ , where  $\kappa > 0$ .

**Data and code availability.** Certificates, verification programs, logs, the Lean formalisations, the re-verification reports and the notes referred to above are available at [32], <https://github.com/anshM123/IQQI-OQP-27>, under the MIT license. The material for Section 9 is in its directory `clauses/`: the file `LEDGER.md` (the wording of Problem 27 and the status of each clause), `noise-cgImp/` (Corollary 9.1; the Lean file is `lean/OQP27/CgImpNoise.lean`), `noise-literal/` (Theorem 9.2, Propositions 9.9 and 9.11, Remarks 9.10 and 9.12) and `kl-divergence/` (Theorem 9.14, Corollary 9.15 and Proposition 9.16), each with its re-verification reports. The exact certificates for  $d \leq 20$  of [31] are available at <https://github.com/anshM123/CGLMP>.

## APPENDIX A. THE MODEL FOR LARGE $d$

This appendix defines the model  $\Upsilon_e$  of Proposition 6.8 and records the decomposition used in its verification. Throughout  $d \geq 2001$ ,  $\varepsilon = 1/d$ , and for real  $b \in (0, d)$  we write  $x = b/d$  and  $w = 1/b$ .



*Ingredients.* Let  $\kappa_n(b) = \frac{d^n}{db^n}[b\psi(b+1)]$  and  $B_{2j}(b) = b^{2j-1}\kappa_{2j}(b)/(2j-2)!$  (so that  $B_{2j}(b) \rightarrow 1$  as  $b \rightarrow \infty$ ),  $k_j = 2(2j-2)!/(2^j j!)$ . Let  $g_r(x) = \sum_{i \text{ odd}} g_i x^i$  (Lemma 6.3),  $a_n = (-1)^n g_r^{(n)}(1)/n!$ ,  $P_n(x, \varepsilon) = \prod_{k < n} \frac{x+k\varepsilon}{1+k\varepsilon}$  (so  $P_n = \mathbb{E}X_b^n$ ),  $\Gamma(x, \varepsilon) = \sum_{i \text{ odd}} g_i P_i(x, \varepsilon) = \bar{\Xi}_r(b)/d^2$ , and  $S(x, \varepsilon) = \Gamma(x, \varepsilon) - \sum_n a_n P_n(x, \varepsilon)$  (the second sum is  $\bar{\Xi}_r(d-b)/d^2$ , since  $X_{d-b}$  and  $1 - X_b$  have the same law). With  $c_j(b) = [\bar{\Xi}^{(2j)}(b) - \bar{\Xi}^{(2j)}(d-b)]/(2^j j!)$ , the decoupled equation (6.2) reads  $\sum_{j \geq 1} c_j(m) t^j + (\text{remainder}) = \tau_m$ , and in the scaled unknown  $u = t/(\varepsilon m^2)$ , after multiplication by  $\frac{\pi}{2m}$ , its coefficients are

$$\text{Ch}_j(b) = \frac{\pi}{2} \varepsilon^j b^{2j-1} c_j(b) = \varepsilon^{j-1} \left\{ k_j \left[ B_{2j}(b) - \left( \frac{x}{1-x} \right)^{2j-1} B_{2j}(d-b) \right] + \frac{\pi}{2} \frac{x^{2j-1} \partial_x^{2j} S(x, \varepsilon)}{2^j j!} \right\}.$$

Let  $p = g_r - \frac{\varepsilon^2}{12} g_r'' + \frac{\varepsilon^4}{240} g_r''''$  (an Euler–Maclaurin interpolant),  $\tilde{P}(b) = d^2[p(x) - x(p(1) - g_r(1))]$ ,  $\tau_e(b) = [\tilde{P} - \bar{\Xi}_r](b) - [\tilde{P} - \bar{\Xi}_r](d-b)$  and  $\hat{\tau}_e = \frac{\pi}{2} \tau_e/b$ .

*The model.*  $u_e(b)$  is the root in  $(0, 1)$  of  $\sum_{j=1}^4 \text{Ch}_j(b) u^j = \hat{\tau}_e(b)$ , and  $\Upsilon_e(b) = \varepsilon b^2 u_e(b)$ ;  $\Upsilon_e(0) = 0$  and  $\Upsilon_e(d-b) = \Upsilon_e(b)$ . The scan of `a10_pointwise.py` certifies that this root exists and is unique, so  $u_e$  is real-analytic; with the exact discrete targets and Gaussian order 7 the model reproduces the certified solutions at  $d = 1000$  to relative accuracy  $10^{-18}$  (numerical check).

*Decomposition.* Expanding  $u_e = U_0(w) + xU_1(w) + x^2 R_2(x, w)$  in  $x$  at fixed  $w$ , one has  $b^2 U_0 = A_0 b^2/B(b)$  and  $b^3 U_1 = \varphi_2(b)$  for an explicit function  $\varphi_2$ , so that

$$\Upsilon_e(b) = \varepsilon A_0 b^2 + A_0 \varepsilon f(b) + \varepsilon^2 \varphi_2(b) + \varepsilon^3 b^4 R_2(x, w),$$

and, since  $\Delta^4$  annihilates quadratic polynomials,  $D_B[\Upsilon_e](m) = \varepsilon^2 A_2(m) + \varepsilon^3 \Delta^4[b^4 R_2](m) - A_0 \varepsilon \Delta^4[f(d-\cdot)](m)$  with  $A_2 = \Delta^4 \varphi_2$ . For  $m \geq 3$ ,  $\Delta^4 h(m) = \int_{-2}^2 h''''(m+s) M_4(s) ds$  with the cardinal B-spline  $M_4 \geq 0$  of mass 1, and  $\frac{d^4}{db^4}[b^4 R_2] = \Lambda(R_2)$  with  $\Lambda = (\vartheta+1)(\vartheta+2)(\vartheta+3)(\vartheta+4)$ ,  $\vartheta = x\partial_x - w\partial_w$ . In the *inner zone* ( $x \leq 0.013$ ) the certified bounds are  $A_2(m) \geq 0$  for  $m \geq 2$  and  $\Lambda(R_2) \geq 6.022$ ; in the *outer zone* ( $x \in [0.011, 0.501]$ ) one uses instead  $\varepsilon^{-3} \Delta^4[\Upsilon_e - \varepsilon A_0 b^2/B](m) = \int Q_O(x_s, \varepsilon) M_4(s) ds$  with  $Q_O = \partial_x^4[x^2(u_e - A_0/B(1/w))]$  at fixed  $\varepsilon$ , certified  $\geq 7.419$ . Since the stencil of  $\Delta^4$  has width  $4/d \leq 0.002$ , every  $m$  is covered by one of the two zones;  $m = 1, 2$  are treated directly. All enclosures are Taylor models in  $(x, \varepsilon)$  or  $(w, \varepsilon)$  with  $\varepsilon \in [0, 1/2001]$ : thin centre jets of the model root (Krawczyk iteration, then implicit differentiation degree by degree), Lagrange remainders from jets over the full box, special functions (polygamma and Binet functions, Pochhammer sums with rigorous tails) by interval series. For boxes with  $x > 0.42$  the exactly antisymmetric coefficients are divided by  $1 - 2x$  before enclosing (they vanish at  $x = \frac{1}{2}$ ).

*Pointwise lemma and slack.* With the exact targets  $\tau_m$  and the Gaussian expansion to order 7 plus remainder, `a10_pointwise.py` bounds the difference between the decoupled solution and the model by  $6.6468 \cdot 10^{-4} \varepsilon^2$ , using analytic bounds for the Euler–Maclaurin defect of  $\tilde{P}$  (at most  $235.6 \varepsilon^6$  for its second differences), for the Gaussian remainder, and for the higher coefficients  $\text{Ch}_5, \dots, \text{Ch}_7$ . For the slack, let  $\text{Dl}_e = \tilde{P} - \bar{\Xi}_r$  (the model of  $\text{Dl}$ , by Lemma 6.3) and  $W_e(b) = \text{Dl}_e(b) - \sum_{j=1}^4 \Upsilon_e(b)^j \bar{\Xi}^{(2j)}(b)/(2^j j!)$ , which is symmetric under  $b \mapsto d-b$  by the model equation. Its leading term is exactly linear,  $-(1 - \frac{2}{\pi})b$ , so  $\Delta^2 W_e = O(\varepsilon)$ ; the density  $\Delta^2 W_e/\varepsilon$  and the perturbation from  $W_e$  to  $W$  are certified as stated in Section 6.6.

## APPENDIX B. A DISTRIBUTION-FREE PROOF OF THEOREM D

The proof of Theorem D in Section 3.5 uses the Paley–Wiener–Schwartz theorem, Fourier inversion along lines and the structure of distributions supported on a line, and the proof of Theorem C in Section 3.6 uses the uniqueness of Fourier transforms. Here we give a second



proof of both theorems, which uses no distributions. Its tools are linear algebra, the Cauchy integral formula on a circle and Hurwitz's theorem, differentiation under the integral sign, dominated convergence, Weierstrass' theorem on locally uniform limits of holomorphic functions, the theorems of Fubini and Tonelli, the identity theorem in one variable, the Weierstrass approximation theorem, and a Gaussian approximate identity in one variable (Lemma B.7). Its key step is a Radon identity for the pencil (Theorem B.1), which we prove through a complex inviscid Burgers equation satisfied by the roots of the pencil. This is the route followed by the Lean formalisation (Section 11.5).

**B.1. The Radon identity.** Throughout this appendix  $H$  is a Hermitian  $M \times M$  matrix,  $B$  an orthogonal projection of rank  $r$ ,  $P = \mathbf{1} - B$ , and  $H_d := BHB + PHP$ . Let  $\lambda_1, \dots, \lambda_M$  be the eigenvalues of  $H$  and  $\lambda_1^0, \dots, \lambda_M^0$  those of  $H_d$ , which consist of the eigenvalues  $\mu_1, \dots, \mu_{M-r}$  of  $PHP|_{\text{ran } P}$  and  $\beta_1, \dots, \beta_r$  of  $BHB|_{\text{ran } B}$ ; note that  $\text{Tr } H = \text{Tr } H_d$ . We write  $\mathbb{C}_\pm = \{z : \pm \text{Im } z > 0\}$ , and  $\text{Log}$  is the principal logarithm on  $\mathbb{C} \setminus (-\infty, 0]$ . For  $0 < \tau < 1$  and  $c \in \mathbb{C}$  put

$$p(y; \tau, c) = \det(H + c - y(P - \tau)), \quad Y(\tau, c) = (P - \tau)^{-1}(H + c) = \left(\frac{P}{1 - \tau} - \frac{B}{\tau}\right)(H + c).$$

Since  $H + c - y(P - \tau) = (P - \tau)(Y(\tau, c) - y)$ , the roots of  $p(\cdot; \tau, c)$ , with multiplicity, are the eigenvalues of  $Y(\tau, c)$ , and the leading coefficient  $\det(\tau - P) = (\tau - 1)^{M-r} \tau^r$  does not depend on  $c$ . Let  $y_1(\tau), \dots, y_M(\tau)$  be the roots for  $c = 0$ , and put

$$\mathsf{L}(H) = \frac{1}{2\pi} \int_0^1 \sum_{i=1}^M |\text{Im } y_i(\tau)| d\tau, \quad \mathsf{R}(H) = \text{Tr } H_+ - \text{Tr } (H_d)_+.$$

**Theorem B.1** (Radon identity). *The function  $\tau \mapsto \sum_i |\text{Im } y_i(\tau)|$  is continuous on  $(0, 1)$  and bounded by  $M\|H\|/\sqrt{\tau(1-\tau)}$ , and  $\mathsf{L}(H) = \mathsf{R}(H)$ .*

If  $r \in \{0, M\}$ , then  $P - \tau$  is a nonzero multiple of  $\mathbf{1}$ , all roots are real and  $H_d = H$ , so both sides vanish. We assume  $1 \leq r \leq M - 1$  from now on. Applied to  $H = g - \xi P - w$ , Theorem B.1 says that the integrals of the density  $\rho$  of Theorem D along the lines  $s = w + \xi\tau$  are the real slices  $U_\xi(w)$  of Lemma 3.5 (Section B.4); this explains the name.

**B.2. The complexified pencil.** In this subsection  $\text{Im } c > 0$  and  $0 < \tau < 1$ , and we put  $\Gamma_c = \|H\| + |c|$ . For a root  $y$  of  $p(\cdot; \tau, c)$  we fix a unit vector  $v$  with  $(H + c)v = y(P - \tau)v$ , and we put  $a = \langle v, (P - \tau)v \rangle = \|Pv\|^2 - \tau \in [-\tau, 1 - \tau]$ .

**Lemma B.2** (location and count). (1)  $\text{Im } c = a \text{Im } y$ . Hence  $a \neq 0$ ,  $y \in \mathbb{C}_+$  if and only if  $a > 0$ ,  $\text{Im } y \geq \text{Im } c/(1 - \tau)$  if  $y \in \mathbb{C}_+$ , and  $\text{Im } y \leq -\text{Im } c/\tau$  if  $y \in \mathbb{C}_-$ .  
(2) No root of  $p(\cdot; \tau, c)$  is real, and exactly  $M - r$  roots, counted with multiplicity, lie in  $\mathbb{C}_+$ .  
(3) For  $H_d$  in place of  $H$  the roots are the numbers  $(\mu_k + c)/(1 - \tau) \in \mathbb{C}_+$  and  $-(\beta_j + c)/\tau \in \mathbb{C}_-$ .

*Proof.* (1)  $\langle v, (H + c)v \rangle = \langle v, Hv \rangle + c$  and  $\langle v, (H + c)v \rangle = ya$ , where  $\langle v, Hv \rangle$  and  $a$  are real; taking imaginary parts gives  $\text{Im } c = a \text{Im } y$ . As  $\text{Im } c > 0$ ,  $a \neq 0$  and  $\text{Im } y = \text{Im } c/a$ , with  $0 < a \leq 1 - \tau$  or  $-\tau \leq a < 0$ . (2) No root is real by (1). For  $\sigma \in [0, 1]$ , part (1) applies to the Hermitian matrix  $\sigma H$ , so no root of  $\det(\sigma H + c - y(P - \tau))$  is real. The leading coefficient does not depend on  $\sigma$ , so the roots, as an unordered  $M$ -tuple, depend continuously on  $\sigma$ , and the number of roots in  $\mathbb{C}_+$  is constant on  $[0, 1]$ . At  $\sigma = 0$  the polynomial is  $(c - y(1 - \tau))^{M-r}(c + y\tau)^r$ , with the root  $c/(1 - \tau) \in \mathbb{C}_+$  of multiplicity  $M - r$  and the root  $-c/\tau \in \mathbb{C}_-$  of multiplicity  $r$ . (3)  $H_d + c - y(P - \tau)$  is block diagonal, equal to  $PHP + c - y(1 - \tau)$  on  $\text{ran } P$  and to  $BHB + c + y\tau$  on  $\text{ran } B$ .  $\square$

**Lemma B.3** (bounds). (1) Every root satisfies  $|y| \leq \Gamma_c / \min(\tau, 1 - \tau)$ .

- (2) If  $y \in \mathbb{C}_+$  and  $\tau \leq \frac{1}{2}$ , or  $y \in \mathbb{C}_-$  and  $\tau \geq \frac{1}{2}$ , then  $|y| \leq \Gamma_c / \sqrt{\tau(1-\tau)}$ .  
(3) Let  $\Sigma_\pm(\tau, c)$  be the sums of the roots in  $\mathbb{C}_\pm$ , and  $\Sigma_+^0(\tau, c) = (\text{Tr}_P(PHP) + (M-r)c)/(1-\tau)$ ,  $\Sigma_-^0(\tau, c) = -(\text{Tr}_B(BHB) + rc)/\tau$  the corresponding sums for  $H_d$ . Then  $\Sigma_+ + \Sigma_- = \Sigma_+^0 + \Sigma_-^0$ , and

$$|\Sigma_+(\tau, c) - \Sigma_+^0(\tau, c)| \leq M\Gamma_c(2 + (\tau(1-\tau))^{-1/2}).$$

*Proof.* We have  $|y|(P-\tau)v = (H+c)v \leq \Gamma_c$ . (1)  $|(P-\tau)v|^2 = (1-\tau)^2|Pv|^2 + \tau^2|Bv|^2 \geq \min(\tau, 1-\tau)^2$ . (2) With  $|Pv|^2 = a + \tau$ ,  $|(P-\tau)v|^2 = (1-\tau)^2(a+\tau) + \tau^2(1-\tau-a) = \tau(1-\tau) + a(1-2\tau)$ , and  $a(1-2\tau) \geq 0$  in both cases by Lemma B.2(1). (3)  $\Sigma_+ + \Sigma_- = \text{Tr } Y(\tau, c)$ , and  $\text{Tr}[(\frac{P}{1-\tau} - \frac{B}{\tau})(H - H_d)] = \text{Tr}[(\frac{P}{1-\tau} - \frac{B}{\tau})(PHB + BHP)] = 0$ , so  $\text{Tr } Y(\tau, c)$  is the same for  $H$  and for  $H_d$ . For  $\tau \leq \frac{1}{2}$ , part (2) gives  $|\Sigma_+| \leq (M-r)\Gamma_c/\sqrt{\tau(1-\tau)}$ , and  $|\Sigma_+^0| \leq (M-r)\Gamma_c/(1-\tau) \leq 2(M-r)\Gamma_c$ . For  $\tau \geq \frac{1}{2}$  write  $\Sigma_+ - \Sigma_+^0 = \Sigma_-^0 - \Sigma_-$  and use part (2) for the roots in  $\mathbb{C}_-$  and  $|\Sigma_-^0| \leq r\Gamma_c/\tau \leq 2r\Gamma_c$ .  $\square$

Let  $\mathcal{L}_+(\tau, c) = \sum \text{Log } y$ , the sum over the roots  $y$  in  $\mathbb{C}_+$  with multiplicity, and  $\mathcal{L}_+^0(\tau, c) = \sum_k \text{Log } \frac{\mu_k + c}{1-\tau}$ , the same sum for  $H_d$ .

**Lemma B.4** (the Burgers identity). *On  $(0, 1) \times \mathbb{C}_+$  the functions  $\Sigma_+$  and  $\mathcal{L}_+$  are continuously differentiable in  $\tau$  and holomorphic in  $c$ , their partial derivatives are jointly continuous, and*

$$\partial_\tau \mathcal{L}_+ = \partial_c \Sigma_+, \quad \partial_\tau \mathcal{L}_+^0 = \partial_c \Sigma_+^0 = \frac{M-r}{1-\tau}.$$

*Proof.* Fix  $(\tau_0, c_0) \in (0, 1) \times \mathbb{C}_+$ , put  $\eta = \frac{1}{2} \text{Im } c_0$  and  $R_0 = 2(\|H\| + |c_0| + 1)/\min(\tau_0, 1-\tau_0)$ , and let  $U$  be a neighbourhood of  $(\tau_0, c_0)$  on which  $\text{Im } c \geq \eta$ ,  $|c| \leq |c_0| + 1$  and  $\min(\tau, 1-\tau) \geq \frac{1}{2} \min(\tau_0, 1-\tau_0)$ . By Lemmas B.2(1) and B.3(1), for  $(\tau, c) \in U$  the roots in  $\mathbb{C}_+$  lie in  $Q = \{y : \text{Im } y \geq \eta, |y| \leq R_0\}$ . Let  $\Omega$  be the open disc with centre  $iT$  and radius  $T - \frac{\eta}{2}$ , where  $T = R_0^2/\eta + 1$ . Its closure lies in  $\{\text{Im } y \geq \frac{\eta}{2}\} \subseteq \mathbb{C}_+$ , and  $Q \subseteq \Omega$ : for  $y = u + iw \in Q$ ,  $|y - iT|^2 = |y|^2 - 2wT + T^2 \leq R_0^2 - 2\eta T + T^2 < (T - \frac{\eta}{2})^2$ , because  $\eta T > R_0^2$ . So for  $(\tau, c) \in U$  the roots in  $\mathbb{C}_+$  lie in  $\Omega$ , the roots in  $\mathbb{C}_-$  lie outside  $\bar{\Omega}$ , and  $p(\cdot; \tau, c)$  has no zero on  $\partial\Omega$ . Writing  $p = \det(\tau - P) \prod_i (y - y_i)$ , so that  $p_y/p = \sum_i (y - y_i)^{-1}$ , the Cauchy integral formula gives, for every  $\varphi$  holomorphic near  $\bar{\Omega}$ ,

$$(B.1) \quad \sum_{y_i \in \mathbb{C}_+} \varphi(y_i) = \frac{1}{2\pi i} \oint_{\partial\Omega} \varphi(y) \frac{p_y(y; \tau, c)}{p(y; \tau, c)} dy.$$

We take  $\varphi(y) = y$  and  $\varphi = \text{Log}$ . The integrand is a smooth function of  $(y, \tau, \text{Re } c, \text{Im } c)$  on  $\partial\Omega \times U$  and is holomorphic in  $c$ , so differentiation under the integral sign gives the regularity statements. Next,  $p(y; \tau, c) = f(y, c + y\tau)$  with the polynomial  $f(y, w') = \det(H + w' - yP)$ , so  $p_\tau = yp_c$ . With  $q = p_c/p$  we get  $\partial_\tau(p_y/p) = \partial_y(p_\tau/p) = \partial_y(yq)$  and  $\partial_c(p_y/p) = \partial_y q$ , hence

$$\partial_\tau \mathcal{L}_+ - \partial_c \Sigma_+ = \frac{1}{2\pi i} \oint_{\partial\Omega} [\text{Log}(y) \partial_y(yq) - y \partial_y q] dy = \frac{1}{2\pi i} \oint_{\partial\Omega} \partial_y [(y \text{Log } y - y) q] dy = 0,$$

because  $\text{Log}(y)(q + yq_y) - yq_y = \partial_y [(y \text{Log } y - y) q]$  is the derivative of a function that is holomorphic near  $\partial\Omega$ . For  $H_d$  the identity follows from the explicit formulas  $\mathcal{L}_+^0 = \sum_k \text{Log}(\mu_k + c) - (M-r) \log(1-\tau)$  and  $\Sigma_+^0 = (\sum_k \mu_k + (M-r)c)/(1-\tau)$ .  $\square$

For a simple root  $y(\tau, c)$ , formula (B.1) on a small circle around  $y$  (in place of  $\partial\Omega$ ) reduces Lemma B.4 to  $\partial_\tau y = y \partial_c y$ , the complex inviscid Burgers equation; the lemma is the form of this equation, summed over the roots in  $\mathbb{C}_+$ , that remains valid at multiple roots.

**Lemma B.5** (the two ends). *Fix  $c \in \mathbb{C}_+$ .*

- (1) As  $\tau \rightarrow 1^-$ ,  $\mathcal{L}_+(\tau, c) - \mathcal{L}_+^0(\tau, c) \rightarrow 0$ .  
 (2) Let  $T_0 = PHP + c - PHB(BHB + c)^{-1}BHP$  on  $\text{ran } P$  (the matrix  $BHB + c$  is invertible on  $\text{ran } B$  since  $\text{Im } c > 0$ ). Then  $\text{spec } T_0 \subseteq \mathbb{C}_+$ , and as  $\tau \rightarrow 0^+$ ,

$$\mathcal{L}_+(\tau, c) - \mathcal{L}_+^0(\tau, c) \rightarrow \theta_0(c) := \sum_{t \in \text{spec } T_0} \text{Log } t - \sum_k \text{Log}(\mu_k + c), \quad e^{\theta_0(c)} = \frac{\det(H + c)}{\det(H_d + c)}.$$

*Proof.* (1)  $(1 - \tau)Y(\tau, c) = (P - \frac{1-\tau}{\tau}B)(H + c) \rightarrow P(H + c)$  as  $\tau \rightarrow 1$ . With respect to  $\text{ran } P \oplus \text{ran } B$ ,  $P(H + c)$  is block upper triangular with diagonal blocks  $PHP + c$  and 0, so its eigenvalues are the numbers  $\mu_k + c \in \mathbb{C}_+$  and 0 ( $r$  times). By continuity of eigenvalues, for  $\tau$  near 1 exactly  $M - r$  eigenvalues  $z_k$  of  $(1 - \tau)Y$  lie near the  $\mu_k + c$ , hence in  $\mathbb{C}_+$ , labelled so that  $z_k \rightarrow \mu_k + c$ , and  $r$  lie near 0. Multiplication by  $1/(1 - \tau) > 0$  preserves  $\mathbb{C}_+$ , and  $Y$  has exactly  $M - r$  eigenvalues in  $\mathbb{C}_+$  (Lemma B.2(2)); so the roots in  $\mathbb{C}_+$  are the  $z_k/(1 - \tau)$ , and since  $\text{Log}(z/(1 - \tau)) = \text{Log } z - \log(1 - \tau)$  for  $z \in \mathbb{C}_+$ ,  $\mathcal{L}_+ - \mathcal{L}_+^0 = \sum_k [\text{Log } z_k - \log(\mu_k + c)] \rightarrow 0$ .

(2) By the Schur complement formula with respect to  $\text{ran } P \oplus \text{ran } B$ ,  $p_0(y) := \det(H + c - yP) = \det_B(BHB + c) \det_P(T_0 - y)$ , a polynomial of exact degree  $M - r$  whose roots form  $\text{spec } T_0$ . For real  $y$  the matrix  $H - yP$  is Hermitian and  $\text{Im } c > 0$ , so  $p_0(y) \neq 0$ :  $\text{spec } T_0$  contains no real number. As  $\tau \rightarrow 0$ ,  $p(y; \tau, c) = \det(H + c - yP + y\tau) \rightarrow p_0(y)$  coefficientwise. By Hurwitz's theorem, applied on a circle that encloses  $\text{spec } T_0$  and on small circles around its points, for small  $\tau$  exactly  $M - r$  roots of  $p(\cdot; \tau, c)$  lie in a fixed disc, and they converge to the points of  $\text{spec } T_0$  with multiplicity. The other  $r$  roots leave every bounded set:  $\tau Y(\tau, c) = (\frac{\tau}{1-\tau}P - B)(H + c) \rightarrow -B(H + c)$ , whose eigenvalues are 0 ( $M - r$  times) and the numbers  $-(\beta_j + c)$ , which lie in  $\mathbb{C}_-$  at distance at least  $\text{Im } c$  from  $\mathbb{R}$ ; so  $r$  eigenvalues of  $Y$  are of the form  $-(\beta_j + c + o(1))/\tau$  and lie in  $\mathbb{C}_-$  for small  $\tau$ . Since  $\mathbb{C}_-$  contains exactly  $r$  roots, the  $M - r$  bounded roots are those in  $\mathbb{C}_+$ . Their limits lie in the closure of  $\mathbb{C}_+$  and are not real, so  $\text{spec } T_0 \subseteq \mathbb{C}_+$  and, by continuity of  $\text{Log}$  on  $\mathbb{C}_+$ ,  $\mathcal{L}_+ \rightarrow \sum_{t \in \text{spec } T_0} \text{Log } t$ . Also  $\mathcal{L}_+^0 = \sum_k \text{Log}(\mu_k + c) - (M - r) \log(1 - \tau) \rightarrow \sum_k \text{Log}(\mu_k + c)$ . Finally  $e^{\theta_0} = \det_P T_0 / \det_P(PHP + c) = \det(H + c) / (\det_B(BHB + c) \det_P(PHP + c)) = \det(H + c) / \det(H_d + c)$ .  $\square$

**Lemma B.6** (boundary values). *Let  $c_0 \in \mathbb{R}$ ,  $0 < \tau < 1$ , and let  $y_1, \dots, y_M$  be the roots of  $p(\cdot; \tau, c_0)$ . As  $c \rightarrow c_0$  within  $\mathbb{C}_+$ ,  $\text{Im } \Sigma_+(\tau, c) \rightarrow \frac{1}{2} \sum_i |\text{Im } y_i|$  and  $\text{Im } \Sigma_+^0(\tau, c) = (M - r) \text{Im } c / (1 - \tau) \rightarrow 0$ . If  $H + c_0 > 0$ , all  $y_i$  are real.*

*Proof.* For real  $y$ ,  $p(y; \tau, c_0)$  is the determinant of a Hermitian matrix, hence real; so the non-real roots of  $p(\cdot; \tau, c_0)$  come in conjugate pairs. The roots depend continuously on  $c$ , since the leading coefficient does not. If  $z$  is a root of  $p(\cdot; \tau, c_0)$  of multiplicity  $m_z$ , then for  $c$  near  $c_0$  exactly  $m_z$  roots lie near  $z$ ; they lie in  $\mathbb{C}_+$  if  $z \in \mathbb{C}_+$ , in  $\mathbb{C}_-$  if  $z \in \mathbb{C}_-$ , and their imaginary parts tend to 0 if  $z$  is real. Summing over the roots in  $\mathbb{C}_+$  gives  $\text{Im } \Sigma_+(\tau, c) \rightarrow \sum_{\text{Im } z > 0} m_z \text{Im } z = \frac{1}{2} \sum_i |\text{Im } y_i|$ . If  $A = H + c_0 > 0$  and  $y$  is a non-real root with eigenvector  $v$ , then  $\langle v, Av \rangle = y \langle v, (P - \tau)v \rangle$  with both inner products real, which forces  $\langle v, (P - \tau)v \rangle = 0 = \langle v, Av \rangle$ , a contradiction.  $\square$

**B.3. Proof of Theorem B.1.** *Step 1.* For  $c \in \mathbb{C}_+$  let  $\Psi(c) = \int_0^1 [\Sigma_+(\tau, c) - \Sigma_+^0(\tau, c)] d\tau$ ; by Lemma B.3(3) the integral converges absolutely, uniformly for  $c$  in compact subsets of  $\mathbb{C}_+$ . For  $0 < \delta < \frac{1}{2}$  the function  $\Psi_\delta(c) = \int_\delta^{1-\delta} (\Sigma_+ - \Sigma_+^0) d\tau$  is holomorphic on  $\mathbb{C}_+$  (Lemma B.4 and differentiation under the integral sign), and by Lemma B.4 and the fundamental theorem of calculus,

$$\Psi'_\delta(c) = \int_\delta^{1-\delta} \partial_c(\Sigma_+ - \Sigma_+^0) d\tau = \int_\delta^{1-\delta} \partial_\tau(\mathcal{L}_+ - \mathcal{L}_+^0) d\tau = (\mathcal{L}_+ - \mathcal{L}_+^0)(1 - \delta, c) - (\mathcal{L}_+ - \mathcal{L}_+^0)(\delta, c).$$

As  $\delta \rightarrow 0$ ,  $\Psi_\delta \rightarrow \Psi$  locally uniformly on  $\mathbb{C}_+$ , so  $\Psi$  is holomorphic and  $\Psi'_\delta \rightarrow \Psi'$  (Weierstrass); by Lemma B.5,  $\Psi'(c) = -\theta_0(c)$ .

*Step 2.* The function  $\theta_1(c) = \sum_j \text{Log}(\lambda_j + c) - \sum_j \text{Log}(\lambda_j^0 + c)$  is holomorphic on  $\mathbb{C}_+$  and  $e^{\theta_1} = \det(H + c)/\det(H_d + c) = e^{\theta_0}$ . Hence  $\theta_0 - \theta_1$  is continuous with values in  $2\pi i\mathbb{Z}$ , so it equals a constant  $2\pi i k$  on the connected set  $\mathbb{C}_+$ . Let

$$\mathcal{E}(c) = \text{Tr}[(H_d + c) \text{Log}(H_d + c)] - \text{Tr}[(H + c) \text{Log}(H + c)],$$

that is,  $\mathcal{E}(c) = \sum_j (\lambda_j^0 + c) \text{Log}(\lambda_j^0 + c) - \sum_j (\lambda_j + c) \text{Log}(\lambda_j + c)$ . Since  $\frac{d}{dc}[(\lambda + c) \text{Log}(\lambda + c)] = \text{Log}(\lambda + c) + 1$  and both spectra have  $M$  elements,  $\mathcal{E}' = -\theta_1$ . Hence  $(\Psi - \mathcal{E})' = -\theta_0 + \theta_1 = -2\pi i k$ , and  $\Psi(c) = \mathcal{E}(c) - 2\pi i k c + \kappa_0$  on  $\mathbb{C}_+$  for a constant  $\kappa_0 \in \mathbb{C}$ .

*Step 3.* Fix  $c_0 \in \mathbb{R}$  and let  $\epsilon \rightarrow 0^+$ . By Lemma B.6, for every  $\tau$ ,  $\text{Im}(\Sigma_+ - \Sigma_+^0)(\tau, c_0 + i\epsilon) \rightarrow \frac{1}{2} \sum_i |\text{Im} y_i(\tau; c_0)|$ , where the  $y_i(\tau; c_0)$  are the roots of  $p(\cdot; \tau, c_0)$ , that is, the roots of Theorem B.1 for  $H + c_0$  in place of  $H$ ; and by Lemma B.3(3) the integrand is dominated by  $M(\|H\| + |c_0| + 1)(2 + (\tau(1 - \tau))^{-1/2})$  for  $\epsilon \leq 1$ . By dominated convergence,  $\text{Im} \Psi(c_0 + i\epsilon) \rightarrow \pi \mathbf{L}(H + c_0)$ . On the other side, as  $z \rightarrow t \in \mathbb{R}$  within  $\mathbb{C}_+$ ,  $z \text{Log} z \rightarrow t \log |t| + i\pi \min(t, 0)$  (with limit 0 at  $t = 0$ ). Since  $\min(t, 0) = t - t_+$  and  $\sum_j \lambda_j = \sum_j \lambda_j^0$ ,

$$\begin{aligned} \text{Im} \mathcal{E}(c_0 + i\epsilon) &\rightarrow \pi \left[ \sum_j \min(\lambda_j^0 + c_0, 0) - \sum_j \min(\lambda_j + c_0, 0) \right] \\ &= \pi \left[ \sum_j (\lambda_j + c_0)_+ - \sum_j (\lambda_j^0 + c_0)_+ \right] = \pi \mathbf{R}(H + c_0). \end{aligned}$$

Taking imaginary parts in  $\Psi = \mathcal{E} - 2\pi i k c + \kappa_0$  gives  $\mathbf{L}(H + c_0) = \mathbf{R}(H + c_0) - 2k c_0 + \text{Im} \kappa_0 / \pi$  for every real  $c_0$ .

*Step 4.* For  $c_0 > \|H\|$  the matrices  $H + c_0$  and  $H_d + c_0$  are positive definite ( $\|H_d\| \leq \|H\|$ ). By Lemma B.6 all roots are then real, so  $\mathbf{L}(H + c_0) = 0$ , and  $\mathbf{R}(H + c_0) = \text{Tr}(H + c_0) - \text{Tr}(H_d + c_0) = 0$ . Hence  $-2k c_0 + \text{Im} \kappa_0 / \pi = 0$  for all  $c_0 > \|H\|$ , so  $k = 0$  and  $\text{Im} \kappa_0 = 0$ , and  $\mathbf{L}(H + c_0) = \mathbf{R}(H + c_0)$  for every real  $c_0$ ; the case  $c_0 = 0$  is the identity. Finally,  $\tau \mapsto \sum_i |\text{Im} y_i(\tau)|$  is continuous on  $(0, 1)$  by continuity of the roots, and every non-real root satisfies  $|y| \leq \|H\| / \sqrt{\tau(1 - \tau)}$ , by the argument of Lemma 3.3(3).  $\square$

One can show in addition that  $\kappa_0 = 0$ , that is,  $\Psi = \mathcal{E}$  on  $\mathbb{C}_+$ ; we do not need this.

**B.4. Theorem D from the Radon identity.** Let  $g$  be Hermitian, let  $\rho$  and the roots  $\xi_i(s, \tau)$  be as in Theorem D, and let  $\lambda_j(\xi)$ ,  $\lambda_j^0(\xi)$  and  $U_\xi$  be as in Lemma 3.5. The properties of  $\rho$  stated in Theorem D are those of Lemma 3.3, whose proof is elementary. Fix  $\xi, w \in \mathbb{R}$  and put  $H = g - \xi P - w$ . For  $s = w + \xi \tau$ ,

$$g - s - x(P - \tau) = H - (x - \xi)(P - \tau),$$

so the roots of  $p_{s, \tau}$  are the numbers  $\xi + y_i(\tau)$ , with  $y_i(\tau)$  the roots of Theorem B.1 for this  $H$ , and  $\rho(w + \xi \tau, \tau) = \frac{1}{2\pi} \sum_i |\text{Im} y_i(\tau)|$ . Moreover  $H_d = g_d - \xi P - w$ . Theorem B.1 therefore gives

$$(B.2) \quad \int_0^1 \rho(w + \xi \tau, \tau) d\tau = \sum_j (\lambda_j(\xi) - w)_+ - \sum_j (\lambda_j^0(\xi) - w)_+ = U_\xi(w),$$

the last equality because  $(l - w)_+ - (w - l)_+ = l - w$  and  $\sum_j \lambda_j(\xi) = \sum_j \lambda_j^0(\xi)$ . Thus the integrals of  $\rho$  along all non-horizontal lines are the real slices  $U_\xi$ .

For real  $a > 0$ ,  $\int_{\mathbb{R}} e^{aw} (l - w)_+ dw = e^{al}/a^2$ , and for real  $a < 0$ ,  $\int_{\mathbb{R}} e^{aw} (w - l)_+ dw = e^{al}/a^2$ . Using the middle expression in (B.2) for  $a > 0$  and the definition of  $U_\xi$  for  $a < 0$ , we get, for real  $a \neq 0$ ,

$$\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = a^{-2} \sum_j (e^{a\lambda_j(\xi)} - e^{a\lambda_j^0(\xi)}) = a^{-2} \mathcal{D}(a, a\xi),$$

as in Lemma 3.5(2). Since  $\rho \geq 0$ , Tonelli's theorem and the substitution  $s = w + \xi\tau$  at fixed  $\tau$  give

$$\int_{\mathbb{R}} e^{aw} U_{\xi}(w) dw = \int_0^1 \int_{\mathbb{R}} e^{aw} \rho(w + \xi\tau, \tau) dw d\tau = \int_0^1 \int_{\mathbb{R}} e^{as - a\xi\tau} \rho(s, \tau) ds d\tau.$$

Every real  $t$  is of the form  $a\xi$ , so  $\mathcal{G}(a, t) = \mathcal{D}(a, t)/a^2$  equals  $I(a, t) := \iint e^{as - t\tau} \rho(s, \tau) ds d\tau$  for all real  $a \neq 0$  and all real  $t$ . Both functions are entire on  $\mathbb{C}^2$ :  $\mathcal{G}$  by Step 0 of Section 3.5, and  $I$  because  $\rho$  is integrable with compact support. For fixed real  $t$ ,  $a \mapsto \mathcal{G}(a, t) - I(a, t)$  is entire and vanishes on  $\mathbb{R} \setminus \{0\}$ , hence identically; then for fixed  $a \in \mathbb{C}$ ,  $t \mapsto \mathcal{G}(a, t) - I(a, t)$  is entire and vanishes on  $\mathbb{R}$ , hence identically. This is (1.6). For the last statement of Theorem D, put  $m(\tau) = \int_{\mathbb{R}} \rho(s, \tau) ds$ , which is continuous on  $(0, 1)$  and integrable. Since  $\mathcal{G} = I$  on  $\mathbb{C}^2$ , Step 4 of Section 3.5 gives  $\int_0^1 e^{-t\tau} m(\tau) d\tau = I(0, t) = \mathcal{G}(0, t) = \|BgP\|_F^2 \int_0^1 e^{-t\tau} d\tau$ , that is,  $\int_0^1 e^{-t\tau} (m(\tau) - \|BgP\|_F^2) d\tau = 0$  for every real  $t$ . Taking  $t = 0, 1, 2, \dots$  and using that the polynomials in  $e^{-\tau}$  are dense in  $C[0, 1]$  (Weierstrass),  $\int_0^1 \varphi(m - \|BgP\|_F^2) = 0$  for every continuous  $\varphi$ ; so  $m = \|BgP\|_F^2$  almost everywhere, hence everywhere on  $(0, 1)$  by continuity.  $\square$

**B.5. Theorem C without distributions.** The proof of Theorem C in Section 3.6 uses Theorem D at  $t = \mp ia$ , Lemmas 3.1 and 3.2 (residues, geometric series, and the fact that two harmonic functions that agree on an open subset of the strip agree everywhere), and Tonelli's theorem. Only step (b) uses distributions and the uniqueness of Fourier transforms; we replace it by the following lemma.

**Lemma B.7.** *Let  $Z: \mathbb{R} \rightarrow \mathbb{C}$  be continuous with  $|Z(w)| \leq Ce^{-c|w|}$  for some  $c > 0$ . If  $\int_{\mathbb{R}} e^{aw} Z(w) dw = 0$  for all  $a$  in a nonempty open interval  $J \subseteq (-c, c)$ , then  $Z = 0$ .*

*Proof.* The function  $A(a) = \int_{\mathbb{R}} e^{aw} Z(w) dw$  is holomorphic on  $\{| \operatorname{Re} a | < c\}$  (differentiation under the integral sign) and vanishes on  $J$ , hence on the whole strip. So  $\hat{Z}(k) = \int e^{-ikw} Z(w) dw = A(-ik) = 0$  for all real  $k$ . For  $\epsilon > 0$  and  $w_0 \in \mathbb{R}$ , Fubini's theorem and  $\int_{\mathbb{R}} e^{iku - \epsilon^2 k^2/2} dk = (\sqrt{2\pi}/\epsilon)e^{-u^2/(2\epsilon^2)}$  give

$$0 = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{Z}(k) e^{ikw_0 - \epsilon^2 k^2/2} dk = \int_{\mathbb{R}} Z(w) \frac{e^{-(w-w_0)^2/(2\epsilon^2)}}{\epsilon\sqrt{2\pi}} dw,$$

and the right side tends to  $Z(w_0)$  as  $\epsilon \rightarrow 0$ , since  $Z$  is bounded and continuous.  $\square$

*Step (b) without distributions.* Let  $\omega, \sigma, V$  and  $W$  be as in the proof of Theorem C. By (3.5),  $\hat{\omega}(a) - \hat{\sigma}(a) = O(a^2)$  as  $a \rightarrow 0$ , so  $\omega - \sigma$  has zero mass and zero first moment, and  $W(w) = \int (w-u)_+ d(\omega - \sigma)(u) = \int (u-w)_+ d(\omega + \sigma)(u)$ . For  $0 < a < \pi$ , the second expression and Fubini's theorem, which applies because  $\iint e^{aw} (u-w)_+ dw d(\omega + \sigma)(u) = a^{-2} \int e^{au} d(\omega + \sigma)(u) < \infty$ , give  $\int e^{aw} W(w) dw = a^{-2}(\hat{\omega}(a) - \hat{\sigma}(a)) = \int e^{aw} V(w) dw$ . The functions  $W$  and  $V$  are continuous and  $O(e^{-\pi|w|})$  (step (b) of Section 3.6), so Lemma B.7 with  $Z = W - V$ ,  $c = \pi$  and  $J = (0, \pi)$  gives  $W = V$ . Steps (c) and (d) are unchanged; step (d) uses only Theorem D and the formula  $\mathcal{G}(0, t) = \|BgP\|_F^2(1 - e^{-t})/t$  of Step 4 of Section 3.5.

## B.6. Remarks.

*Remark B.8* (arbitrary Hermitian  $B$ ; Stahl's theorem). Let  $B$  be a Hermitian matrix with distinct eigenvalues  $b_1 < \dots < b_n$  ( $n \geq 2$ ) and spectral projections  $\Pi_k$ , let  $H_d = \sum_k \Pi_k H \Pi_k$ , and let  $G_B = (b_1, b_n) \setminus \{b_2, \dots, b_{n-1}\}$ . For  $\tau \in G_B$  let  $y_i(\tau)$  be the eigenvalues of  $(B - \tau)^{-1}H$ . The proof of Theorem B.1 extends to this setting: in a gap  $(b_l, b_{l+1})$  exactly  $\sum_{k>l} \operatorname{rank} \Pi_k$  roots lie in  $\mathbb{C}_+$ , the bounds of Lemma B.3 hold with constants that depend on the gap and are integrable

over it, and at an interior eigenvalue  $b_k$  the boundary terms coming from the two adjacent gaps cancel. The result is

$$\frac{1}{2\pi} \int_{G_B} \sum_i |\operatorname{Im} y_i(\tau)| d\tau = \operatorname{Tr} H_+ - \operatorname{Tr}(H_d)_+.$$

The argument of Section B.4 then shows that, for Hermitian  $A$  and  $B$ ,

$$\operatorname{Tr} e^{aA-tB} - \sum_k e^{-tb_k} \operatorname{Tr}_{\Pi_k} e^{a\Pi_k A \Pi_k} = a^2 \iint e^{as-t\tau} \rho_{A,B}(s, \tau) ds d\tau,$$

where  $\rho_{A,B}(s, \tau) = \frac{1}{2\pi} \sum_i |\operatorname{Im} x_i(s, \tau)|$  for  $\tau \in G_B$ , with  $x_i(s, \tau)$  the roots of  $\det(A - s - x(B - \tau))$ , and  $\rho_{A,B} = 0$  otherwise. At  $a = 1$  this says that  $t \mapsto \operatorname{Tr} e^{A-tB}$  is the Laplace transform of the positive measure  $\sum_k \operatorname{Tr}_{\Pi_k}(e^{\Pi_k A \Pi_k}) \delta_{b_k} + (\int_{\mathbb{R}} e^s \rho_{A,B}(s, \tau) ds) d\tau$  on  $[b_1, b_n]$ , which is Stahl's theorem [40], the former conjecture of Bessis, Moussa and Villani [5], here with a proof that uses neither distribution theory nor Fourier analysis. We make no priority claim about this by-product; for the history of the theorem and of the Fourier-slice mechanism see [40, 13, 28, 17]. The details are in the accompanying notes [32], and they were checked in the re-verification of Section 11.3; they are not used in this paper.

*Remark B.9 (formalisation).* The Lean theorem `QBP27.StripL3b.hyp_RI` is Theorem B.1, and `QBP27.StripL3b.theorem1_bmv2` is Theorem D, proved from it as in Section B.4. In the uniqueness step of the proof of Theorem C, the formal proof uses the Fourier inversion theorem of Mathlib in place of Lemma B.7.

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